Subset sums and four summands
A cardinality–85 exclusion and the upper bound 84

Proof supplement to the supplied upper–85 report

5 October 2026

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1 The new upper bound and its proof boundary

For a finite real set A, let $FS(A)$ include the empty subset sum, and let $4A$ allow repeated summands. We continue the supplied report on the condition $FS(A) \subseteq 4A$. Its established upper bound and its explicitly stated general mathematical inputs are retained.

Theorem 1. Under the mathematical inputs of the supplied upper–85 report, every feasible finite real set has at most 84 elements. The supplied 71–element construction remains feasible. Consequently

$$71 \le N \le 84.$$

The previously established $N \le 86$ theorem is an explicit input in the original report. The supplied attachments do not contain its complete proof or an identifiable citation for that global theorem. It is not reproved here; the improved global conclusion retains that dependency. The new contribution is the complete exclusion of cardinality 85, in addition to the original report's exclusion of cardinality 86.

The new work excludes cardinality 85. It does not assume that arbitrary subsets of a feasible set are feasible. If a feasible 85–element set omitted zero, adjoining zero would produce a feasible 86–element set, contrary to the retained bound. Thus zero is present. The rationalization lemma preserves both cardinality and this zero coordinate. The fresh structural argument below then reduces to an integral source D and integral core E with parameters

$$|D| = 77, \quad |E| = 69, \quad r_s + r_t = 68, \quad I = (r_s - 34)^2, \quad \Lambda = 1191 + I, \quad \Omega = 600 - I.$$

The complete 85–element argument is given in the following supplements. All references to original covering, peeling, defect and inverse lemmas are to the supplied report and its bundled mathematical sources; they are not new claims certified merely by a program run.

1.1 What makes the improvement possible

There are three changes. First, all structural boundary cases are recomputed at the new cardinality, including the $13 : 55$ and $14 : 54$ boundaries. Second, the original fixed-label gap inventory is sharpened to 11, 8, and 15 labels for the first gap, beta-active second gap, and their union. Third, the two source envelopes are compared using the actual distinct fringe sums.

For the four offsets $1 \le a_v < b_v < c_v < e_v$ beyond each core maximum, let P_v count the distinct pair sums of $\{a_v, b_v, c_v\}$ above c_v, and B_v the distinct pair sums of $\{a_v, b_v, c_v, e_v\}$ above e_v. The elementary inequality

$$P_v + B_v - a_v - b_v \le 5$$

improves the source mass parameter to

$$T = 77j - 280 - w + I + \tau + \delta \le 165, \qquad I + \tau + \delta \le 445 - 77j + w.$$

The one-maximum envelope also yields $e_v \ge \max(4, T - 89)$ on each physical sign. These are bounds on actual source sets, not formal endpoint descriptors.

1.2 Exhaustive case assembly

Write z for the core hull-hole count and j for the number of genuinely empty ordinary whole units. Paired units retain a single capacity throughout. The physical signs are never reordered after an owner is selected.

Actual family	Indices	Supporting proof below
Zero core holes	Z_0, Z_1, Z_2	Adaptive forests and raw gaps
Zero core holes	Z_3	Endpoint cards and physical side union
Zero core holes	Z_4	Retention cards and ownerless certificate
Zero core holes	Z_5	Generic rank cards and final tail arguments
Positive low holes	P_1, P_5	Common localization and terminal/source bounds
Positive low holes	P_2	Fresh activity and rooted forest proof
Positive low holes	P_3	Dense seed, two actual forms, endpoint proofs
Positive low holes	P_4	Fresh source profiles and owner certificates
High core holes	H, $z \geq 15$	Paired profile floors and full localization

The no-empty-unit obstruction covers the zero-index positive-hole alternatives arising in the structural reduction; the fresh no-hole reduction covers Z_0. The scalar budgets exclude indices beyond those shown. These statements are proved in the supplements, including the actual activity classifications.

1.3 Verification and dependencies

The new verifier checks explicit finite arithmetic, profile minima, the dense seed lemma, small offset inventories, and the final scalar certificates. Its coverage manifest identifies every new written proof and every checked script. The inherited verification program and mathematical sources are bundled intact under `audit/verification_program`. The new driver can also run their full verification suite.

The verifier does not purport to formalize the infinite inverse theorems, the passage from actual sets to forest inequalities, or the complete prose proof. Those remain written mathematical arguments, with their hypotheses and original input boundary retained. The package therefore supplies both the proof and a reproducible check of its specified finite components.

2 Complete SIZE85 structure: integration and dependency check

The fresh structural argument is complete relative to the general mathematical inputs already stated and audited in the supplied upper85 foundation sources. It does not import a SIZE86 inverse row or finite exception census. The two companion proofs are `structure85_ap_bootstrap.md` and `../inverse/proof85_inverse.md`. The additional strict-inverse a13 closure is now included in the latter and its exact checker.

2.1 Entry: a prescribed zero and rationalization

Assume the already established upper bound `|A|<=85`. If a hypothetical 85-element feasible real set did not contain zero, adjoining zero would keep every subset sum feasible: its subset sums are unchanged and its fourfold sumset only grows. This would give a feasible 86-element set, a contradiction. Thus a hypothetical size85 set contains zero.

To pass from its existence to an integral feasible example, enumerate the 85 coordinates and choose one fourfold representation of every subset sum. These finitely many identities form a homogeneous linear system with integer coefficients. Include the equation setting the distinguished zero coordinate equal to zero. Its real kernel has a basis over the rationals; rational points are dense in that kernel. Take a rational kernel point sufficiently close to the original point to preserve all strict coordinate orders and nonzero signs. The chosen identities preserve feasibility, and the added zero equation preserves the prescribed zero exactly. A positive common-denominator scale then gives 85 distinct integers. This argument transfers existence; it does not claim that the original real coordinates themselves lie on a lattice.

2.2 Exact resulting theorem

After the outer four-on-each-sign deletion and two further two-on-each-sign peels, a positive normalization gives integral source sets of sizes

$$|D| = 77, \quad |D_1| = 73, \quad E = P \cup \{0\} \cup (-Q), \quad |E| = 69, \quad a + b = 68, \quad 15 \le a, b \le 53.$$

Both positive core sides are individually primitive. With `T_r=r(r+1)/2`, `k=34-min(a,b)` and `I=k2`,

$$\Lambda = 1191 + I, \quad \Omega = 600 - I, \quad 0 \le k \le 19.$$

On this common integer lattice the hull-hole counts satisfy

$$h_P, h_Q \le \kappa, \quad z = h_P + h_Q - \kappa, \quad 0 \le z \le \min(h_P, h_Q) \le \kappa, \quad 0 \le w \le 4z.$$

The actual extreme subset-sum translates in both peels intersect. If `calT=kappa+ell1+ell2+g`, then

$$c + \rho + \mathcal{T} \le 600 - I, \quad \mathcal{T} \ge \kappa + 144 - 3z, \quad \mathcal{T} \ge \max(48, z, 144 - 2z).$$

Each `rho_X>=0`. A non-AP side support obeys `mu_X>=r-3` and `rho_X>=r-14` for `15<=r<=20`, or `rho_X>=r-5` for `r>=21`. Maximum removal gives

$$R_X \le r + \rho_X + h_X + 2, \qquad R_P + R_Q \le 72 + \rho + \kappa + z.$$

In particular, if `z>0`, both supports are non-AP and

$$\mu \ge 62, \quad \tau = \mu + h_P + h_Q \ge 62 + 2z, \quad \rho \ge \begin{cases} 58, & \min(a,b) \ge 21, \\ 49, & 15 \le \min(a,b) \le 20. \end{cases}$$

No outer-integrality or leading-activity assertion is part of this theorem.

2.3 Exhaustive branch checklist

Core branch	Alignment and pre-lattice argument	Boundary or primitivity closure
Initial deficient sign	Provisional available-sign peel; use `24*85+77=2117` before fixing `d=77`	One-sided 70-point lower count `2416` excludes it
Sign disappearing in later peel	Fixed `d=77`, one-sided 69-point core	Lower `2347` exceeds upper `1925`
`a=10`	Larger support must be AP, residue count and strict layer rows	`G144` contradicts budget `Omega+8=32`
`a=11`	AP branch or strict simultaneous inverse, with small-side `+16` correction	`G144`, or `T48+rho52-8=92>71`
`a=12`	AP branch or strict inverse, with `+16` correction	`z>=36`; effective `(56,5,132)` forces `rhoQ125>124`
Larger support AP, `a>=13`	AP-plus-set formula, one residue, unclipped gaps, `z=0`; layers then `G144`	Any nonprimitive smaller side has `c<=59<77`, `w=0`; `j0` obstruction; boundaries `a13,a14` also have `c<77`
Only smaller support AP, one residue, clipped gap or unique off-scale coefficient	Final disjoint supports cost `T_a+T_(b-2)+2-2b+rhoP`	Exact budget contradiction throughout
Only smaller support AP, one residue, no gap	`z=0`, `hQ=kappa`; both primitive	`a13: 192>159`; `a14: c<=7<77,w=0`, `j0` obstruction
Smaller AP, two residues	Order-two residue, exact parity-fiber identity before hole bound; odd coefficient `W<=15`; strict AP-specific layers	`G144` gives `c<=61<77,w<=56`; `j0` obstruction
Both supports non-AP, sufficient inverse test fails, `k3..20`	Fresh 18 counted-endpoint repairs, including strengthened non-AP prefix bounds; short-diameter theorem	`z<=16`; strict layers before `G144`; nonprimitive `P` gives `c<=28<77,w<=64`, hence `j0`
Both supports non-AP, failed-test `a13`	Intrinsic primitive Q effective `(55,2,52)` gives `hQ<=4`; residue and short diameter	`G144: T+rho>=185>159`
Both supports non-AP, sufficient inverse test holds	Strict common lattice and hole bounds; all32 large-hole final-overlap orientations repaired before `G144`	Strict hole bounds imply individual primitivity
Both supports non-AP, strict-test `a13`	`kappa<=110`; corrected layer bound `-811<0`; then `G144`, `z>=17`	Effective `(55,4,119)` gives `rhoQ>=113`; `48-1+113=160>159`
Both supports non-AP, repaired `a14`	Endpoint `H<=8`; corrected strict pre-layer	`c<=23<77,w<=32`, `j0`
Both supports non-AP, strict-test `a14`	Corrected strict pre-layer; `G144`	If `z>=13`, effective `(54,6,168)` gives `rhoQ153`; otherwise `c<=31,w<=48`; both impossible

Reflection is used only to order the scalar core sizes; both physical sign orientations are included in every actual-owner statement. AP support type, actual strict-test success/failure, and the boundary rows partition the entire bootstrap range `a>=10`; no actual test is replaced by its sufficient floor.

2.4 Dependency ordering: no use of integer holes before its lattice

1. The covering, peeling, `mu/rho` estimates, exact budget, inverse tests, residue counts, and counted-endpoint repairs are cardinality statements and valid before common-source integrality.
2. The short-diameter theorem is used on the proved common core lattice, and does not require unproved individual primitivity of the other side.
3. The pre-G comparisons use `C+Omega=-165-2I`, plus `16` exactly when the small-side maximum-deletion prefix has fewer than14 elements. Large-hole overlap is supplied independently of `G144`.
4. A strict comparison proves lattice alignment of the next layer and the intersection of its two extreme support translates. This is applied to both layers before counting their integer restricted-sum hull holes.
5. Only then use the sharp integral four-extreme inequality `G=144`, derive `w<=4z` and the structural T bounds, and invoke the conditional `j0` obstruction where the already proved caps supply `c<77,w<=104`.

6. Individual primitivity follows after these steps in the repaired branches; it is available immediately from strict hole bounds in strict-inverse branches. Thus neither residue repair nor `G144` assumes primitivity of P.

The remaining retained inputs are the general inverse/stability/defect, restricted-sum and cover lemmas, not fresh finite cardinality-specific rows. The exact checkers `ap_bootstrap85.py` and `../inverse/check85_inverse.py` both pass; their arithmetic checks supplement the actual-set applicability arguments in the written companion proofs.

3 Fresh structural deductions at cardinality 85

These deductions concern a hypothetical zero-containing **integral feasible** set A of size 85. Zero follows only after the preceding exclusion of all sizes at least 86 has been accepted: a zero-free size-85 example could otherwise be extended by adjoining zero. The general inverse/stability/defect theorems are the imported mathematical inputs identified in the audited upper-85 package. This is an intermediate proof supplement, not yet the complete STRUCTURE85 theorem; the both-non-AP inverse and overlap repairs are supplied separately.

3.1 1. Bootstrap, derived from the new cardinality

Four-on-each-sign outer deletion and two four-element peels give d=77, e=69, m=68. If an outer sign had fewer than four elements, provisional peeling from the available sign gives a one-sided core with at least 70 points, whose subset-sum count is at least 1+T_69=2416. Before the outer deletion size has been fixed, use the safe upper bound s<=24|D|+77<=24*85+77=2117, not a premature substitution |D|=77. If a sign disappears under the further peels, a one-sided 69-point core gives s>=1+T_68=2347, whereas s<=24*77+77=1925. Thus both physical signs actually support all the required extreme deletions.

The unpaired-absolute-value estimate gives q^2<=96e+1073-e^2=2936. Here q is even, so q<=54; at least (68-54)/2=7 opposite pairs remain in E. In particular zero is not a diagonal exception and both core sides have at least 7 elements.

Reflect only in scalar analysis and write a=34-k, b=34+k. Then Lambda=1+T_a+T_b=1191+k^2 and Omega=22*77+97-Lambda=600-k^2. The larger-side adjusted defect is nonnegative; the smaller-side uniform bound is rho_P>=-8. The exact slack identity therefore implies k^2<=608, so k<=24 and a>=10. The initial small rows are:

a	b	k	Omega	Omega+8
10	58	24	24	32
11	57	23	71	79
12	56	22	116	124
13	55	21	159	167
14	54	20	200	208

At a=14 the threshold-14 theorem improves the last slack bound to Omega=200. At a=13, the actual threshold-14 induction table gives Phi_13<=4, hence rho_P>=-2; if its support is non-AP, the general non-AP theorem gives the stronger rho_P>=-1.

3.2 2. Exact layer constants

On a common core lattice, maximum removal gives R_s+R_t<=72+rho+kappa+z when both core sizes are at least 15. A smaller side requires adding 8 to this maximum bound, or 16 to a two-span comparison. The conservative pre-lattice D1 comparison has constant

```
C=2m+4e+14-Lambda=-765-k^2,
C+Omega=-165-2k^2.
```

Thus F1-2F0<=-165-2k^2+2z, with an additional +16 if a<=14. After the source layers and their actual overlaps have been established, the sharp restricted-sum inequality is G=2d-10=144. It gives

```
T>=kappa+144-3z, T>=144-2z, T>=z, T>=48,
c+rho+T<=600-k^2, w<=4z.
```

The quantitative layer rows, using |R(E)|<=2R-1 instead of the conservative 2R+1 bound, are freshly derived as follows:

```
F2-2F1=-293-2k^2-c-rho-mu-2kappa-2theta-2ell2-2g;
V0-C1+delta+2z<=-292-2k^2-77j-mu-rho;
F1-2F0<=-167-2k^2+rho-mu+2z-c-2theta-ell1-g;
U0-C0+delta-2z<=-166-2k^2-77j, U0-C0<=0.
```

The last two rows assume the maximum-removal bound with core sizes >=15; they are not silently applied on small boundaries.

3.3 3. Boundaries 10, 11 and 12

At a=10 a non-AP larger Q would give rho>=-8+(58-5)=45>Omega=24. Hence Q is AP. Its intrinsic step forces P onto the same lattice by residue counting, because kappa<=32<T_58. The AP-plus-set formula precludes a clipped gap and gives h_Q=z=0,h_P=kappa. The conservative D1 comparison with +16 is strict, as is the outer comparison Lambda>4d-10+(Omega+8). Consequently D is integral on this lattice and G144 applies. But T>=144>Omega+8, impossible.

At a=11 and non-AP Q, rho_Q>=52, so kappa<=71+8-52=27<T_11-1=65. The strict simultaneous inverse supplies the common lattice and both hole bounds. The layer comparisons are strict even with z<=79 and +16; G144 is therefore available. Then T+rho>=48+52-8=92>71. If Q is AP, the preceding z=0 argument gives T>=144>79.

At a=12 and non-AP Q, rho_Q>=51 and kappa<=116+8-51=73<T_12-1=77. The strict inverse again supplies the common lattice and strict Q upper-hole bound; both layer comparisons are strict. From G144 and rho>=43 we obtain 144-2z+43<=116, hence z>=36. Therefore h_Q>=36. The parameterized effective hole witness (r,beta,C)=(56,5,132) has

```
K=17,D=149,E=150,M=20,J=20,
(r-J+1)beta-D=36>0,
T_(J-1)-1-D=40>0.
```

It gives mu_Q>=189; the retained flexible defect inverse gives rho_Q>=125. But the budget gives rho_Q<=116+8=124. The AP-Q branch has z=0 and is already excluded by T>=144>124. Thus every surviving core has a>=13.

3.4 4. A general no-empty-unit obstruction

This uses only the original 55-fixed-label raw gap bounds, not any newly strengthened gap count. Once D is integral, suppose j=0 and w<=104. All four own centers on both signs are active, so all outer magnitudes are integral. The source maxima satisfy M>=m+8=76; active C7 gives t_v=d_v-M>=5, hence d_v>=81. Choose a genuinely contributing beta sign s (the beta pair supplies one in branch A, and both beta singletons contribute in branch B). Then

```
t_s<=(57+eta_s)/2<=(57+w)/2<=80.5,
t_t<=56+eta_t<=56+w<=160.
```

An active C4_s forces t_s>d_t>=81. An active C4_t forces t_t>h_s+d_s>2d_s>=162. Both alternatives are impossible, although pair1 must contribute when j=0. Thus every aligned configuration with j=0,w<=104 is excluded. In particular this applies whenever c<77 and w<=104.

3.5 5. Complete AP-branch simplification for a>=13

The larger-AP case forces one common residue and gives z=0 as before. If its smaller side is nonprimitive, h_P>=T_a+mu_P, and hence c<=Omega-144-T_a-rho_P. The maximum cap over all a>=13 is 59; the a=13 universal floor rho_P>=-2 is included. Therefore c<77 and the no-empty-unit obstruction excludes every such branch, without 152 auxiliary certificate systems. The larger unit-step AP side is intrinsically primitive. A smaller primitive side is the remaining legitimate z=0 structural alternative. At a=13, Q AP also gives c<=159+2-144=17, and at a=14, Q AP gives c<=200-144=56; both boundaries are excluded by the same obstruction.

Suppose only the smaller support is AP. The residue-count inequality kappa>= (#residues-1)(u-1), with u>=T_a+1, and Omega<2T_a over all the present rows permits at most two residue classes. A one-residue clipped gap or a unique off-scale generator forces final disjoint supports of Q and mu_Q>=T_(b-2). The necessary budget cost

```
T_a+T_(b-2)+2-2b+rho_P
```

strictly exceeds Omega in every row; the executable checks these margins. If there is one residue and no clipped gap, z=0,h_Q=kappa,h_P=0. The larger side is primitive because h_Q<=Omega-rho<T_b<=|FS(Q)|-1, and the smaller unit-step AP support is primitive. This is another legitimate z=0 alternative. Its a=13 boundary is impossible because T+rho>=144+50-2=192>159. At a=14, c<=200-144-49=7<77 and w=0, so the no-empty-unit obstruction excludes it.

In the remaining two-residue case, at least two nonzero-residue generators force that residue to have order two. Normalize U=2[0,u-1]. Let W be the least odd Q coefficient and t>=2 the number of odd Q generators. The parity-fiber identity, before using a hole bound, is

```
kappa=u-1+eta, h_P=u-1,
h_Q=kappa-u+W, z=W-1,
mu_Q+kappa=sum(Q)+u-W-T_b.
```

Distinct even and odd generators give

```
sum(Q)>=(b-t)(b-t+1)+tW+t(t-1).
```

The exact finite minimization in ap_bootstrap85.py proves that W>=17 contradicts mu_Q+kappa<=Omega+2b-2-rho_P, with minimum contradiction margin 22. It uses the floor rho_P>=0 for a>=14 and rho_P>=-2 for a=13. Since W is odd, W<=15, hence z<=14; also W<u, so h_P,h_Q<=kappa.

For this half-step branch, distinct-generator bounds give

```
R_P+R_Q<=m+a+mu+kappa-T_a+14.
```

The conservative D1 comparison therefore gives

```
F1-2F0<=2367+2a-4T_a-2T_b<=-761<0.
```

The outer-layer strict comparison likewise holds. D is integral and its extreme support translates intersect; w<=4z<=56 and G144 apply. Using rho_Q>=b-5 then gives

```
c<=Omega-T_a-144+42-(b-5)-rho_P<=61<77.
```

The no-empty-unit obstruction excludes every half-step branch. This includes the smaller-AP a=13 boundary. The only remaining a=13 or a=14 possibilities have both supports non-AP; their inverse repairs are treated separately.

3.6 6. Executable scope

ap_bootstrap85.py checks every scalar row k=0,...,21, the exact odd-coefficient mass minimum, half-step and nonprimitive caps, lattice-comparison constants, and one-residue contradiction margins. It supplies no computational proof of the general inverse theorem or the actual-set parity-fiber identities. The result file is ap_bootstrap85_results.json.

4 Fresh SIZE85 inverse alignment and final-support overlap

This companion proves the inverse portion of a SIZE85 structural argument. Its general mathematical inputs are the previously established covering and peeling identities, prefix defect recurrence, counted-hole lemma, effective hole lemma, flexible defect inverse, and Stanchescu's two distinct-set theorems. It additionally invokes the SIZE85 no-empty-unit obstruction after lattice forcing and G144; that obstruction must be proved in the common source. The checker `check85_inverse.py` checks only the explicit finite arithmetic.

4.1 1. Parameters and exact pre-lattice budget

Use m=68,d=77,e=69,G=144. After reflection a=34-k,b=34+k,I=k². Put T_r=r(r+1)/2, Lambda=1+T_a+T_b=1191+I, Omega=600-I, and

```
C=2m+4e+14-Lambda=-765-I, C+Omega=-165-2I.
```

For U=FS(P),V=FS(Q), write u=1+T_a+muP,v=1+T_b+muQ and kappa=|U+V|-u-v+1. The exact budget is

```
rhoP+rhoQ+kappa+2theta+c+ell1+ell2+g=Omega.
```

The non-AP support floors used below are n(r)=r-5 for r>=21 and n(r)=r-14 for 14<=r<=20. At r=13, the sharper non-AP floor is -1. All mu values are nonnegative, and non-AP mu_X>=r-3. Also rho<=mu. The main range is a>=15, k0..19. The a14 and a13 boundaries are included and closed explicitly below; smaller boundaries are treated in the bootstrap.

The larger-side sufficient strict inverse condition is automatic because Omega-n(a)-n(b)<T_b+b-4=v_min-2 for every k0..21. Do not replace this with Omega<T_b-1: that unnecessarily stronger inequality is false at balanced k0, where Omega=600,T34-1=594.

The sufficient small-side condition is

```
muP+rhoP+rhoQ>Omega-T_a+2.
```

Its floor margin is positive exactly at k0,1,2 and nonpositive at k3..21. Whenever the actual condition fails, the following fresh cardinality bounds hold:

```
rhoQ<=R=Omega-T_a+2-(a-3)-n(a),
kappa<=Kmax=Omega-n(a)-n(b), u<=Umax=Kmax+3,
v>=Vmin=T_b+b-2, u>=Umin=T_a+a-2.
```

4.2 2. Two counted-endpoint repairs

The original repair starts with rho14>=0. Given rhoQ<=R, fix beta>=3 with (beta-1)(b-14)>R. Some index j>=15 has y_j<=beta; take the last one. The recurrence rho_i>=rho_(i-1)+y_i-2 and nonnegative prefix defects give

```
j>=j0=b-floor(R/(beta-1)).
```

The counted-hole lemma applies if j0>=2beta+5. It supplies, on an intrinsic prefix lattice, a deleted interval with at most beta lower holes in [1,2beta] and their reflections. That prefix is primitive on its own lattice because j elements among j+beta positive positions cannot all be divisible by 2.

Assume T_j0-b-1>R. A first later coefficient off this lattice has disjoint support translates and raises rho above R; the subsequent increments, all at least beta+1, cannot undo this. Thus every coefficient is on that lattice. Every later increment is at most R+2. If a current m-prefix has sum L and the next coefficient x satisfies x>=L-4beta, the new translate adds above L at least min(x,L+1)-2beta>=L-6beta>=T_m-6beta points. When T_m-6beta>m+R+3, this exceeds its allowed increment. Therefore x<=L-4beta-1. Also x>max(prefix)>=m>2beta. The central intervals join, the lower holes remain unchanged, and reflection supplies the upper holes. Induction gives final hQ<=H=2beta and a central interval [2beta+1,L_Q-2beta-1].

The strengthened repair uses a non-AP prefix floor, which is essential in several SIZE85 rows. Suppose the final Q support is non-AP, rhoQ<=R, and q>=20 satisfies T_q-b-1>R. Then every prefix of size at least q is non-AP. Indeed, if a last AP-support prefix had size l>=q, the next support must break the progression by disjoint translates. At that step y>=T_l-l. Every later increment is positive by the zero-increment equality theorem. The prefix recurrence, starting from rho_l>=0, gives

```
rhoQ>=T_l-b-1>=T_q-b-1>R,
```

a contradiction. Hence rho_q>=n(q), and rho_j>=j-5 at j>=max(q,21).

If n(q)+(beta-1)(b-q)>R, there is a last j>=q+1 with y_j<=beta. The later increments yield

```
R>=rho_j+(beta-1)(b-j)
  >=(j-5)+(beta-1)(b-j),
```

and thus

```
j>=j0=max(q+1,21,ceil(((beta-1)b-R-5)/(beta-2))).
```

Now apply exactly the same counted-hole, off-scale, and bridge arguments. This strengthens the index bound without imposing a new hypothesis on Q.

4.3 3. All failed-inverse rows acquire a common lattice

The explicit choices (k,b,R,beta,q,j0,H) are:

k	b	R	beta	q	j0	H
3	37	43	3	14	16	6
4	38	69	4	14	15	8
5	39	92	6	14	21	12
6	40	112	6	14	18	12
7	41	129	7	14	20	14
8	42	143	8	14	22	16
9	43	154	8	14	21	16
10	44	162	8	20	24	16
11	45	167	8	14	22	16
12	46	169	8	14	22	16
13	47	168	7	21	22	14
14	48	173	7	21	22	14
15	49	166	7	14	22	14
16	50	156	7	14	24	14
17	51	143	6	14	23	12
18	52	127	5	14	21	10
19	53	108	5	14	26	10
20	54	86	4	14	26	8

Every strict tail/off-scale/bridge margin and every intermediate m in j0..b-1 is checked with integers. For k10, T20-b-1-R=3; for k13,k14, T21-b-1-R=15,9. These are the three uses of the strengthened repair.

Residue counting gives kappa>=(t-1)(v-1) if U occupies t residue classes modulo Q's prefix lattice. Since Kmax<Vmin-1, U has only one class, and 0 in U places P on the same lattice. Set alpha=4beta+2. Q's central interval has length l=L_Q-4beta-1=v+hQ-alpha>=v-alpha. Apply the AP-plus-set formula to that interval and U. Since it is a subset of V, its clipped gap sum is at most kappa+alpha. The checked inequality

```
Vmin-Kmax-2alpha-1>0
```

makes kappa+alpha<l-1, so no gap can be clipped; hence hP<=kappa+alpha. The further positive inequalities

```
Vmin-Kmax-alpha-2>0, Umin-H-2>0
```

give both diameters at most u+v-3. Stanchescu's short-diameter theorem therefore gives the exact hP,hQ<=kappa, and z=hP+hQ-kappa<=H. No individual primitivity of P has been assumed.

The pre-G D1 comparison is bounded by -165-2I+2H, with the required +16 correction at a14; every row is strict. The outer comparison is strict since 1191>4*77-10+600. This proves D1, D integral and the extreme support intersections before any integer-hole G144 estimate is used.

At a13/b55, the actual failed sufficient test gives rhoQ<=61. Normalize Q by its intrinsic gcd. The primitive effective-hole witness (r,beta,C_h)=(55,2,52) has (K,D,M,J)=(11,63,13,13), strict margins23,14, and forces muQ>=108, rhoQ>=62 whenever hQ>4. Hence hQ<=4. Residue counting, endpoint hulls and the same short-diameter theorem give hP,hQ<=kappa and z<=4; all required dense-diameter margins are positive. The D1 comparison with +16 is -1023. After G144, rhoP+rhoQ+T>=-1+50+144-8=185>159, so this boundary is impossible.

The remaining a13/b55 branch is the actual sufficient strict inverse branch. Its strict hole bounds give individual primitivity on the joint lattice. Here kappa<=159-(-1)-50=110, so the rough D1 comparison is at most -165-2*441+2*110+16=-811<0; the outer comparison is strict as above. Thus lattice forcing and G144 are established without using any later hole estimate. The post-G budget gives

```
-1+50+144-2z<=159,
```

so z>=17 and hQ>=z>8. The fresh primitive effective-hole witness (r,beta,C_h)=(55,4,119) has (K,D,M,J)=(17,136,19,19), strict margins 12,34, and forces muQ>=175 whenever hQ>8. The flexible defect inverse then gives rhoQ>=113 (the g7,8,9,10 rows each give 113). Since T>=48,

```
rhoP+rhoQ+T>=-1+113+48=160>159.
```

Consequently both actual inverse branches at a13 are excluded.

4.4 4. Large-hole final-support overlap: all 32 orientations

For unrepaired strict-inverse main branches, D1 is immediately strict when z<=82+I. Therefore only z>=Z=83+I needs an extra argument. Every repaired branch has z<=16<Z and cannot enter this large-hole case. The strict inverse upper-hole bounds hX<|FS(X)|-1 are retained here. The budget kappa>=z and both non-AP floors exclude k16..19 directly. The remaining k0..15 have 32 oriented possible disjoint-final-support sides.

If the side X has r coefficients and its final prefix translates are disjoint, muX>=T_(r-2), so rhoX>=T_(r-2)+2-2r. The elementary contradiction margin is

```
T_(r-2)+2-2r+n(s)+Z-Omega.
```

It is positive in 18 orientations. The remaining 14 comprise P at k0..12 and also Q at balanced k0. Put R=Omega-(T_(r-2)+2-2r)-Z; the budget forces the other side's defect at most R. Apply the strengthened repair with q20 using the following exact (k,s,R,beta,j0) choices:

k	s	R	beta	j0
0	34	55	5	26
1	35	83	7	25
2	36	106	8	24
3	37	124	8	22
4	38	137	9	24
5	39	145	9	24
6	40	148	9	24
7	41	146	8	23
8	42	139	8	25
9	43	127	7	26
10	44	110	6	27
11	45	88	5	29
12	46	61	4	36

All strict margins hold, including T20-s-1>R; the final support has at most 2beta<=18 holes, whereas h_other>=z>=Z>=83, a contradiction. Thus both sides' actual final prefix support translates intersect. This is a fresh endpoint argument; the weaker effective-hole tests alone do not close k2..8 at SIZE85.

Let Y remove the largest coefficient from an r-side X. Intersecting final support hulls cover the full side hull, so hX<=2hY. The exact maximum-deletion identity, with rhoY>=0 and destroyed-defect count nu>=0, gives

```
max X<=r+rhoX+hX/2+2.
```

The common D1 comparison becomes

```
|FS(D1)|-2s<=C+2rho-mu+z+ell2
          <=C+Omega+rho-mu<=-165-2I<0.
```

Here the one budget is used via rho+kappa+ell2<=Omega and z<=kappa. Together with the outer comparison, this finishes pre-G non-AP lattice forcing without circular use of G144.

4.5 5. Post-G individual primitivity and a14 closure

Only after the proved lattice and overlap steps use

```
T>=kappa+144-3z, c+rhoP+rhoQ+T<=Omega, w<=4z.
```

In a strict simultaneous-inverse branch, an individual gcd>=2 would imply hX>=|FS(X)|-1, contradicting the strict inverse upper-hole bound. In an endpoint-repaired branch, Q is primitive on the constructed prefix lattice. If P had gcd g>=2, its intrinsic support is non-AP and has h0>=1. Rescaling gives

```
hP=(g-1)(T_a+muP)+g*h0>=T_a+a-1.
```

Thus

```
c<=Omega-n(a)-n(b)-T_a-a+1-144+3H.
```

The exact caps for k3..20 are

```
-119,-88,-54,-35,-13,6,16,23,27,28,20,24,16,5,-15,-38,-58,-87.
```

Negative caps exclude immediately. In the remaining rows c<=28<77, so all 18 ordinary covering units contribute (j0), and w<=4H<=64. The SIZE85 no-empty-unit obstruction, valid for w<=104, excludes these. Hence P is individually primitive in every surviving repaired main row.

For a14/b54, the repaired branch has z<=8, so c<=200-49-(144-16)=23<77 and w<=32; j0 excludes it entirely. In the strict-inverse branch, if z>=13, the witness (r,beta,C_h)=(54,6, 168) has (K,D,M,J)=(19,187,22,22), strict margins11,43. It yields muQ>=223, rhoQ>=153. Since T>=48, the budget would require 200>=153+48=201. Therefore z<=12. Then c<= 31<77,w<=48, and j0 again excludes the branch. This closes a14 without using P primitivity first.

4.6 6. Verification boundary

Run `python -I -S -B check85_inverse.py`. It reconstructs all SIZE85 parameters, all 18 endpoint inverse rows, both a13 effective boundary, all 32 overlap orientations and 14 endpoint repairs, nonprimitive caps, and the a14 effective closure. The mathematical passage from actual sets to the displayed inequalities, the retained published general theorems, the bootstrap/AP branches and the no-empty-unit obstruction remain written proofs. No SIZE86 row combination or old exception census is reused.

5 A sharper elementary source envelope at size 85

This deduction concerns the **77-point source** in the fresh size-85 reduction. It does not use the computed fixed-label tables. The 69-point inner source has zero, positive maximum `R_s`, negative absolute maximum `R_t`, and four further entries on each sign

$$R_v + a_v, \quad R_v + b_v, \quad R_v + c_v, \quad R_v + e_v, \qquad 1 \le a_v < b_v < c_v < e_v.$$

Write

$$R = R_s + R_t, \quad U = a_s + b_s + a_t + b_t, \quad C = c_s + c_t, \quad J = (e_s - c_s) + (e_t - c_t).$$

Let `K` be the number of distinct pair sums of this 77-point source, with repetition. The previously derived mass identity, with `T=77j-280-w+I+tau+delta`, is

$$2K = T + 4R + U + 2C + J. \tag{1}$$

For a positive integer quartet `a<b<c<e`, define the distinct fringes

$$P = |\{x + y > c : x, y \in \{a, b, c\}\}|, \qquad B = |\{x + y > e : x, y \in \{a, b, c, e\}\}|.$$

The elementary quartet fringe lemma already used in the source reduction says

$$2B \le a + b + 8 - p, \qquad p = (a + b) \bmod 2. \tag{2}$$

5.1 The single-sign improvement

For every such quartet,

$$P + B - a - b \le 5. \tag{3}$$

Here is a short proof, including its two exceptional small cases. Put `u=a+b`. There are at most six unordered pair sums of the triple, so `P<=6`. For `u>=9`, (2) gives

$$P + B - u \le 6 + \lfloor (u + 8)/2 \rfloor - u \le 5.$$

For `u=8`, necessarily `a<=3`, hence `2a<=c`; at least this triple pair lies outside its upper fringe. Thus `P<=5`, and (2) gives `B<=8`. For `u=7`, `P<=5` and `B<=7`; for `u=5`, `P<=4` and `B<=6`; for `u=3`, `P=3` and `B<=5`. These stated triple counts are the complete small cases tabulated below.

For `u=4`, the only way `P=4` is `(a,b,c)=(1,3,4)`. Its possible values of `B` at `e=5,6,7` are `5,5,4`; at all `e>=8=2c`, `B=4`. Thus `P+B-u<=5`. For `u=6`, the only way `P=5` is `(a,b,c)=(2,4,5)`. Its values of `B` at `e=6,7,8,9` are `6,6,5,5`; at all `e>=10=2c`, `B=4`. This also gives (3). If these respective triple maxima do not occur, (2) already gives (3).

The complete triple counts at `u<=7` are finite: when `c>=2b`, exactly `a+c,b+c,2c` exceed `c`, so `P=3`; the remaining values are:

(a,b)	c below 2b and corresponding P
(1,2)	3:3
(1,3)	4:4, 5:3
(1,4)	5:4, 6:4, 7:3
(2,3)	4:4, 5:4
(1,5)	6:4, 7:4, 8:4, 9:3
(2,4)	5:5, 6:3, 7:4
(1,6)	7:4, 8:4, 9:4, 10:4, 11:3
(2,5)	6:5, 7:4, 8:3, 9:4
(3,4)	5:5, 6:5, 7:4

In particular the weaker triple fringe inequality `2P<=u+4-p` also holds universally: for `u>=8` use `P<=6`, and otherwise use this table, followed by rounding to an even integer.

5.2 Two exact source envelopes

Remove the largest entry on each sign from the 77-point source. The remaining source has 75 points. Its pair sums lie in

$$[-2R_t - c_t,\ 2R_s + c_s]$$

except for distinct same-sign sums between the three respective tails whose offset sum exceeds `c_v`. The interval has `2R+C+1` integer positions, and there are at most `P_s+P_t` distinct exceptions. Reinserting two entries creates at most

$$2 \cdot 75 + 3 = 153$$

unordered pair labels, including repetition. Hence

$$K \le 2R + C + 154 + P_s + P_t. \tag{4}$$

The analogous interval for the full source is

$$[-2R_t - e_t,\ 2R_s + e_s],$$

with `2R+C+J+1` integer positions. Only pairs between the four same-sign tails can lie outside it, yielding

$$K \le 2R + C + J + 1 + B_s + B_t. \tag{5}$$

Combining (1) with (4) and (5) gives

$$T \le 308 + 2(P_s + P_t) - U - J, \qquad T \le 2 + 2(B_s + B_t) - U + J.$$

Add these inequalities and divide by two. Using (3) separately on both signs,

$$T \le 155 + (P_s + B_s - a_s - b_s) + (P_t + B_t - a_t - b_t) \le \boxed{165}. \tag{6}$$

Consequently the source budget improves to

$$I + \tau + \delta \le 445 - 77j + w. \tag{7}$$

This deduction is uniform in the source core, all offsets, and the outer coefficients. No integrality of the outer coefficients is used beyond the existing integer source reduction.

5.3 A sharper one-maximum envelope

Remove only the maximum on sign `v`; write `o` for the opposite sign. Its 76-point remainder has a pair-sum central interval with `2R+c_v+e_o+1` integer positions, and its exceptions have at most `P_v+B_o` distinct values. Reinsertion creates at most77 unordered pair labels. Therefore

$$K \le 2R + c_v + e_o + 78 + P_v + B_o. \tag{9}$$

Put `j_v=e_v-c_v`, so `J=j_s+j_t`, and `p=p_s+p_t`. Combining (9) with (1), `2P_v<=a_v+b_v+4-p_v`, and (2) on sign `o` gives

$$T + j_v - j_o \le 168 - p.$$

The reflected inequality and `J>=T-18+p` consequently give

$$j_v \ge T - 93 + p.$$

For every positive strictly increasing triple, `c_v+p_v>=4`: the sole possibility with `c_v=3` is `(a_v,b_v)=(1,2)`, for which `p_v=1`. Since `p>=p_v`, this proves the uniform endpoint row

$$e_v \geq \max(4, T - 89). \tag{10}$$

For `Z5, w=0,j=5` and `T=105+I+tau+delta=105+A`, so this is `e_v>=A+16` on **both physical signs**. This improves the earlier crude one-maximum floor `e_v>=A+8` without an activity, normality, or ownership assumption.

5.4 Fringe bound at the possible endpoint `T=165`

The original weaker two-maximum envelope `K<=2R+C+166`, combined with (1), gives `T+U+J<=332`. Summing (2) over both signs, with `p=p_s+p_t`, and using (5) gives `J>=T-18+p`. Therefore

$$2T + U + p \leq 350.$$

If `T=165`, this says `U+p<=20`, so

$$B_s + B_t \leq (U + 16 - p)/2 \leq 18. \tag{8}$$

Thus the endpoint fringe saves four in a scalar estimate which had used `B_s+B_t<=20`. The local scalar computation supplied by the separate audit agent has raw margin at least `1` for `T<=164` and margin `-2` at `T=165`; (8) increases the latter margin to at least `2`.

5.5 Exact verification

`source_triple_fringe85.py` checks every finite small case needed by (3), the two exceptional quartet tables, and the displayed algebra. The unbounded domains are reduced by the explicit `c>=2b` and `e>=2c` observations above. It also performs a larger independent numerical sanity check; that numerical check is supplementary, not the justification for the unbounded statement.

6 Sharp fixed-label inventories in the raw gaps

This proof does not depend on cardinality 85. It applies to the original four-extreme cover with positive outer magnitudes $v_1 > v_2 > v_3 > v_4 > M_v > 0$, opposite outer magnitudes $o_i > M_o > 0$, and the same 55 raw fixed labels on target v.

Write

$$d = v_1 - v_2, \quad h = v_2 - v_3, \quad x = v_3, \quad y = v_4, \quad M = M_v + M_o.$$

The ordinary centre identities are

$$\lambda = d - x - y, \quad \alpha = h - y, \quad \beta = h - d - y, \quad H = d + h - y, \quad O = H + d.$$

The raw fixed labels are C labels $\lambda + b_i + b_j$ for unordered pairs in $\{v_1, v_2, v_3, v_4, -o_1, -o_2, -o_3, -o_4\}$; U labels $\alpha + b_i$ and V labels $\beta + b_i$, excluding $b_i = v_1$; and the five F labels of the original cover.

All statements below count formal labels. Colliding or nonintegral values only reduce the ability to cover integer target points.

6.1 First gap: at most eleven labels

Assume the first gap

$$(H + M_v, O - M_o)$$

is guarded inside the positive target. Its length is $t = d - M$, and the lower guard gives $H + M_v > 0$.

Subtract λ. A C label in the interval must be either one of the six positive pairs among v_2, v_3, v_4, or one of the four cross pairs $v_1 - o_i$. All other pairs are below or above the interval. A U or V label can use only v_2, v_3, v_4, giving three candidates of each kind.

The seventeen internal candidates split into the following five chains, each with consecutive difference d:

$$\{C22, U2, V2\}, \quad \{C23, U3, V3\}, \quad \{C24, U4, V4\}, \quad \{C33, F1, F3\}, \quad \{F2, F4\},$$

and three remaining labels $C34, C44, F5$. Since $t < d$, each chain meets the gap at most once.

After subtracting H, the last two labels have values

$$C44 - H = y - h - (x - y), \qquad F5 - H = y + (x - y) - 3h - 2d.$$

The guard is $y < d + h + M_v$. If C44 is above the lower endpoint, then $y > M_v + h + (x - y)$, hence $x - y < d$. Consequently

$$F5 - H < M_v + (x - y) - 2h - d < M_v.$$

Thus C44 and F5 cannot both meet the gap. There are at most seven internal labels and four cross labels: eleven in total. The original covering gap argument therefore gives

$$t \leq 12 + \eta_v. \tag{1}$$

6.2 Beta-active second gap: at most eight labels

Assume beta is active. Since $\beta + M_v \geq 1$,

$$h \geq d + y - M_v + 1 > d.$$

After subtracting alpha, the second gap is

$$J = (M_v, d - M_o).$$

Its internal candidate labels, expressed relative to alpha, are:

Chain or singleton	Relative values
D: C33,F1,F3	$d+x-h,\ x-h,\ x-h-d$
G: F2,F4	$x-2h,\ x-2h-d$
B: U3,V3	$x,\ x-d$
C: U4,V4	$y,\ y-d$
V2	$x+h-d$
F5	$x-3h-d$

The label U2 has value $x+h>d$, so it cannot occur. C34 and C44 are below the lower endpoint because

$$C34-\alpha=d+y-h<M_v, \qquad C44-\alpha<C34-\alpha.$$

The only cross candidates are C labels using v_1-o_i or v_2-o_i. For each i the two values differ by d, so their total contribution to J is at most four.

Each of the four internal chains contributes at most one. If V2 lies in J, then $x+h<2d$, hence $x<2d-h$; G and F5 are below zero. This leaves V2 and the three other chains, at most four labels. If F5 lies in J, then $x>3h+d+M_v$; B,D,G and V2 are above J, leaving only chain C and F5, at most two. If neither singleton occurs, the four chains give at most four. Thus the second gap has at most four internal and four cross labels, eight in total.

When H is active its upper guard is valid and the second gap therefore gives

$$t\le 9+\eta_v. \tag{2}$$

6.3 Beta-active union: at most fifteen labels

After subtracting alpha, the union of the two raw gaps is $J\cup(J+d)$. Its whole hull has length $2d-M<2d$. The candidate internal labels are the five d-spaced chains

$$\begin{aligned}
A:&\quad C22,U2,V2=(d+x+h,\ x+h,\ x+h-d),\\
B:&\quad C23,U3,V3=(d+x,\ x,\ x-d),\\
C:&\quad C24,U4,V4=(d+y,\ y,\ y-d),\\
D:&\quad C33,F1,F3=(d+x-h,\ x-h,\ x-h-d),\\
G:&\quad F2,F4=(x-2h,\ x-2h-d),
\end{aligned}$$

and singleton F5, with value $x-3h-d$. Each chain has at most two values in the union.

- If $x<d$, A,B,C have at most two each; D has at most one; G and F5 are below zero. Total at most seven.
- If $d\le x\le 2d$, A has at most one; B,C,D at most two each; G and F5 are below zero. Total at most seven.
- If $x>2d$, A is absent; B has at most one; C,D,G at most two each. If F5 is absent, the total is seven. If F5 occurs, then $x>3h+d+M_v>4d$, which makes B absent; the total is again seven.

The cross pairs v_1-o_i, v_2-o_i give at most eight labels across the union. All other candidates are excluded exactly as above. Thus both disjoint gaps together have at most fifteen labels.

When O and beta are active, both intervals are guarded inside the target, regardless of H activity. They must share these fifteen labels and the same η_v holes. Therefore

$$2(t-1)\le 15+\eta_v, \qquad 2t\le 17+\eta_v. \tag{3}$$

Nonpositive gap lengths make all three bounds automatic. Leading activity, or the other established lower guards from the original common lemma, licenses (1) whenever O is active. These inventories imply the uniform cap $t\le 12+\eta_v$ for the predicate “O active or beta,H active.”

7 Cardinality-85 localization of the entire high-hole family

Assume the fresh size-85 structural theorem: d=77, e=69, a+b=68, 15<=a,b<=53, I=(34-min(a,b))^2, c>=0, individually primitive integer cores, and

```
kappa+rho+c+I<=600,
T>=kappa+144-3z, T>=z, T>=144-2z, w<=4z.
```

Put H=h(P)+h(Q)=kappa+z, mu=mu(P)+mu(Q). The retained verified profile certificates give the following **two** lower bounds, not just adjusted defect floors:

z range	mu floor	rho floor	scalar cap
15--16	426	294	c+I<=194
17--20	470	338	c+I<=158
21--48	514	382	c+I<=170
>=49	554	422	c+I+z<=178

These are minima over every oriented pair a+b=68. The mu values are direct profile outputs; they do not follow merely from rho<=mu. Keeping the mu information fixes the failure of the weaker single-maximum comparison in the last band.

7.1 1. Source width versus every peeled coefficient

The core's absolute-multiset support has mass

```
C0=1190+I+tau=1190+I+mu+H.
```

If each core mass exceeds H, the elementary sumset-hole lemma fills [H,C0-H]. Write W=C0-2H=1190+I+mu-H for its endpoint displacement.

Use source offsets alpha_v<beta_v<gamma_v<epsilon_v, one quadruple on each physical sign. Write

```
u=sum(alpha_v+beta_v), C=gamma_s+gamma_t,
J=sum(epsilon_v-gamma_v), L=C+J=M-R.
```

The two-maximum source envelope, freshly at d=77, is

```
K<=2R+C+166,
J+u<=612+w-I-tau-c.
```

Here the central interval, twelve exceptional peeled-pair labels, and reinsertion of two maxima contribute 1+12+(2*75+3)=166. The freshly derived outer-layer inequality is

```
2C+J<=C0+u-292-2I-c-mu-rho-2z.
```

Adding the J upper bound gives

```
2L<=1510+w-2I-2c-mu-rho-2z.
```

Maximum removal gives R<=72+rho+H. Thus

```
M<=827+H+(rho-mu)/2+w/2-I-c-z
  <=827+H+(rho-mu)/2+z-I-c,
```

using only the original w<=4z, before any localization.

Subtract from W and use H=kappa+z and the original single budget:

```
W-M >= 363+2I+c+3mu/2-rho/2-2kappa-3z
     >= -837+4I+3c+3(mu+rho)/2-3z.                (1)
```

Every coefficient has the correct sign; no slack is spent twice. Substituting the paired mu and rho floors gives minimum margins

```
195, 315, 363
```

in the first three bands. In the tail, (1) gives

```
W-M>=627+4I+3c-3z
     >=93+7I+6c>0,
```

because z<=178-I-c. Therefore every one of the eight absolute peeled coefficients is at most M<W. The enlarged interval has even greater width after adjoining each entry, so all eight entries obey the propagation criterion x<=current_mass-2H+1.

7.2 2. Each core mass exceeds H

The original budget and T>=kappa+144-3z give

```
H<=456-rho-I-c+4z.
```

In the first three bands this yields respectively

```
H<=226-I-c, 198-I-c, 266-I-c.
```

Their scalar I caps give k<=13,12,13 respectively, where a=34-k is the smaller side. Exact minimization of T_a-H over those integer ranges gives lower bounds 171,199,131. No use of side relabeling occurs in the covering argument; these are symmetric mass bounds.

In the tail, rho>=422 gives kappa<=178-I-c; also z<=kappa. Hence

```
H<=356-2I-2c.
```

Since z>=49, I+c<=129, so k<=11. The exact minimum of T_(34-k)-(356-2k^2) for 0<=k<=11 is 120, attained at k=7. Thus both individual core masses exceed H in every high-hole band.

7.3 3. Localization conclusion

The core sumset-hole lemma supplies [H,C0-H]. Section 1 propagates this same boundary strip through all eight absolute entries. Complement symmetry and inclusion of the old absolute support in the new one then give

```
z and w are even,
0<=w<=z,
eta_s=eta_t=w/2.
```

This proves localization on the **entire** size-85 high-hole domain. It uses only original structural bounds before establishing these stronger conclusions. It in fact does not need the covering index to be zero; the full slack c is kept throughout.

7.4 4. Verification scope

highhole_localization85.py reads the already verified retained profile table and recomputes every paired mu/rho minimum over all 39 oriented sizes. It independently checks the new scalar and core-mass minima and the exact integer source constants. This arithmetic accompanies the written interval propagation proof; it does not replace it.

8 Local elementary exclusions at size 85

All statements below assume a valid fresh structural reduction for a size-85 configuration. Put d=77, |E|=69, r_s+r_t=68, Lambda=1191+I, I=(34-min(r_s,r_t))^2. The structural scalar budget is Omega=600-I and the restricted-sum constant is 144. These are conditional local deductions, not a proof of the imported reduction.

8.1 General source formulas

S=2K+1470-77j+w-delta. With A=I+tau+delta and m0=77j-280-w,

2K=m0+A+4R+U+2C+J.

Deleting the two maxima leaves 75 points. The core interval and 12 exceptional three-peel pair labels, followed by 153 reinserted pair labels, give

K <= 2R+C+166, J+U <= 332-m0-A, A <= 445-77j+w, K-M >= R+77j-440-w+A, K-delta >= 77j-142-w+I+tau.

When z>0, primitive side supports are non-AP, so R_v>=r_v+1 and R>=70. The last bound improves to

K-delta >=77j-138-w+I+tau.

Quantitative layers:

V0-C1+delta+2z <= -292-2I-77j-mu-rho, M0+delta-2z <= -166-2I-77j.

The final source cap follows from the triple and quartet fringes. For one sign let P count sums of the first three peel offsets above the third offset and B count sums of all four offsets above the fourth. The proved elementary inequality P+B-a-b<=5 combines the two-max envelope K<=2R+C+154+P_s+P_t with the full-source envelope. Their average gives m0+A<=165 and therefore A<=445-77j+w. The complete unbounded proof is in foundations/source_triple_fringe85.md.

8.2 Sharp raw-gap counts

The exact inventories are eleven labels in the first gap, eight in the beta-active second gap, and fifteen in the beta-active union. Thus

O_v active => t_v=d_v-M<=12+eta_v, beta_v,H_v active => t_v<=9+eta_v, beta_v,O_v active => 2t_v<=17+eta_v.

The first implication uses the original lower guard; leading activity, H activity, or own2 activity licenses it in the corresponding applications. The other two guards are supplied by the stated active resources. The proof by d-spaced fixed-label chains and three ranges for v3/d is written in full in common/raw_gap_inventory.md. These counts imply the uniform cap t_v<=12+eta_v for the predicate “O active or beta,H active.”

8.3 Star and two-cap constants

Four full-core star edges give 2X+4R+delta>=548. The four-top-tail count has 4*83+5=337 labels per target, so

d_s+d_t+G+delta-2K>=796-77j.

With both sharp t_v<=12+eta_v caps, the two-cap proof, using Y=2M-2K+24+w+delta, gives

4I+4tau+delta+231j <=464+6w.

At j=1, with tau>=62+2z and w<=z, the two-cap inequality would require

15+2z+4I+delta<=0,

which is impossible. In particular the P1 two-cap exclusion needs no additional four-edge estimate.

8.4 General first-gap and terminal lemma

If the absolute core has a hull hole, both side sizes>=16, and I+tau<=395, its first missing integer f is <=8. Write k=k_s+k_t for the low coefficients. Then k<=4,k_v<=3. The high baseline maxima are B_v=r_v+f-k_v, B=68+2f-k, with baseline excess b, residual sigma=tau-b>=0. Direct summation gives

I+b=69f-1-(f+34.5)k+k^2/4+(r_s-34-(k_s-k_t)/2)^2, I+b-5B>=-282.

Proof of f<=8 follows exactly the old argument: f<max R_v<=52+tau<=447, so k<=41; for f>=9,k<=4 the lower bound is >=65f-139>=446. For k>=5, use f>=k(k+2)/4+1 to get [-k^3+68k^2-4k+272]/4>=1827/4>395 (the polynomial increases over k=5,...,41). The baseline-minus-5B minimum is attained at f=1,k=0; f=2,3 and f=4,...,8 give larger lower bounds.

Delete six core entries per sign: the retained set has 57 points, and its certified overlap parameter satisfies

6q=750-7B+R-sigma.

The old rooted terminal forest and the source estimate at j=1 give

2t_u >=389-7B+2R+I+b-w >=107-w.

The sharp double gap gives 2t_u<=17+w/2. Thus the terminal form forces

90<=3w/2.

8.5 P1 localization

The scalar range is 16<=r_v<=52: otherwise I>=361 while delta<=407-I-rho and rho>=49. H=h_s+h_t obeys H<=306+w-I-delta<=362-I-delta. For every physical size, T_r+I>=397, ensuring each core mass>H. Initial width >=890+2I+delta. Every inner entry is <=637, and U0>=142, so the next width is >=1032. The outer-layer row and the same H bound show C1+Q0-4H>=788+6I+4delta+2z>0, where Q0=369+mu+rho+2I+delta+2z. The last-shift gap J<=585 is smaller than the width. Thus the original common-strip argument propagates and gives z,w even, w<=z, eta_v=w/2.

Leading activity follows from K-delta>=-61-w+I+tau>=1+I+2z-w>0. At most one missing own singleton gives full outer integrality. The sharpened two-cap and terminal contradictions exclude P1.

8.6 P5: complete source-only contradiction

The scalar range gives rho>=58, delta<=41-I, 28<=r_v<=40. Also H<=13+4z-I-delta<=69. Quantitative layers give C0-U0>=551+2I+delta-2z and C1-V0>=797+2I+delta+2z. Subtracting 2H leaves >=385 and >=687, so common-strip localization applies: w<=z<=14 and parity holds.

The final source envelope gives A<=60+w<=60+z, whereas positive core holes imply tau>=62+2z. Thus

2+z+I+delta<=0,

an immediate contradiction. This excludes P5 without any outer-coordinate integrality or paired-owner assumption.

8.7 The entire high-hole family

Use the retained paired (mu,rho) floors (426,294), (470,338), (514,382), (554,422), respectively for z bands15-16,17-20,21-48, and z>=49. The whole-family localization proof is in foundations/highhole_localization85.md. It keeps the independently certified mu information as well as rho and requires no extension of the density threshold.

Writing W=1190+I+mu-H for the original core central-strip width, the fresh source and layer envelopes give

M<=827+H+(rho-mu)/2+z-I-c, W-M>=-837+4I+3c+3(mu+rho)/2-3z.

The retained paired floors give W-M margins195,315,363 in the first three bands. In the tail, c+I+z<=178 gives margin at least93+7I+6c. The two individual core masses exceed H with margins171,199,131,120 in the four bands. Thus the same central strip propagates through every absolute source entry. This proves parity, eta_v=w/2 and w<=z on the entire high-hole domain, including H0.

At j>=2 the source envelope gives I+tau+delta<=291+w; with tau>=rho+2z, this would require rho+z+I+delta<=291, already impossible at rho294,z15. All H_j with j>=2 are excluded (larger j only improves the bound).

At j=1, the source envelope leaves only even z=16,18,20 and I+tau<=395. Both-leading activity follows directly after localization from K-delta>=-61-w+I+tau>=233+z+I>0. At least three own centers per sign imply outer integrality. The two-cap contradiction forces the terminal form, and 90<=3w/2<=30 is impossible.

After the same whole-domain localization, at j=0 the lower bound K-delta>=-142-w+I+tau>=152+z+I>0 makes both leadings active. All own centres are active, so outer integrality follows. The two-cap inequality gives 4I+4tau+c<=464+6w, while tau>=294+2z and w<=z would require 712+2z+4I+c<=0. Thus H0 is excluded as well.

9 Conditional cardinality-85 zero-hole reduction

These notes assume the common integral, primitive-core structural conclusion at cardinality 85. They do not claim to prove that conclusion. The original outer eight entries may initially be real. The actual-cover grouping rule and the raw 55 fixed-label inventories are unchanged.

Write $d=77$, $|E|=69$, $r_s+r_t=68$, $I=(r_s-34)2$, $T_r=r(r+1)/2$, and $\tau=\sum(P_s)+\sum(P_t)-T_{r_s}-T_{r_t}$. Put $R=R_s+R_t$, $\xi=R-68$, so $0\le\xi\le\tau$. The four inner peels on side v have magnitudes R_v+a_v, R_v+b_v, R_v+c_v, $R_v+e_v=M_v$, with integral positive strictly increasing offsets. Define $L=e_s+e_t=M-R$, $U=a_s+b_s+a_t+b_t$, $C=c_s+c_t$, $J=L-C$, $A=I+\tau+\delta$.

9.1 Fresh source envelope (all zero-hole j)

The ordinary whole-unit capacity is 77, not 78. The fixed allowance remains 84. Hence

$$S=2K+1470-77j-\delta,$$
$$2K=77j-280+A+4R+U+C+L.$$

The second equality follows from the actual source mass $S=1190+I+\tau+4R+U+C+L$.

Deleting both maxima leaves 75 values. Its pair-sum hull has at most $2R+C+1$ positions and 12 exceptional same-sign peeled-pair labels. Reinsertion costs $2*75+3=153$ labels. Therefore $K\le 2R+C+166$ and $J+U\le 612-77j-A$.

The full source envelope is $K\le 2R+L+1+B_s+B_t$. The four-offset inequalities $2B_v\le a_v+b_v+8$ and $2B_v-a_v-b_v-c_v\le 3$ consequently give

$$J\ge A+77j-298,$$
$$L\ge A+77j-288,$$
$$A\le 452-77j,$$
$$K-\delta\ge 77j-142+I+\tau,$$
$$K-M\ge R+77j-440+A.$$

For the cap on A use $U\ge 6$ and the two bounds on J. For $K-\delta$ combine its exact mass with the lower bound on L and $4R+U+C\ge 272+12$. For $K-M$ use the upper bound on $L=C+J$ and $U\ge 6$. In particular $j\ge 6$ is already impossible.

9.2 Adaptive core supports

The full core has 69 points and span R, so its overlap at integral displacement q is at least $137-R-q=69-\xi-q$.

After deleting the largest k positive core values on each sign, the source has $69-2k$ points. The sum of deviations excluding each maximum is $\tau-\xi$. Monotonicity gives span at most $68-2k+(\tau-\xi)/k$, so its overlap is at least f_k-q, where

$$f_k=69-2k-(\tau-\xi)/k.$$

Every use of $\max(f_0,f_k)$ means choosing the actual one of these two subsets that attains the maximum, before constructing the forest. It does not add two overlap bounds or spend two deficit budgets.

9.3 Z0 is excluded

All own centres are active, making all outer coordinates integral. $M\ge 76$; active C7 gives $d_v\ge M+5\ge 81$. Choose an actual active beta contributor on side s. The raw two-gap bound gives $t_s=d_s-M\le 57/2$; the opposite O gives $t_t\le 56$. Active C4_s would imply $t_s>d_t\ge 81$. Active C4_t would imply $t_t>h_s+d_s>2d_s\ge 162$. Both contradict their caps, so pair1 has no actual contributor despite being retained.

9.4 Z1 is excluded with a stronger adaptive endpoint

Outer integrality is proved from the same actual one-hole argument as the report: a side with all four own centres determines its outer values and lambda; opposite C6/C7 determine the

other lower values; its remaining own centres determine its top values. At least one of C6/C7 is active on each side, so d_v>=81.

If pair1 were missing, an actual beta contributor opposite an actual C5 contributor would contradict the raw O cap. Therefore pair1 is retained. Its actual contributor forces its own O to be the unique whole hole; its beta is inactive. Branch B is impossible. In A the opposite side u has beta, H, O all active, so 2t_u<=57. All alpha, C5, U, V, C7 resources below are retained.

For the six-edge trees L_v-V_v-U_v and L_v-C7_v, across both signs, choose

f=max(137-R,57-(tau-xi)/6).

Designate the full leading on u as root. Its native loss is ell_u. The actual-cover forest inequality yields 6f-2X-G+ell_u<=delta. Active C7_u supplies a positive leading maximum, so d_u>=K+2X_u+g_u-ell_u. Active alpha_v and C5_u also give t_u>=2X+G+2. Adding produces

```
2t_u>=K-M-delta+6f+2
     >=I+max(121-5xi+tau,49+2xi)
     >=I+max(121-4xi,49+2xi)
     >=73+I.
```

The last minimum is attained at xi=12. This contradicts 2t_u<=57. The six-rank support exists because the structural/source size bounds give at least 15 entries on each core side (and the source cap alone gives at least 15 for j=1).

9.5 Z2 common activity reductions

Here A<=298 and K-delta>=12+I+tau, so both leading resources are active. The same two-witness residue argument proves outer integrality.

The four-edge physical arm bound is freshly 2X_v+X_o+g_v>=252-tau-delta: trim three ranks per side, whose span uses the fourth terminal deviation, giving each edge the lower overlap 63-tau/4.

Suppose both cap predicates T_s,T_t hold. The four-top tail contains at most 4*(77+6)+5=337 values on each side, so d_s+d_t+G+delta-2K>=796-77j. Four full-core stars give 2X+4R+delta>=548. With the exact mass and source envelope these two bounds give, at j=2,

4I+4tau+delta<=178.

If any C7_v is active, the four-edge activity implication gives t_v>=253-tau-delta>=75, contradicting the cap t_v<=56. If neither C7 is active, those are the exact two holes; both lower C6 are active, making each missing C7 HIGH. Each corresponding full leading has at least r_v+81 off-target points: the entire shifted set D_v union (M_v+D_v) is HIGH and has at least 77+(r_v+4) distinct points. Thus delta>=r_s+r_t+162=230, contradicting the bound 178. Hence T_s,T_t cannot both hold.

An active C4_v forces C5_v activity in branch A. Otherwise that missing C5 is HIGH above C4; its leading and at least five retained higher singletons lose at least r_v+81 and five copies of r_v+5, respectively. This gives delta>=6r_v+106; but I+6r_v>=195 for integral r_v, so A>=301>298. The actual-owner grammar follows from the unchanged literal centre identities and the active lower bound u>-M_v.

These statements reproduce the eight-row actual Z2 cover classification, with no new owner selected for empty or split paired resources.

9.6 Z2 beta/O simultaneous-hit cards

Assume beta_v and O_v are active. The double raw gap gives t_v<=57/2, and opposite O_o is the first hole. Let W be the other whole hole. Use the report's four retained-resource cards, but every forest now selects f=max(137-R,61-(tau-xi)/4).

1. If W is not V_v,C7_v,C7_o, the retained unused C7_v has at most 20 core target hits and all four negative peel images miss. Its native deficiency is therefore at least 69-20+4=53. The rooted three-edge arm gives

   t_v>=I+max(42-2xi+tau,18+7xi/4+tau/4)>=I+max(42-xi,18+2xi)>=34+I.

2. If W=V_v, its HIGH status makes alpha_v,H_v,O_v each lose at least r_v+5 native points. Two retained shortcut edges and the full-core support give t_v>=-65+I+3r_v>=35; the integer minimum of I+3r_v is 100.

3. If W=C7_v, it is HIGH above retained C6_v. The three unused U_v,H_v,O_v losses give t_v>=max(104-xi,80+2xi)>=96.

4. If W=C7_o, its HIGH leading loses at least r_o+81 points and V_o,U_o,H_o lose at least r_o+5 each. Two edges with both leading roots yield t_v>=16+I+4r_o>=148.

Every lower bound exceeds 57/2. All roots and unused native losses belong to distinct original retained resources; no paired capacity is added in these cards.

9.7 Z2 final six-edge contradiction

After that exclusion the same three literal hole families remain: {O_v,B}, {O_s,O_t}, and {O_v,H_v} with their actual forced owners. Every alpha,C5,U,V,C7 singleton is retained. At least one T_u is true. The unchanged identity alpha_o+C5_u=d_u-2(M+X)-G gives 2X+G<=54.

Use the retained six-edge trees and the adaptive support f=max(137-R,57-(tau-xi)/6). The actual cover gives 2X+G>=6f-delta. Hence

```
A=I+tau+delta>=I+tau+6f-54
 >=I+max(414-5xi,342+xi)-54
 >=300+I,
```

because the minimum of that maximum is 354, at xi=12. This contradicts A<=298. Thus every actual Z2 configuration is excluded, with margin 2 at the balanced worst case.

9.8 Sharper source interface and the later families

The independently proved joint triple/fringe source theorem in ../foundations/source_triple_fringe85.md gives T<=165 and A<=445-77j when there are zero holes. Thus the current bounds are A<=214,137,60 for Z3,Z4,Z5. The Z3 argument retains the harmless weaker cap221 for its arithmetic. Fresh complete-target rank constants are73-r and S+r_other-72.

A BAD ordinary centre has loss at least 73-max(r_s,r_t). The old Z5 initial fixed ledgers with credits13 or14 do not by themselves justify normality at cardinality85. The retained-P cards and two empty-P cards below handle these endpoints directly. Only the alpha-hole card obtains normality, after establishing its actual credit27 independently of normality.

Z4 and Z5 calculations and literal endpoint lists are in zero85_z4_z5_cards.md. The common Z3 A1 calculation is summarized at the end of this file.

9.9 Further result: the Z3 A0 family is excluded

At j=3 the source cap is A<=221 and the source-only rank rows, freshly derived from deletion of one maximum or two same-sign maxima, are

```
L>=A-57,
e_v>=max(4,A-146),
e_v<=378-A.
```

The one-maximum envelope is K<=2R+c_v+e_o+94, because its 76-point remainder has 16 exceptional pairs and reinsertion cost 77. It yields |j_s-j_t|<=237-A-U. Combining with J>=A-67 and U>=6 gives the lower e bound. The two-same-sign-maximum envelope is K<=2R+b_v+e_o+167: 75 remaining points, 13 exceptional labels, reinsertion 153. Combining with J+U<=381-A gives e_v<=382-A-a_v-a_o-b_o<=378-A.

The common j3 activity grammar still follows without any new raw-gap strengthening. Two cap predicates violate 4I+4tau+delta<=640-231j=-53. The five-edge physical-arm bound is now X_o+2X_v+G>=305-tau-delta; active C6 gives t_v>=85+I>56. The four-edge bound gives active C7 t_v>=32+I>57/2. A HIGH C6 above retained C5 has native and higher-singleton losses at

least 4r_v+96; its sum with I has minimum 228, exceeding the source cap 221. These prove the same exhaustive two A families: A0 with holes B,O_s,O_t and pair1 owner s; A1 with holes O_s,O_t,H_s and beta owner s, pair1 owner t.

For A0 use the five endpoint resources C4_s,alpha_s,alpha_t,C5_t,H_s with weights 1,2,1,1, 1. The complete-target lower endpoint is now Z>=73-r_v-k-b; the upper is Z<=S_v+r_o-72+k+b. Thus the endpoint identity has constant 233. Use six retained edges with f=61-(tau-xi)/4, giving gap 2X+G. The source mass constant 1239 and the forest baseline 12f=732-3tau+3xi cancel to give the same master constant:

```
T_r+4r+tau_t+2L-3tau+3xi_s+7xi_t <= 290+delta+Bstar,
Q3=T_r+4r+3I >=626 (r=28,29).
```

Use the common eight O HIGH fixed labels, so epsilon=delta-8. Define heavy selected clips as k>=72. For k<=71 the error is at most both xi_source and k, by (73-k)b<=tau_source and tau+epsilon<=213-I; at k=71 equality in k(74-k)>=213 is allowed. Heavy tail error is at most xi_source+e_source-4, including k=72 where the stronger core-maximum error xi_source is available.

The five nonexceptional scalar finishes (the r-polynomial is the new Q3) are

Endpoint pattern	Upper inequality	Lower bound
alpha_s light, none heavy	Q3+tau_t+2delta<=625	642
alpha_s light, one heavy c	Q3+tau_t+2delta+c<=781	858
alpha_s light, two heavy H	Q3+tau_t+2delta+H<=945	1074
alpha_s heavy, other source t	Q3+tau_t+2delta+c<=866	1002
alpha_s heavy, other C4_s	Q3+tau_t+2delta+c<=948	1002

For the last two lower bounds it is enough to allow clips 72, so delta>=152 and c>=72. Each table row follows from the displayed master and, respectively, 2L>=2A-114, 2L-e_v>=A-53, 2L-e_v-e_w>=0, e_t>=A-146, and 2e_t-e_s>=3A-670.

The sole-heavy-alpha row needs a new fixed ledger. If its clip is exactly 72, the endpoint is the core maximum and has error at most xi_s; absorb it radially and use the first row, which is already contradictory. If its clip is at least 73, alpha_s<=0. The eleven fixed labels

```
U5_s,U6_s,U7_s,U8_s,V5_s,V6_s,V7_s,V8_s,F3_s,F4_s,F5_s
```

are LOW. The U labels are negative because alpha_s<=0. The V labels are negative because beta_s=alpha_s-d_s<0. The F labels are negative because the leading bound and K-delta>=89+I+tau imply d_s>g_s. Those 11 labels touch exactly 9 baseline groups: U7 has no baseline group, and U8/F5 share one. Together with the eight O HIGH labels, the compatible credit is 10. Replacing delta-8 by delta-10 lowers the master right-hand constant from 290 to 286. Cancel the heavy alpha error and use 2e_t>=2A-292; the resulting exact row is

```
Q3+tau_t+2delta<=791.
```

But the native clip requires delta>=10+73=83, so its left-hand side is at least 626+166=792. This strictly excludes the last A0 pattern. It is a new argument, not an automatic translation of the old report.

9.10 Z3 A1 is excluded by the actual physical-side union

The full derivation and exact four-pattern checker are ../common/z3a1_side_union.md and ../common/check_z3a1_side_union.py. The actual holes are O_s,O_t,H_s; beta is sole on s and pair1 sole on t. The missing H_s is HIGH because H_s=beta_s+2d_s and beta_s is active. Twelve compatible HIGH labels give epsilon=delta-12. Grouping the leading and its seven actual native resources on s gives S_s>=K+539-epsilon.

Let H be the native deficiency of active H_t, so H<=76. Lower P_t and upper H_t are the only endpoints. Their actual clips and the two separate forest rebates, on one chosen actual source with 2tau+6f>=398, give

```
M+M_s+6f-H/2<=47+2delta+E,
E<=xi_s+xi_t+p(e_t-4)+h(e_s-4), p,h in {0,1}.
```

Use L>=A-57 and e_s>=A-146. For `Z=r_s+2I+2tau+xi_s+6f-H/2`, the four endpoint positions yield upper bounds182,267,A+32,A+117. Even with A<=221 their largest bound is338, whereas `Z>=398+34-38=394`. This excludes all actual light/heavy positions of both endpoints; no endpoint or error assumption is made on the beta translate. Together with the preceding A0 proof it completes Z3.

10 Excluding the fresh zero-hole Z3 A1 form

This is a conditional local proof for a cardinality-85 configuration after a valid integral primitive-core structural reduction. It uses only the actual Z3 A1 retention form and the common source/rank/forest lemmas. No scalar cell is assumed to be an actual set.

10.1 Actual form and source quantities

The original source has 77 points and the core has 69 points. Write

$$r_s + r_t = 68, \quad I = (r_s - 34)^2, \quad \tau = \tau_s + \tau_t, \quad R_v = r_v + \xi_v, \quad R = 68 + \xi_s + \xi_t.$$

The three missing ordinary whole units are exactly O_s, O_t, H_s. The beta pair is retained and has sole actual owner s; pair1 $\{C4_s, C4_t\}$ has sole actual owner t. Every unlisted ordinary singleton is retained. Both leading resources are active and all outer coordinates are integral, by the preceding actual-activity reduction.

Let e_s, e_t be the fourth peel offsets, $L = e_s + e_t$, $M = R + L$, and $M_s = r_s + \xi_s + e_s$. Put $A = I + \tau + \delta$. The following already proved source bounds suffice:

$$A \le 221, \qquad L \ge A - 57, \qquad e_s \ge A - 146.$$

The stronger new cap $A \le 216$ is unnecessary here.

10.2 Fixed credit and a whole physical side

An empty integral ordinary centre on a complete target is either at most $-M_v$ or at least $S_v + 1$. Since β_s is active and

$$H_s = \beta_s + 2d_s, \qquad d_s > 0,$$

the missing H_s satisfies $H_s \ge S_s + 1$. The missing O_s, O_t are likewise HIGH. The eight $O_v + v_i$ labels and the three $H_s + s_i$, $i = 2, 3, 4$, are HIGH. A twelfth is $\alpha_s + s_2$: beta activity gives $s_2 > d_s$, so $\alpha_s + s_2 = H_s - d_s + s_2 > H_s$. Thus the compatible fixed credit is 12 and $\epsilon = \delta - 12 \ge 0$.

On target s, group the full leading resource with these seven actual ordinary native resources:

$$\beta_s,\ \alpha_s,\ U_s,\ V_s,\ C5_s,\ C6_s,\ C7_s.$$

The beta capacity is licensed because its opposite paired contribution is empty. The group has original capacity $K + 7 \cdot 77 = K + 539$; its capacity deficiency is at most ϵ. Its actual union lies in the complete target, hence

$$S_s \ge K + 539 - \epsilon. \tag{1}$$

This is an actual union bound, not an assumption that the seven native resources are disjoint.

10.3 Two endpoints and two forest inequalities

Let $P = C4_t$, let $\Gamma_P, \Gamma_{\beta_s}$ be the whole-unit native deficiencies, and put

$$B = \Gamma_P + \Gamma_{\beta_s}, \qquad H = \Gamma_{H_t}.$$

The active 77-point singleton H_t has $0 \le H \le 76$. On complete targets its native loss is exactly its lower plus upper clips. Use the lower endpoint of P and the upper endpoint of H_t:

$$P \ge 73 - r_t - l_P - b_P, \qquad H_t \le S_t + r_s - 72 + h_H + b_H.$$

Their error sources are respectively t, s. Define $E_c = l_P + h_H \le B + H$ and $E = b_P + b_H$.

Choose one actual full or trimmed core subset attaining

$$f = \max\left(69-\xi,\ 65-\frac{\tau-\xi}{2},\ 57-\frac{\tau-\xi}{6}\right), \qquad \xi=\xi_s+\xi_t.$$

In particular,

$$f \ge \frac{199-\tau}{3}, \qquad 2\tau+6f \ge 398. \tag{2}$$

The ten retained-resource edges are

$$\begin{aligned} s:&\quad L_s-V_s-U_s,\quad L_s-C7_s-C6_s-C5_s,\\ t:&\quad L_t-V_t-U_t-H_t,\quad L_t-C7_t-C6_t. \end{aligned}$$

Their gaps sum to $2Q$, with $Q=X_s+X_t+g_s+g_t+h_t$. The forest omits both paired endpoint resources, so the actual grouping budget gives

$$Q \ge 5f-\frac{\epsilon-B}{2}.$$

The two X_s star edges L_s-V_s and L_t-C7_t omit these two paired resources and also H_t, giving

$$X_s \ge f-\frac{\epsilon-B-H}{2}.$$

Both are valid statements about the same unchanged deficit. Adding their inequalities gives

$$Q+X_s \ge 6f-\epsilon+B+\frac{H}{2}. \tag{3}$$

10.4 Use the missing H directly

The literal centre identity is

$$H_t = 2d_s+\beta_s+P+2s_4+t_4+g_s+g_t+h_t.$$

Since $\beta_s+2d_s=H_s\ge S_s+1$, the two endpoint inequalities imply

$$S_t \ge S_s+M+M_s+(Q+X_s)+78-E_c-E.$$

Use (1) and the exact source mass $S_s+S_t=2K+1239-\delta$. The result is

$$M+M_s+(Q+X_s) \le 59+\delta+E_c+E.$$

Apply (3) and $E_c\le B+H$. The paired-loss variable B cancels and the unchanged residual deficit gives the crucial inequality

$$M+M_s+6f-\frac{H}{2} \le 47+2\delta+E. \tag{4}$$

No endpoint for beta is used, so all positions of its actual native translate are included automatically.

10.5 All four endpoint positions

For a selected clip at most 72, the core rank bound gives $b_P\le\xi_t$ or $b_H\le\xi_s$. A selected clip at least 73 is a peeled rank and gives respectively

$$b_P \le \xi_t+e_t-4, \qquad b_H \le \xi_s+e_s-4.$$

Let $p,h\in\{0,1\}$ record the latter two alternatives. Thus

$$E \le \xi_s + \xi_t + p(e_t - 4) + h(e_s - 4).$$

Put

$$Z = r_s + 2I + 2\tau + \xi_s + 6f - \frac{H}{2}.$$

Substitute $M + M_s = 68 + r_s + 2\xi_s + \xi_t + 2e_s + e_t$ into (4). The offset rows $L \ge A - 57$, $e_s \ge A - 146$ and $A = I + \tau + \delta$ give:

Lower P endpoint	Upper Ht endpoint	Consequence
core rank	core rank	$Z \le 182$
peeled rank	core rank	$Z \le 267$
core rank	peeled rank	$Z \le A + 32 \le 253$
peeled rank	peeled rank	$Z \le A + 117 \le 338$

All these cancellations are exact. For the four rows the residual offset expressions after subtracting the endpoint error are respectively $L + e_s$, $2e_s + 4$, $L + 4$, $e_s + 8$.

But (2), $H \le 76$, $\xi_s \ge 0$, and the integer identity $r_s + 2(r_s - 34)^2 \ge 34$ give

$$Z \ge 398 + 34 - 38 = 394.$$

Every upper bound is strictly smaller. Therefore every actual Z3 A1 configuration is impossible, in either physical owner orientation, including every heavy or light position of all three original endpoint resources.

11 Fresh Z4/Z5 reductions at cardinality 85

Conditional structural notation and basic cover are in zero85_reduction.md. The stronger source theorem now supplies A<=137 for Z4 and A<=60 for Z5. Below proofs sometimes keep the weaker Z4 cap143, so they remain valid.

All forests/groupings are original full resources. Repeated inequalities may use separate copies of the same deficit; endpoint rebates are made only when the endpoint resources are disjoint from that particular forest/group.

11.1 Z4 source and classification

Fresh rows: S=2K+1162-delta, L>=A+20, e_v>=max(4,A-69), e_v<=301-A; f=61-(tau-xi)/4. A complete-target rank endpoint has lower constant73-r_v and upper constant S_v+r_o-72. Light clip<73 has error<=xi_source; a heavy clip>=73 has error<=xi_source+e_source-4. At most one heavy endpoint is possible in any selected native loss budget<=delta-8<=135. For light outside-hull clips k, b<=k follows from tau_v<=135-k and k(74-k)>=144 on 28<=k<=72.

Both O are absent. Two cap predicates already contradict the five-/four-top tail and source depth. If beta_v,O_v were simultaneously active, their opposite O and their own C7,C6 are holes, since active C7 gives t_v>=110+I while O's raw cap is12. The negative physical-v arm L_o-C7_o-C6_o is fully retained. A retained positive star L_v-V_v plus that arm, rooted at L_v, gives t_v>=K-M+3f-delta+4. With full-core support this is >=147-xi+I; with the four-rank support it is >=123+2xi+I, so it exceeds the double-gap cap27/2. If V_v is the fourth hole, it is HIGH; alpha_v,H_v,O_v are three retained HIGH resources. Use the two shortcuts L_v-U_v,L_o-C7_o, root L_v, and their three unused losses. Full-core overlap gives t_v>=93+I+3r_v>=193. Thus beta/O coexistence is impossible. The remaining one-active-O grammar forces a retained beta/H/C7 triple on the opposite sign, again impossible. Both O are holes and eight HIGH fixed labels give epsilon=delta-8.

The exact A-E/A-H/B retention table follows from the actual cross chains and beta/H incompatibilities. Put `N={U_s,U_t,V_s,V_t,alpha_s,alpha_t}` and `W=N union {P,H_s,H_t,C5_s,C5_t}`. Every unlisted unit is retained.

Type	Exact whole holes	Actual paired status
A-E	B,O_s,O_t,Z; Z in W	P empty when Z=P; owner opposite v when Z=C5_v; either actual owner otherwise
A-H	O_s,O_t,H_v,Z; Z in W except H_v	B sole on v; P empty when Z=P, owner v when Z=C5_opposite, owner opposite otherwise
B	O_s,O_t,X_r,Y_opposite; X_r in {beta_r,H_r}, Y_opposite in {beta_opposite,H_opposite}	Both paired units retained, P sole on r

For completeness the two-beta-hole B exclusion has a fresh scalar check. Suppose the beta opposite the active pair1 owner were HIGH. The exact centre/leading bounds give `2S_v+K+3g_v+6X_v<=M_o+1161+delta`. All the relevant ordinary vertices are retained, so the actual three-edge arm gives `2X_v+g_v>=189-3tau/4-delta`. Use `K-M>=R-132+A`, `S_v>=T_r+4r+10+tau_v`, `M_v>=r+4`, the conservative `A<=143`, and R>=68. The necessary consequence is `2T_r+9r+4I+7tau/4+2tau_v<=1063`. But the integral minimum of `2T_r+9r+4(r-34)2`ís1192, at r=26. Hence that beta is LOW. On the pair1-owner side, HIGH beta and retained H would give2d<=M_o-1, whereas actual pair1 activity gives d>M. That beta is LOW too. Their sum cannot then have branch-B's positive sign, excluding both beta holes. The literal dispatch is: A-E retained P uses F for the four U,V holes or H_t, E-H for H_s, E-alpha for either alpha, E-C5 for C5_t after naming its owner s. A-H uses F for U,V or empty P, O for opposite-owner choices H_t,C5_s,alpha_s, alpha_t, and S for C5_t. The three B cases HH,beta_rH_opposite,H_rbeta_opposite use O,O,S respectively. The only ownerless residual is A-E with Z=P.

No empty pair is assigned an owner. Exact endpoint/forest predicates are in Appendix D of the retained report, labels sec:z4-cardF, sec:z4-cardO,sec:z4-cardS,sec:z4-cardEH,sec:z4-cardEA, sec:z4-cardEC and the immediately following dispatch (source lines 1962–2158). The

calculations below use these literal lists with fresh capacity77 and source mass1162; their formulas are derived again from actual target union bounds.

11.2 Z4 Card F

The five weighted endpoint identity has constant233. Its six-edge forest gives the fresh master

`T_r+4r+tau_res+2L-3tau+2e <=229+delta+B`.

For all light endpoints B<=2epsilon. For one heavy endpoint of weight q=1,2 and clip c>=73, B<=2epsilon-qc+q*tau+q*e_v. Since e_v<=L, source L>=A+20 and A<=143 give respectively

```
Q3+tau_res+4e <=332,
Q_(3+q)+tau_res+q*delta+4e+q*c <=332+306q.
```

The integer minima are Q3>=626,Q4>=649,Q5>=665. At e=8 the light lower bound658 exceeds332; q1 lower835 exceeds638; q2 lower1005 exceeds944. All Card F predicates close.

11.3 Z4 Card O

Use the common actual opposite-owner predicate; e=12, epsilon=delta-12. Source-side s has at least six native ordinary resources. The independent common-agent route uses H_s HIGH and V_t=H_s+P_t+2a+b+g_s. Together with the physical side union and four-star/double-arm forest it proves all four endpoint-heavy flags; see the common-agent proof file. A direct new endpoint master is also

`T_r+4r+5xi_s+tau_s+R+L+2e_s+10f+60+4B+(7/2)H <=1006+4delta+Eerr`.

With f=max(137-R,65-(tau-xi)/2,57-(tau-xi)/6), 4tau+10f>=1990/3. Heavy source-s gives upper1091.5, heavy source-t gives1142, all-light A<=73 gives1137, all-light A>=73 gives1207, against respective minima1380+1/3 or1312+1/3. Thus Card O is closed independently.

11.4 Z4 Card S: stronger physical union proof

Same owner of beta_s and P_s forces C5_t inactive. Target t has six actual ordinary native resources H,U,V,C6,C7,alpha, in either branch. Grouping them with its leading gives S_s<= K+700-e, with e=11.

The endpoint identity alpha_s=P_s+beta_s+alpha_t+d_t+2(M+X)+G has constant155. Leading t and the disjoint six- and three-edge forests therefore give

`2M+9f+4e+2N <=545+3delta+B`.

Baseline errors B<=2xi_s+2xi_t. In the all-light case the radial residual is nonnegative, and L>=A+20 implies

`(9/4)I-(3/4)delta+4e+2N <=-577/4`.

Since delta<=143-I-tau and e=11, its left side is at least `-253/4`, larger by81. In a heavy case cancel e_v-4 from 2L, retaining L+e_o>=A+24. The resulting row is

`(9/4)I-(3/4)delta+4e+2N <=43/4`.

Since N>=73, delta<=143-I-tau and e=11, its left side is at least331/4, exceeding43/4.

11.5 Z4 empty-beta Cards E-H, E-alpha, E-C5

All side caps come from the actual opposite physical target union, rather than a syntactically chosen set of centres.

11.5.1 E-H

Physical t has seven native resources, so S_s<=K+623-e. Five endpoints and the two forests give

R+2L+xi_t+e_t+10f+3e <=416+2delta+Jstar.

If no lower endpoint is heavy, Jstar<=xi_t+e_t-4. Source L>=A+20 gives (5/2)I+delta/2+3e<=-469/2, impossible. If a lower endpoint has heavy clip c, Jstar<=xi_t+xi_v+e_v-4-c; 2L+e_t-e_v>=A+28 gives (5/2)I+delta/2+3e+c<=-159/2, also impossible.

11.5.2 Owner alpha_s missing

Physical t again has seven native resources, so S_s<=K+623-e. The raw-order identity with alpha_s>=1-a and the endpoint/ten-edge forests gives

L+2e_t+2xi_t+10f+3e+N <=419+2delta+B.

All-light baseline B<=xi_s+3xi_t leaves 610-(5/2)tau radially; L+2e_t>=A+28 yields (5/2)I+delta/2+3e+N<=-9/2.

A heavy error on source t cancels e_t-4; L+e_t>=A+24 yields the same row with RHS -1/2. A heavy error on source s can only be P_s; cancelling e_s leaves3e_t>=3A-207 and yields 3I+tau/2+delta+3e+N<=12. Here delta>=e+73 and N>=73, so it is impossible.

11.5.3 Opposite alpha_t missing

Physical t has six native resources, so S_s<=K+700-e. The weighted endpoint identity and eleven edges give

R+2L+xi_s+e_s+11f+3e <=492+2delta+B.

Baseline errors are3xi_s+2xi_t, leaving739-(11/4)tau. All light offsets satisfy2L+e_s>=2A+44 and give a negative upper bound. A heavy endpoint of weightq=1,2 satisfies2L+e_s-q e_v>=A+24; its remaining inequality is (11/4)I+(3/4)delta+3e+4q<=-83/4, impossible.

11.5.4 E-C5

Physical t has six native resources H,U,V,C6,C7,alpha, so S_s<=K+700-e. Five endpoints and eleven edges give

R+2L+xi_t+e_t+11f+3e+N <=493+2delta+B.

Baseline errors2xi_s+3xi_t leave739-(11/4)tau. Light offsets2L+e_t>=2A+44 yield a negative upper bound. A unique heavy error leaves2L+e_t-e_v>=A+28 and gives (11/4)I+(3/4)delta+3e+N<=-111/4. Thus this card is closed.

All displayed Z4 retention rows are excluded. The final ownerless family {P,B,O_s,O_t} has seven actual native ordinary resources on each physical sign. Its complete independent geometric and scalar certificate is ../scalar/z4_ownerless_certificate.tex, checked by ../scalar/check_z4_ownerless_certificate.py. The fresh four-edge arm gives115+I rather than the old131+I; this still yields X+g>=58 and d-h-X>=3>0, enough for the literal ten-label gap. The original eighteen-credit ledger is therefore valid. Deleting the two largest core values gives the actual67-point support with f=67-(tau-xi); this stronger support closes all four H core-hit statuses. Both empty pairs remain ownerless throughout that proof.

11.6 Z5 common activity and endpoints

Here S=2K+1085-delta, A<=60, K-M>=R-55+A, K>=153. Every retained singleton has at least17 hits, exceeding all eight source peels together; thus it has an actual core hit. Once sole pair ownership is proved, the same assertion holds for that whole paired capacity. All actual retained endpoint errors are<=xi_source because their selected clips are<73.

Both O are empty before outer integrality: an active O has raw t<=12, whereas leading activity gives t>=R-55+I+tau+2X+g>13.

If beta_v,H_v were both active, its raw cap21 would give2X_v+g_v<=8-I-tau, so tau<8. If either U_v or C6_o were active then q=X_v+g_v would be integral. The two opposite leading/-core shortcut intersections give2q>=2(137-R)-delta>=78+I-tau>70, contradiction q<8. If either V_v or C7_o were active the same argument with q=X_v contradicts the cap. Therefore

these four ordinary singleton units are all holes; together with the two O holes they exceed j= 5. So beta/H cannot both be active.

The report's literal one- or two-coordinate residue classifier then proves outer integrality; the actual cross chains and HIGH own-chain rules prove extreme-leaf retention and the A0/A1/B hole lists. The four outer-coordinate identities used there are unchanged. An empty cross centre is LOW because a HIGH cross loses at least r_v+81>delta leading points.

An empty alpha below a retained H is LOW without a normality bootstrap. A retained H has a core hit, so H<=S_v+R_o. The leading bound gives d_v>=153-delta+g_v+2X_v>R_o (since R_o<=r_o+tau_o, r_o<=41 and delta+tau<=60). Therefore alpha=H-d<S_v. Its emptiness forces alpha<=-M_v, giving h_v<=X_v.

Eight O HIGH fixed labels give epsilon=delta-8. After the outer-integrality classifier every extreme star leaf is retained. On every chosen core support with overlap f, weighted stars give ca*X_s+cb*X_t>=(ca+cb)f-max(ca,cb)*epsilon/2. The four disjoint ordinary groups, including the one-empty form when U or C6 is missing, give cg*g_s+dg*g_t>=(cg+dg)f-max(cg, dg)*epsilon/2. These are separate valid scalar inequalities, not a claim that stars and pairs are disjoint.

Source-only sharpenings needed below are

```
M>=R+99+A,
e_v>=A+16.
```

The first uses the quartet Q<=3 equality case and central-gap correction, with fresh mass coefficient105: L>=103+A-Q_s-Q_t; both Q=3 would force L<=12, so L>=98+A. Equality implies a huge offset with c<=15, leaving a central gap of at least11+I+delta; hence L>=99+ A. These are the same actual exceptional-pair counts and equality conditions of Appendix E, with the new source mass.

For the second, the sharpened one-maximum envelope is `K<=2R+c_v+e_o+78+P_v+B_o`. The actual distinct exceptional triples and pairs satisfy `2P_v<=a_v+b_v+4-p_v` and `2B_o<=a_o+b_o+8-p_o`, where p_v records quartet equality. Consequently `T+J_v-J_o<=168-p`, with `T=105+A`. Combined with `J>=A+87+p`, this gives `J_v>=A+12+p`. Since `c_v+p>=4`, it follows that `e_v>=A+16`. The endpoint tables below deliberately use the weaker `e_v>=A+8` to preserve their independently audited residuals. The alpha-hole terminal proof uses the stronger floor16.

11.7 All thirteen Z5 A0 retained-P templates

Keep the report's literal endpoint lists, which are exhaustive after its now freshly proved retention grammar. The fresh constants and baseline rank-error weights are:

Here P=C4_s is the actual sole pair1 owner, a=s4, b=t4, and G=g_s+g_t. These are the literal endpoint identities; each upper list is summed with unit weight and each number in the lower list is its coefficient. No empty pair appears as a resource.

Card	Upper endpoints	Lower endpoints with weights	Centre identity body
A	V_s,V_t	P,2C5_t,2alpha_s,alpha_t	5a+5b+2G
B	V_s,H_t	P,2C5_t,2alpha_s	5a+4b+2g_s+3g_t+2h_t
C	H_s,U_t	P,2C5_t,3alpha_s	6a+4b+3G+h_t
D	H_s,H_t	P,2C5_t	3a+4b+3G+3h_s+2h_t
E	H_s,U_t	P,2C5_t,alpha_t	3a+5b+3G+3h_s
F	U_s,H_t	P,2C5_t,2alpha_t	3a+6b+3G+2h_s
G	H_s,H_t	P,2C6_t,2alpha_t	3a+6b+3G+h_s
H	H_s,H_t	P,2C6_t,alpha_s	4a+4b+3G+2h_t
I	H_s,U_t	P,2C6_t,alpha_s,alpha_t	4a+5b+3G
J	H_s,H_t	P,2C7_t,alpha_s,2alpha_t	4a+6b+g_s+3g_t
K	U_s,H_t	P,2C6_t,2alpha_t	3a+6b+3G

All templates have base holes B,O_s,O_t. The two extra holes are respectively: D(alpha_s, alpha_t); A(U_s,H_s), A(U_t,H_t), A(H_s,H_t); B(alpha_t,H_s); C(alpha_t,H_t); E(alpha_s,H_t); F(alpha_s,H_s); G(C5_t,alpha_s); H(C5_t,alpha_t); I(C5_t,H_t); J(C5_t,C6_t); K(C5_t,H_s). This is all thirteen templates, including both physical owner orientations under reflection.

Card	Constant	Error weights (s,t)	Maximum native clip weight
A	310	4,4	2
B	305-r	4,3	2
C	378-2r	5,3	3
D	159+r	2,3	2
E	164+2r	2,4	2
F,G,K	169+3r	2,5	2
H	232	3,3	2
I	237+r	3,4	2
J	242+2r	3,5	2

Each constant is constructed as76+sum(weight*(73-r_side)), with r_t=68-r. Use f=61-(tau-xi)/4 and the mass lower bounds M>=R+99+A, M_v>=r_v+xi_v+A+8. Drop positive h terms, except in D. Use the independent weighted star/pair rows and Eclip<=maxweight*epsilon. The residual lower-minus-upper polynomial, after applying delta<=60-I-tau, is bounded below as follows:

Card	Strict positive lower residual
A	356+(9/2)I+tau+xi/2
B	166+5I+(3/2)tau+xi/2
C	351+(13/2)I+(5/2)tau+xi
E	2+5I+(3/2)tau+xi/2
F,G,K	178+(11/2)I+(7/4)tau+(3/4)xi
H	51+(9/2)I+tau+xi/2
I	227+5I+(5/4)tau+(3/4)xi
J	281+(11/2)I+2tau+xi/2

For D both alpha resources are missing and LOW. The two retained HUV triples and cross g_t pair are distinct, with native group deficiencies Psi_s,Psi_t,Omega. Their common f support gives

`3G+3h_s+2h_t>=305+6f-3Psi_s-2Psi_t-Omega>=305+6f-3epsilon`.

This replaces the weaker3G bound. The final residual is at least `53+6I+(11/4)tau+xi/4>0`. Thus all thirteen literal A0 retained-P templates are excluded without assuming any retained endpoint is normal.

11.8 A0 both pairs empty, H_s or C5_s extra hole

Physical side s has six native resources; the other side has seven. Grouping side s gives S_t<=K+623-e. Let A0=alpha_s, B=alpha_t, D=C5_t. The identity H_t=2(a+b)+G+A0+B+D, the generic upper endpoint, and K<=M+R+21 give

`M+2X+G+A0+B+D<=R+572+r_s-e+u_Ht+xi_s`.

Alpha_s is retained. Lower it generically; pack B+D in its actual two-resource group Gamma, giving B+D>=2(73-r_t)+f-2Gamma-2xi_t. Hence

`M+2X+G+f<=R+489-e+u_Ht+l_alpha_s+2Gamma+2xi`.

Let N=Gamma+Gamma_Ht+Gamma_alpha_s. All these endpoints avoid the stars and the four ordinary g groups. Their independent rebates give2X>=4f-(epsilon-N), G>=2f-(epsilon-N)/2. Since the endpoint weighted loss is<=2N, combination yields

`M+7f<=R+489-e+2epsilon+2xi`.

Substitute M>=R+99+A, f=61-(tau-xi)/4, xi<=tau and A<=60 to obtain `2I+3e<=23`. Already e=8 gives24<=23, impossible. Thus both cases close without the old normality assumption or extra fixed ledgers.

11.9 A0 both pairs empty, alpha_s extra hole

Initial fixed ledger e=14 remains valid without normality because both paired components are LOW, the alpha hole is LOW, and d>g proves all F3,F4,F5 labels LOW. With epsilon=delta-14, the leading/stars/pairs give d>=107+g+2X, X_v,g_v>=38, and h_s>=15.

The exact32-type upper-tail classification can therefore reproduce the HIGH21,LOW29, touched23 ledger and e=27 before normality. Two places previously using C5_v>0 are replaced by actual core-hit lower C5_v>=1-R_v: C5_s+s3+h_s>26 follows from e_s+X_s+g_s+h_s+1>26, and C5_t+M_t+2X_t+g_t>16 follows from e_t+2X_t+g_t+1>16. The tail argument only needs h_s>2, not29, to exclude d_s+1,d_s+2. Once e=27, the BAD native-loss floor73-max(r_s,r_t) exceeds epsilon=delta-27 uniformly, so all retained centres are normal.

At e27, the target caps are S_s<=K+546-e and S_t<=K+623-e. If B=alpha_t>=a or D=C5_t>=a, the normal H_t endpoint and lower endpoint of the other centre give M+2X+G+h_s<=R+567-e+endpointloss. Rebate those unused endpoint losses N in both the four-star and triple/pair gap bounds to get A+6f+N<=407+delta-3e. With adaptive f=max(137-R,65-(tau-xi)/2) one has2tau+6f>=398; A<=60 then forces2I+N+3e<=69, impossible at e27. Thus B, D<a. The nine further LOW labels give e34. The fixed-only prefix argument, now with X_t>=38, gives C=C5_s<b and the four further LOW labels give e38.

Five additional distinct labels are LOW:

```
C25_s=C-D-2b-g_t<0,       C25_t=D-C-2a-g_s<0,
C35_t=C25_t-h_t<0,
C45_s=C35_s-g_s<0,        C45_t=C35_t-g_t<0.
```

Normality gives C,D>=1, while the proved bounds C<b,D<a give the two strict signs. C25_s,C25_t occupy their one previously untouched baseline group; the other three labels lie in groups already occupied respectively by V3_t,V8_s,V8_t. Thus the ledger has HIGH21,LOW47, touched26 and genuine compatible credit42. This counts the eight O HIGH labels once. The labeled incidence is recomputed against the original26 baseline groups by the checker.

For completeness the actual endpoint/group transport is as follows. Pack B=alpha_t and D=C5_t in their two-resource group Gamma. The retained H_s,U_s,V_s form a three-resource group Psi. The two g_t pairs U_t,V_t and C6_s,C7_s form Omega. The separate C=C5_s and H_t endpoint losses are unused by these groups. All these are original77-point native resources. The source Q chosen below is common to their packing and the stars. If its size is q>=61 and its overlap is f, then

```
2B+D>=3(73-r_t)+f-3Gamma,
G+h_s>=q+f-Psi-Omega/2.
```

Use the two normal H upper endpoints, the normal C lower endpoint, and the exact H centre identities. Rebate the distinct native losses in the group bounds and in the separately valid weighted-star bound `2X_s+4X_t>=6f-2epsilon`. Their combination gives

```
2M_t+2q+11f<=2R+827+2r_t-delta+5epsilon.
```

In particular q>=61 permits the conservative constant122. Substitute M_t>=r_t+xi_t+A+16, epsilon=delta-42, and delta<=60-I-tau. The necessary scalar inequality becomes

```
4I+4tau+11F-2xi_s+210<=929.
```

For each sign choose independently whether to retain all core values or delete its largest j=1,2,3,4 magnitudes. Define `phi_v=min(xi_v,min_j(j+floor((tau_v-xi_v)/j)))` and `F=69-phi_s-phi_t`. The minimizing choices give one actual common Q of size at least61 with overlap at least F at zero displacement; all displacements subtract their integral gap. Hence this is an actual source selection, not a combination of different budgets. Credit42 and A<=60 restrict tau<=18. Exact enumeration of every nonnegative integer tau_s,tau_t,xi_s,xi_t with xi_v<=tau_v and tau<=18 has minimum

```
4tau+11F-2xi_s+210=935>929.
```

Dropping the nonnegative4I only enlarges the domain, so this finite certificate excludes every size and every residual mass. The independent derivation, labeled ledger, all7315 relaxed scalar states, and exact checker are in `../foundations/z5_alpha_final85.md` and `../foundations/z5_alpha_final85.py`. The scalar checker verifies the displayed finite consequence; the geometric prerequisites above remain written actual-resource arguments.

11.10 Z5 A1 and branch B

The activity/outer-integrality classifier is the literal Appendix E classifier following equation(z5-crosschain). It is valid with the fresh base reductions above. For split beta the whole paired unit alone is charged: its hit count N obeys77-delta<=N<=77 and beta_s+beta_t>=N-M. There is no selected-component capacity. For sole beta, normalize s to its actual owner; its base holes are O_s,H_s,O_t. When pair1 is empty, the final extra hole is exactly alpha_s,H_t,alpha_t,C5_s,C5_t,or U_s. The independent fresh exclusions for all split, retained-pair1, empty-pair1 and branch-B alternatives are in `zero85_z5_a1_b.md`, with exact arithmetic in `check_zero85_z5_a1_b.py`. The final C5_t case's cutoff and32-type tail received the independent receipt `../scalar/z5_c5t_cutoff_audit.md`, checked by `../scalar/check_z5_c5t_cutoff_audit.py`.

11.11 The cardinality-85 ownerless Z_4 family

Assume the actual activity classification has left exactly the four missing whole units $\{P, B, O_s, O_t\}$ in branch A. Both paired units are empty; no component is assigned an owner. The fourteen retained singletons are $H_v, U_v, V_v, C_{5v}, C_{6v}, C_{7v}, \alpha_v$ on the two physical signs. Use the cardinality-85 structural source

$$|D| = 77, \quad r_s + r_t = 68, \quad I = (r_s - 34)^2, \quad \xi_v = R_v - r_v, \quad \xi = \xi_s + \xi_t, \quad A = I + \tau + \delta.$$

The source results used here, without changing the meaning of δ, are

$$A \le 137, \qquad L = e_s + e_t = M - R \ge A + 20, \qquad e_v \ge \max(4, A - 69), \quad 0 \le \xi \le \tau. \tag{1}$$

The deliberately weaker last depth bound is enough; the sharper one-maximum bound $e_v \ge A - 61$ is unnecessary for this proof.

The fixed credit and the two H gaps. The same actual gap-label inventories as in the ownerless Z_4 proof give fixed credit $\eta = 18$. Here are the fresh numerical checks needed to use those inventories at cardinality 85. The four-edge physical arm gives

$$2X_v + X_o + g_v \ge 252 - \tau - \delta \ge 115 + I.$$

Consequently the literal active-center identities imply

$$d_v \ge M + 116 + I, \qquad X + g_v \ge 58, \qquad h_v \ge X_v + 1.$$

The raw gap first gives $h_v \le M + 56 < d_v$; the complete ten-label inventory improves this to $h_v \le M + 11$. Its only possible fixed labels are

$$C_{33}, C_{34}, C_{44}, C_{15}, C_{26}, C_{27}, C_{28}, U_2, U_3, V_2.$$

Here indexed U_i, V_i are fixed labels, not ordinary resources. The paired centers are LOW. Per sign the sixteen LOW labels

$$C_{55}, C_{56}, C_{57}, C_{58}, \quad V_3, V_4, V_5, V_6, V_7, V_8, \quad U_5, U_6, U_7, \quad F_3, F_4, F_5$$

occupy 25 of the 26 baseline groups over both signs. Three additional LOW witnesses are supplied by the reflected pairs V_2, C_{35}, C_{45} in already occupied groups. In particular $V_2^s + V_2^t \le 20 - 3X - G < 0$, since $3X + G \ge 230 + 2I$. The untouched C_{25} group supplies another LOW witness. Together with the eight HIGH O labels this is 44 misses against the baseline 26. Thus

$$\varepsilon = \delta - 18 \ge 0.$$

For the truncated core gap below H_v, the original counts 37 and 30 are unchanged. The two strict eliminations used to pass from 37 to 30 remain valid: freshly

$$a - (\alpha_v + M_v) \ge 79 + I > 0, \qquad a - U_3 \ge X + g_o + h_o - 36 > 0.$$

Therefore

$$h_v \le R + 31 \quad (H_v \text{ has a core hit}), \qquad h_v \le R_v + M_o + 30 \quad (H_v \text{ has no core hit}). \tag{2}$$

In the second case its actual native upper clipping loss is at least 73: the 69 core values and four positive source peels are all above the target. This uses the complete target $[1, S_v]$, not its hull.

A source obtained by deleting just the two maxima. Put $D_0 = \tau - \xi \ge 0$. After the maximum on each core side is deleted, its remaining rank deviations sum to $\tau_v - \xi_v$. Hence the resulting signed source Q has 67 values and span at most $66 + D_0$. At every integral displacement q,

$$|Q \cap (Q + q)| \ge 67 - D_0 - q.$$

Write $f = 67 - D_0$. Even when this lower bound is negative it is a valid intersection bound, so no positivity assumption on f is needed. Each leading-to-V edge contains the actual copies $\lambda_v + M_v + Q$ and $\lambda_v + v_4 + Q$ at gap X_v; the reflected leading-to-C_7 edge has the same gap. Designate the two full leadings as roots, with summed native loss $\ell \geq 0$. These stars avoid both alphas and both H resources.

Let Γ_v be the native alpha deficiency, k_v its lower clip, $N = \Gamma_s + \Gamma_t \geq k_s + k_t = k$, and let B denote the sum of the two actual rank errors. The fresh complete-target endpoints give

$$\alpha_s + \alpha_t \geq 78 - k - B.$$

For a light clip $k_v < 73$ its error is at most ξ_v. At most one clip is heavy, because $2 \cdot 73 > \varepsilon$. The rooted stars give separately valid inequalities

$$2X_v \geq 2f + N + \ell - \varepsilon, \qquad 2X \geq 4f + N + \ell - \varepsilon. \tag{3}$$

If an H has no core hit, its unused native upper clip $u \geq 73$ rebates the second row:

$$N + u \leq \varepsilon, \qquad 2X \geq 4f + N + u + \ell - \varepsilon. \tag{4}$$

These are deficiencies of distinct original retained resources.

Both H have core hits; both alpha clips are light. Now $B \leq \xi$ and $h_s + h_t \leq 2R + 62$. The identity $\alpha_s + \alpha_t = h_s + h_t - M - X$ gives $L + X \leq R - 16 + k + B$. Insert $L \geq A + 20$ and (3), then use $N \geq k$, $k \leq \varepsilon$, $B \leq \xi$. This yields

$$4f + 2I + 2\tau + 2\eta + \ell \leq 64 + 4\xi.$$

Since $f = 67 - \tau + \xi$ and $\eta = 18$, it follows that $2\tau \geq 240 + 2I + \ell$. But $\tau \leq 137 - I - \delta \leq 119 - I$. Thus $2 + 4I + \ell \leq 0$, a contradiction.

Both H have core hits; one alpha clip is heavy. Say $k_v \geq 73$. The opposite alpha is light. Its endpoint and $h_o \leq R + 31$ imply $e_o + X_o \leq 26 + \xi_v + k_o + \zeta_o$, where $\zeta_o \leq \xi_o$. Using the individual star, $N \geq k_v + k_o$ and $k_o \leq \varepsilon - k_v$ gives

$$2f + 2e_o + 2k_v + \ell \leq 52 + 2\xi + 2\varepsilon.$$

Consequently $2e_o + 264 + \ell \leq 2(\tau + \delta) \leq 274 - 2I$, so $e_o \leq 5$. On the other hand $\delta \geq 18 + 73 = 91$, and the source floor gives $e_o \geq A - 69 \geq 22$. This is impossible; the heavy endpoint error has not been bounded or discarded.

Exactly one H has no core hit. Its clip $u \geq 73$ leaves $N \leq \varepsilon - u \leq 46$, so both alphas are light and $B \leq \xi$. If v is the core-hit side, the two caps in (2) give $h_s + h_t \leq 2R + e_o + 61$, hence $e_v + X \leq R - 17 + k + B$. Use (4) and $k \leq \varepsilon - u$ to get

$$2e_v + 4f + 2u + \ell + 34 \leq 2R + 2B + 2\varepsilon.$$

Thus $2e_v + 202 + 2u + \ell \leq 4\tau + 2\delta$. Since $\delta \geq 18 + u \geq 91$ and $A \leq 137$, the right side is at most $366 - 4I$. Hence $e_v \leq 9$, whereas $e_v \geq A - 69 \geq 22$.

Neither H has a core hit. The two distinct native deficiencies total at least $146 > 137 \geq \delta$. All four H statuses are therefore impossible. This excludes the conditional cardinality-85 ownerless Z_4 family without assigning an owner or capacity to either empty pair.

12 Complete exclusion of the positive-hole SIZE85 stratum P2

This is a fresh argument for j=2 and 1<=z<=14 at cardinality85. It assumes the proved SIZE85 integral primitive structural theorem, the common covering and source envelopes, native-loss/forest lemmas, and the sharp raw-gap counts. All occurrences on opposite signs remain distinct labelled occurrences. No canonical endpoint model or old SIZE86 activity census is used.

12.1 1. Source parameters

Set d=77, |E|=69, r_s+r_t=68, I=(r_s-34)², G144=144. Let rho=rho_s+rho_t, mu=mu_s+mu_t, tau=sum(P_s)+sum(P_t)-T_rs-T_rt, H=h_s+h_t=tau-mu. Write C_v=sum(P_v), R_v=max(P_v), R=R_s+R_t, M_v=max(D_v), M=M_s+M_t, K=|2D|, and S=sum(D_s)+sum(D_t). The source peel magnitudes are R_v+a_v,R_v+b_v,R_v+c_v,R_v+e_v, with 1<=a_v<b_v<c_v<e_v. Put

```
U=a_s+b_s+a_t+b_t, C=c_s+c_t,
J=(e_s-c_s)+(e_t-c_t), A=I+tau+delta, m0=-126-w.
```

The source-only identities and inequalities are

```
S=2K+1316+w-delta,
2K=m0+A+4R+U+2C+J,
J+U<=458+w-A, J>=A-144-w, U>=6, J>=2,
A<=298+w, K-M>=R-286-w+A,
K-delta>=16-w+I+tau.
```

The quantitative layers, for U0 the first four peeled masses, V0 the last four, C0=C_s+C_t=1190+I+tau, and C1=C0+U0, give

```
C0-U0>=320+2I+delta-2z,
C1-V0>=446+2I+mu+rho+delta+2z.
```

Initially w<=4z<=56. The structural budget gives delta<=302-I-rho+2z. If the smaller side were at most17, then I>=289 and rho>=49, giving delta<=-8. Hence 18<=r_v<=50; throughout this range rho>=49 and mu>=62. Non-AP primitivity gives R_v>=r_v+1 and tau_v>=r_v-3.

12.2 2. Common-strip localization using the actual four inner occurrences

The source envelope gives H<=236+w-I-delta<=292-I-delta. For every integer r, T_r+(r-34)²>=397; hence each C_v>H. The common-strip lemma fills [H,C0-H] in the absolute core support. Its width L=C0-2H satisfies

```
L>=1016+2I+delta-w>=960.
```

Sort the four inner occurrences as t1<=t2<=t3<=t4. All sixteen subset choices are required. The largest consecutive gap among their subset sums is at most max_i(t_i-sum_(h<i)t_h). The first three terms are at most t3<=U0/2. The layer and source inequalities give

```
U0<=1168+w-2I-2delta+2z<=1252,
```

so these three terms are at most626<L.

For the last term, t4 is necessarily one side's second occurrence R_v+b_v; the other three contain R_v+a_v and both opposite occurrences. Thus

```
t4-t1-t2-t3=b_v-a_v-2R_o-a_o-b_o
            <=U-8-2R_o.
```

Crucially, use the offset sum U rather than the much larger layer mass U0. Since J+U<=458+w-A and J>=2, U<=512. Since R_o>=19, this last gap is at most466<L. Therefore every

translate of the core central interval by an actual inner subset sum overlaps or abuts the next, filling [H,C1-H].

For the outer occurrences, selecting none or all already suffices because

```
C1-V0-2H>=-26+4I+mu+rho+3delta+2z-2w
         >=85-6z>=1.
```

This fills [H,S-H]. Complement symmetry in the same fixed boundary strip now gives z,w even, w<=z<=14, and both targets have eta_v=w/2 holes. The old coarse all-sixteen inner estimate is not needed.

After localization, H<=250-I-delta and R<=72+rho+H<=72+tau<=384-I-delta. Thus T_r-R>=13+delta. The first absolute-core gap f is below both R_v: if one side had all its entries below f, its total would be at most f-1<R, contradicting C_v>=T_r>R. All peeled occurrences exceed f, so f survives and w>=2.

Every integral ordinary center in [1-M_v,S_v] is active. To prove this without an endpoint assumption, use R_o<=r_o+tau_o and tau_v>=r_v-3:

```
S_v-H-R_o>=T_r+7r+2I-594-2w+2delta>=39+2delta.
```

The last integer quadratic has minimum39 at r26. A target hole u lies within H of its upper endpoint, so all u-P_o are positive hull integers; at most eta_v<=7 are missing and r_o>= 18 guarantees a target hit. For a nonpositive center u>=1-M_v, the value u+M_v lies in the complete lower target interval, since S_v-M_v>=C_v>H. This proves the activity claim.

Finally K-delta>=16-w+I+tau>0 makes both leading translates active and their shifts integral. A nonintegral v3 or v4 would force respectively {own2_v,C6_o} or {own3_v,C7_o} to be the two missing singleton units. With only these holes H_v and alpha_v remain active, contradicting the integrality of H_v-alpha_v=lambda_v+v3+v4. Two nonintegral lower coordinates would require four holes. A nonintegral v2 would make O_v, H_v,alpha_v three missing singletons. All outer coordinates are integral.

12.3 3. The first-gap profiles and two-cap exclusion

Write t_v=d_v-M, where d_v=v1-v2, and define T_v to mean that O_v is active or both beta_v, H_v are active. Sharp raw gaps give

```
T_v => t_v<=22+w/2<=29,
beta_v and O_v active => 2t_v<=27+w/2<=34.
```

Both cap predicates cannot hold. The four-star estimate is 2X+4R+delta>=548, and the high-tail count gives d_s+d_t+G+delta-2K>=642. With Y=2M-2K+44+w+delta, two caps and the leading counts imply

```
2Y>=1190-4R-delta,
Y<=616+3w-2R-2I-2tau-delta.
```

Consequently 4I+4tau+delta<=42+6w, contrary to tau>=62+2z and w<=z. At least one O singleton is missing, and at least one cap predicate holds.

Let k_v count core entries below the first gap f and k=k_s+k_t. The greedy first-gap rule gives their total f-1 and interval completeness below f. The common rank lower bound is

```
I+tau>=69f-1-(f+34.5)k+k²/4
            +(r_s-34-(k_s-k_t)/2)².
```

Since f<R/2<=192 and k(k+2)/4<=f-1, k<=26. For f>=7,k<=4 the displayed lower bound is at least320>312. For k>=5 it is at least (-k³+68k²-4k+272)/4>=1827/4>312. Thus f<=6.

The complete low multisets are empty, {1}, {1,1}, {1,2}, {1,1,2}, and at f6 either {1,1,3} or {1,2,2}. These are fixed by the total f-1, interval completeness and multiplicity at most two. Put b for the mass excess of the coordinatewise minimal side profiles, B for their two maxima, and sigma=tau-b>=0. If one distinguished side has r=34+x, their constants are:

f	I+b	B	minimum tau
1	I+68	70	68
2	101+x(x-1)	71	85
3	I+132	72	132
4 split	I+199	74	199
4 together	199+(x-1)²	74	168
5	228+x(x-1)	75	212
6 both low multisets	294+x(x-1)	77	278

In particular f>=2 implies tau>=85, and I+b-2B>=-41 in every such row.

12.4 4. No C7 singleton is empty

If C7_v were empty, it and at least one missing O would be the two holes, so C6_v is active. Integral activity localization makes C7_v HIGH, hence lambda_v>S_v+M_o. In the leading resource the entirely HIGH set lambda_v+(D union(M_v+D)) has at least d+(r_v+4)=r_v+81 points. At least three of the four higher native singletons V_v,U_v,H_v,O_v are retained, and each loses r_v+5 points. Thus delta>=4r_v+96.

If f>=2, A>=I+4r_v+96+tau>=132+96+85=313>312, impossible. Here I+4r=(r-32)²+132.

At f1 write tau=68+sigma. The source bound yields

```
(r_v-32)²+sigma<=2+w<=16.
```

Thus 28<=r_v<=36, the opposite size is32..40, and sigma<=16. Each side contains the consecutive prefix 2,...,r_v-sigma_v+1, at least 2,...,13; every remaining coefficient is at most r_v+1+sigma_v<=57. The prefix support [0,L] minus{1,L-1}, with L>=90, keeps exactly those two reflected holes under every remaining coefficient. Therefore both side hole counts are2, the absolute core has exactly two holes, z=2, and the persisted gap1 gives w=2.

If the missing O were opposite C7_v, all four higher v singletons would be retained and delta>=5r_v+101. Then I+5r_v+sigma<=129+w<=131, whereas the integer minimum of I+5r is164. Therefore O_v is the other hole. It is HIGH, and its four fixed labels O_v+v_i are HIGH; they supply four additional fixed credits beyond the baseline nonpositive witnesses. Hence delta>=4r_v+100. With w2 the source gives (r_v-32)²+sigma<=0. Thus r_v32,I4,sigma0,delta228, A300.

Now m0=-128,T=m0+A=172. Source envelopes give J+U<=160,J>=154,U>=6, so U6,J154. Both signs have (a,b)=(1,2), for which the four-offset fringe counts F_s,F_t are at most5 per sign. These F values are distinct from the baseline core-maximum sum B in the profile table. The full fringe comparison gives

```
T+U<=J+2+2(F_s+F_t),
```

and would require178<=154+2+20=176. This excludes the final equality. Therefore every C7 singleton is retained.

12.5 5. Rooted forests reduce to f1 and exclude simultaneous beta/O activity

Trim the four largest core entries on each sign, leaving an actual61-point set Q. Its rank-certified overlap constant q satisfies

```
4q=516-5B-sigma+R.
```

On a cap side v use the positive arm L_v-V_v-U_v and the negative arm L_o-C7_o-C6_o. If V_v is retained, C7_o is also retained and at most one of U_v,C6_o can be empty. Thus choose one full two-edge arm and the first edge of the other; every selected resource is retained. Designate L_v as a root. The same original deficiency budget then gives

```
t_v>=K-M+3q-delta
   >=101-w+I+b-2B+sigma/4.
```

For every f>=2 this is at least60-w>=46>29, contradicting the cap.

If V_v is empty, the other hole must be O_o: choosing O_v instead would leave both O_o and beta_v,H_v active, giving two caps. Both H singletons are retained. Branch B would retain both beta singletons and again give two caps; in A the retained beta pair is sole on v. Hence beta_v,O_v are active and the sharp double-gap upper bound is t_v<=17. The empty V_v is HIGH, since leading activity implies V_v>1-M_v. Both H_v,O_v are retained HIGH resources and each loses r_v+5 native points. Use edges L_v-U_v and L_o-C7_o, root L_v and include those two unused losses. This gives

```
t_v>=-18-w+I+b-B+2r_v+sigma/2.
```

The f1 integer minimum is47-w>=33; every f>=2 is at least48-w>=34. Both contradict17. Thus every cap side has V_v retained, and f=1.

At f1 the rooted lower bound is t_v>=29-w+I+sigma/4. If beta_v and O_v were simultaneously active, the double gap would force

```
4I+sigma<=5w-62<=8.
```

For w<=12 this is already impossible. For w14, I<=2 and sigma<=8; the consecutive core prefixes and small final coefficients then preserve exactly the two side holes as in the preceding dense-profile argument. This gives z=w2, which contradicts the same inequality. No side has both beta_v and O_v active.

12.6 6. Exhaustive activity forms and the terminal contradiction

If both O singletons are missing, B is impossible because both beta,H caps would hold; in A the retained beta pair has a sole actual owner. If exactly O_v is missing, O_o is active, hence beta_o is inactive. B would make beta_o the other hole and leave beta_v,H_v active, giving two caps. In A, either H_v is the other hole or the whole beta pair is empty; every other choice would again leave both cap sides active. Thus exactly the following A whole-hole forms remain:

```
{O_v,beta pair}, {O_s,O_t}, {O_v,H_v}.
```

All unlisted whole units are retained. In the first and third forms choose the side u opposite the missing O_v; in the middle choose the sole beta owner. That side has the cap predicate. The retained singleton centers alpha_o,C5_u and their exact identity imply t_u>=2X+G+2. Hence

```
2X+G<=20+w/2.
```

At f1 write sigma_v=sum_i(x_vi-i-1), xi_v=R_v-r_v-1>=0, and sigma=sigma_s+sigma_t,xi=xi_s+xi_t. Source mass gives

```
I+sigma+delta<=230+w.
```

Trim six largest core entries per sign, leaving57 actual points. Its span is at most58+(sigma-xi)/6, so the certified intersection lower bound at gapq is55-(sigma-xi)/6-q. All U_v,V_v,C7_v are retained in every form. On each target the retained tree L_v-V_v-U_v with edge L_v-C7_v has three gaps X_v,g_v,X_o. Both targets together give, in one budget,

```
2X+G>=330-sigma+xi-delta.
```

Combining these three inequalities forces

```
80+I+xi<=3w/2<=21,
```

a contradiction. This excludes the whole P2 stratum, including all positions of empty endpoints and all paired-unit ownership choices.

12.7 7. Exact finite arithmetic replay

`check85_p2.py` checks the range, width, activity quadratics, first-gap polynomial and profile minima, C7 equality, rooted-forest scalar margins, and terminal contradiction. It is an arithmetic checker of these written lemmas, not an enumeration of actual source sets or activity descriptors.

13 Fresh n=85 positive-case prefix lemmas

This file contains the common positive-case notation, the P3 localization and activity classification, and its first-gap4 exclusion. The completed first-gap1/2/3 P3 terminals are in p3_terminal.md. The complete P4 source-seed argument, classifier, and terminals are in p4_fresh.md. There are no open transport branches in these files.

The new structure and source results are proved in inverse/proof85_inverse.md and foundations/source_triple_fringe85.md (paths relative to progress85). Throughout, the source has77 points, its core E has69 labelled points including zero, and the two positive/negative core sizes satisfy a+b=68. Its triangular baseline is1190 and I=(a-34)^2. Put A=I+tau+delta, m0=77j-280-w, T=m0+A, R=R_s+R_t, M=M_s+M_t, and L=M-R. The first two peel offsets on both signs total U; the third offsets total C and the last-minus-third offsets total J, so L=C+J.

The exact source identity and the source envelopes are

- S=2K+1470-77j+w-delta.
- 2K=m0+A+4R+U+2C+J.
- K<=2R+C+166, J+U<=332-m0-A.
- J>=m0+A-18, L>=m0+A-8.
- K-delta>=77j-142-w+I+tau.
- K-M>=R+77j-440-w+A.
- T<=165, equivalently A<=445-77j+w.

The deliberately weaker envelope A<=452-77j+w is used below where it already suffices; it follows from the same source bounds and is not a provisional assertion. Positive-core mu>=62. Use rho>=49 initially, and rho>=58 once both sizes are>=21. The new inverse identities give

- rho+kappa+delta<=456-77j+3z-I.
- U0-C0+delta-2z<=-166-2I-77j.
- V0-C1+delta+2z<=-292-2I-77j-mu-rho.

13.1 P3 localization before a first-gap profile

Here j=3. Initially 1<=z<=14,w<=4z. The weaker source bound gives A<=221+w<=277 and H=tau-mu<=215-I-delta. The integer inequality T_r+(r-34)^2>=397 implies that each core side mass exceeds H, so the core central strip exists.

Its width L0=C0-2H is at least1037+2I+delta. The total U0 of the first two peeled magnitudes on each sign is at most1098-2I-2delta, using C0<=1467 and the inner row. For their four sorted weights, the first three gap terms are<=U0/2<=549. The other three weights at the final gap total at least72+r_min. Since A>=I+mu, r_min>=20. Thus

```
L0-(t4-t1-t2-t3)>=1037-1098+2(72+20)=123.
```

All sixteen inner subset translates overlap. At the outer peel the none/all translates overlap because

```
C1-V0-2H>=204+4I+3delta+2z>0,
```

using mu>=62,rho>=49. Consequently w,z are even, 0<=w<=z<=14, and the two targets have equal hole counts and localized boundary holes.

13.2 P3 finite dense seed and boundary activity

After localization tau<=235, C0<=1425, and r_min>=21. Six absent values in[1,17] force tau>=276, so at least twelve distinct values occur there. The exact finite lemma checked by dense_seed_check.py is:

Every twelve-element subset B of[1,17], with total T_B, satisfies [11,T_B-11] subset FS(B), and its lower half has at most five holes.

The checker covers all C(17,12)=6188 subsets by integer bitset addition. To adjoin the remaining labelled core occurrences, each magnitude is<=47+235=282. For n<=33 selected occurrences, a failed join x>T_B-21 implies x>=58; at least35 remaining entries would give C0>=78+35*58>1425. For n>=34 the selected total is>=306 and x<=282<=T_B-21. Every join succeeds, and hence z<=10. The weaker source bound then gives A<=231, H<=169-I-delta, tau>=62+2z, and delta<=159-z-I.

Five low holes among[1,10] omit five singleton values of sum<=40 and force tau>=5*68+30-80=290. Therefore z<=8. For the terminal first gaps m<=3, four low holes have total<=m+8+9+10<=30 and force tau>=232. The weaker consequence T<=167 of the proved T<=165 gives A<=216+w<=224, excluding this possibility. Thus at m<=3, z,w<=6.

The source row K-delta>=89-w+I+tau proves both leadings active. From H<=169-I-delta and R<=68+tau<=299-I-delta,

```
S_v-H-R_o>=T_r+5r-468+2I+2delta>=141+2delta.
```

Also T_r-R>=98+delta. A target-hole center has at least r_o>=21 distinct positive images under opposite core entries, all within its hull, whereas it has at most w/2<=5 target holes. Nonholes use zero; a center in[1-M_v,0] uses M_v and the complete lower target interval. Every integral center in[1-M_v,S_v] is therefore active, and an empty integral center above an active lower center is HIGH>=S_v+1.

At the first failed greedy inequality, the entries below the gap m total m-1 and generate[0,m-1], while every remaining entry exceeds m. If R_v<=m, that entire side would lie in the prefix, giving C_v<=m-1<R<T_r, impossible. Thus m<min(R_s,R_t); it survives both peels, w>=2, and the target holes are exactly S_v-h for the lower source holes h.

13.3 P3 outer integrality and exhaustive activity forms

Write U_v=own2_v, V_v=own3_v, C_iv=lambda_v-o_(i-3) for4<=i<=7, X_v=v4-M_v, X=X_s+X_t, G=g_s+g_t. The active leadings make lambda_v integral. Each lower coordinate v3 has singleton witnesses U_v,C6_o, and v4 has V_v,C7_o; the eight witnesses are distinct. Two nonintegral lower coordinates require four holes. A sole nonintegral lower coordinate requires its two witnesses and one of H_v,alpha_v missing. If alpha_v is missing, retained O_v,H_v force both top coordinates integral, a contradiction. If H_v is missing, retained O_v,alpha_v and C5_o in A or beta_v in B force v2 and then the lower coordinate integral. A nonintegral top pair would make O_v,H_v,alpha_v and C5_o in A or beta_v in B missing. All eight outer magnitudes are integral.

The branch-independent retained-resource forest rows are

```
2X_v+X_o+g_v>=252-tau-delta,
2X_v+X_o+G>=305-tau-delta,
2X+4R+delta>=548.
```

The two candidate arms are leading_v--V_v--U_v and leading_o--C7_o--C6_o, both with gaps X_v,g_v. At most one vertex of one arm is empty. Restrict core translates using trim3 for the first row and trim4 for the second. Positive and negative restricted arms on one target are disjoint since M>R. Inside a target every vertex at a point is retained; outside it at most one is empty. Thus edge counts are bounded by retained-resource union deficiency, with each full leading capacity counted once. Four full-core stars give the third row.

The following five assertions exhaust the activity restrictions:

1. O_v,C6_v cannot both be active.
2. O_s,O_t cannot both be active.
3. beta_v,H_v,C6_v cannot all be active.
4. Active C5_v implies active C6_v.
5. beta_v,O_v,C7_v cannot all be active.

For1 and3, actual C6_v activity gives d_v-M>=306-tau-delta>=85-w+I; O_v activity or beta_v,H_v activity gives a raw cap<=27, impossible. For5 the lower bound is253-tau-delta>=32-w+I, exceeding the double cap<=16. These conservative raw caps follow from the proved O22+eta, beta/H21+eta, and double27/2+eta/2 counts with w<=10. For2, with Y=2M-2K+44+w+

delta, the high-tail count565 and star count548 yield189+4I+4tau+delta<=6w<=60, impossible. For4, active C5_v and empty C6_v make C6_v HIGH. The leading then loses all77 points of lambda_v+D_v, and at least three retained higher singletons among C7,V,U,H,O each lose r_v+5. This forces delta>=92+3r_v, but delta<=169-I and min_r(I+3r)=100 give a contradiction.

The literal owner identities, with W=v3+v4+o3+o4>2M, are

```
C4_s+C4_t=-(s2+s3+s4+t2+t3+t4),
beta_s+beta_t+C5_s+C5_t=-2W,
C4_v+beta_v+C5_o=-2v3-2v4-o2-o3-o4,
C4_v+C5_o=-(beta_s+beta_t)-2W-d_o.
```

Comparison with the actual lower activity endpoint1-M_v proves that retained pair1 has a sole owner, the displayed triple cannot be active, and the two retained B pairs have matching sole owners.

To exhaust whole-unit holes, at least one O is missing by2. If both are missing, neither C6 can be the third A hole, because4 would also delete its C5 singleton. Both C6 are retained. An empty beta pair gives A0. Otherwise3 forces H missing on the active beta's side; the opposite beta is inactive and owner grammar puts pair1 on the opposite side, giving A1. In B, both O missing would require one further hole in each disjoint triple{beta_v,H_v,C6_v}, exceeding three holes.

If just O_o is missing, active O_v forces empty C6_v and C5_v by1 and4. In A these are all three holes: the beta owner is v and pair1 owner o; active beta_v,O_v,C7_v contradict5. In B the holes are{O_v,C6_o,Z} with Z in{beta_v,H_v,C6_v}; pair owners are v, and4 excludes Z=C6_v. In either remaining choice beta_o,O_o,C7_o are active, contradict5. Exactly these two forms remain:

form	whole-unit holes	actual owners
A0	beta pair,O_s,O_t	pair1 sole owner s or t
A1	O_s,O_t,H_v	beta sole owner v, pair1 sole owner o

All unlisted whole units are retained. The fixed HIGH/LOW label counts give credits8 for A0 and11 for A1; the terminal proofs retain their exact native/grouped deficiencies.

13.4 P3 first-gap profiles and the gap4 exclusion

The first-gap5 low multiset must be{1,1,2}. If k is the side containing1,2, its profiles are(1,2,6,...,k+3) and(1,6,...,q+4), q=68-k. Therefore

```
I+tau>=228+(k-34)(k-35)>=228>A<=224.
```

For m6, the total count below m is<=3, giving tau>=4(68-3)=260. The dense seed puts every lower hole in[1,10], so the first gap satisfies m<=10. Thus T_(k_v)<=m-1<=9 gives k_v<=3. For7<=m<=10, each side floor (m-k_v)(r_v-k_v) is at least4(r_v-3), yielding tau>=4(68-6)=248, again impossible. Hence m<=4.

At m4 the low multiset is{1,2}, either split or together. The minimal I+tau is199+chi+sigma, with chi=I if split and chi=(k-35)^2 if k is the together side. A value x in[5,12] absent from both cores adds at least76-2x>=52 to the baseline, exceeding even the weaker true-excess cap32. Hence the actual seed{1,2,5,...,12} is present; it has exactly the hole4 and its reflection. Every remaining magnitude is<=83. While selected total is below92, a failed join x>T_B-9 implies x>=63 and at least54 remaining occurrences would force mass>1425; once T_B>=92, x<=83<=T_B-9. All joins succeed, so z=w=2. The weaker T<=167 cap gives chi+sigma+delta<=19; below we retain the still weaker24, since it already suffices. Charged endpoints are normal because delta<=24<r_min+5.

In A0 let s be sole pair1 owner. The identity H_s=P+alpha_s+alpha_t+d_t+2V+G gives S_s>=d_t+2V+G+155-E. The opposite leading and K<=M+R+21 give

```
S_t+M+(2X+G)+(2X_t+g_t)<=R+1105+w-e.
```

The six- and three-edge forests use f=(146-sigma+R)/4 and yield

```
S_t+M+5R/4-9sigma/4-2delta+3e<=1553/2+w.
```

The physical baseline tau_t>=2r_t-4, R_t>=r_t+2 gives S_t+M>=T_r+6r+R+L+14. With R>=74, L>=142-w+chi+sigma+delta, and chi+sigma+delta<=24,

```
T_r+6r+9chi/4+3e<=488.
```

As a function of physical r, chi is(r-34)^2 in the split placement and(r-35)^2 or(r-33)^2 in the together placements. All three integer quadratic minima exceed600, contradicting488.

In A1 let s be beta owner and t pair1 owner. Normal endpoints at beta_s,P_t,V_t give S_s+ 3g_s+4X_s+2s4+t4<=1089+w+r_s+delta-2e. Source mass, the ordinary gap pair and stars imply

```
T_r+6r+chi+sigma/4+283/2-2w+2e<=21R/4+9epsilon/2.
```

The profile radial estimate is2R+chi+delta<=178+w, derived from K<=C+2B+174+sigma with B74 and J>=A-67-w. Using w2 and chi+sigma+delta<=24 yields

```
T_r+6r+11chi/2+17sigma/8+13e/2<=380.
```

Each physical orientation has a larger global integer quadratic minimum. Thus m4 is excluded in both forms. Together with the complete first-gap1/2/3 proof in p3_terminal.md, this excludes P3.

13.5 P4 proof reference

p4_fresh.md contains the full seven-seed availability and propagation proof, all six resulting profiles, the exact CORE8 classifier, and every terminal card. check_p4_cards.py checks the stated exact seed supports, propagation mass bounds, fixed-label incidence ledger, adaptive scalar combinations, integer quadratics, and terminal arithmetic.

14 n=85 P3 terminal exclusions using the fresh universal source cap

This proof uses the new structure and the weaker consequence T=m0+A<=167 of the proved universal source theorem T<=165, where m0=-49-w and A=I+tau+delta. Thus

```
A<=216+w.
```

The source is77 points, the core69 labelled points including zero, and a+b=68. The baseline I is(a-34)^2. Source mass is S=2K+1239+w-delta, and full-source central-fringe bound K<= M+R+21. The exact source offset bound is L>=A-57-w.

The localization/classification in `fresh_positive_notes.md` supplies the two forms A0 and A1. The dense seed lemma first gives z<=10. Five low holes among[1,10] would avoid five values of sum at most40, forcing tau>=5*68+30-80=290, impossible. Hence z<=8. For a first gap m<=3, four low holes include m and have total at most m+8+9+10<=30. They force tau>= 4*68+20-60=232, whereas A<=224. Therefore z<=6, w<=6.

At m=5, the profile I+tau>=228 contradicts A<=224. At m>=6 the generic low-rank floor also contradicts the cap. At m=4 the seed lemma gives z=w=2 and sigma+delta+chi<=19 (the former notes used the weaker24); the separate normal-endpoint m=4 exclusions in the notes apply unchanged. It remains to treat m=1,2,3.

14.1 Profile and endpoint constants

In the following table sigma is excess over the displayed minimal profile and B the sum of baseline maxima. Q61 is obtained by deleting four largest core magnitudes on each side.

m	I+tau	B	4f	L lower bound	mass cap
1	68+I+sigma	70	166-sigma+R	11-w+I+sigma+ delta	sigma+delta+ I<=148+w
2	101+chi+sigma	71	161-sigma+R	44-w+chi+ sigma+delta	sigma+delta+ chi<=115+w
3	132+I+sigma	72	156-sigma+R	75-w+I+sigma+ delta	sigma+delta+ I<=84+w

For m=2, k is the physical unique-1 owner's size and chi=(k-34)(k-35)>=0. It need not equal the paired-resource owner. If that physical owner is side s then k=r_s; otherwise k=68-r_s.

For a native retained ordinary/sole-pair center Z on target v the nominal endpoint rows are

```
Z>=73-r_v-l_Z-A_Z,
Z<=S_v+r_o-72+h_Z+D_Z.
```

Normal endpoints use zero error. Write c for the selected lower or upper clipping count. An outside lower endpoint has c=73-r_v+j, with the next positive core rank j+1 if c<73; reflection gives the same upper argument. If its baseline displacement is d, then

```
(73-c)d<=sigma_v.
```

The baseline endpoint error is at most b+d, where b=1 at m1 and b=2 at m2,m3. For a light c<73, the licensed sigma+epsilon bound gives d<=c. For c<=71 this follows from c(74-c)>=213 on the relevant interval26..71, and sigma+epsilon<=146; smaller actual side lower bounds only strengthen it. For c=72:

- A1 uses credit11, so sigma+epsilon<=148+6-11=143 at m1; the other profiles have still smaller caps.
- A0 m1 with z2 has sigma+epsilon<=142. With z4 it is<=144. With z6, the two additional low holes imply tau>=176 and delta<=46, so c72 cannot occur. A0 m2,m3 again have smaller caps.

Thus every light error is at most b+c<=b+Gamma_Z.

For a heavy c>=73, at most one relevant selected resource is heavy. In A1 epsilon<=143 at m1, and <=110,79 at m2,m3, so two73 clips are impossible. In A0, m2/m3 also have epsilon<146. At m1, z2 gives epsilon<=142; z4 implies tau>=120, delta<=100, epsilon<=92; z6 gives epsilon<=38. Thus again two73 clips are impossible.

The heavy next source magnitude is peeled. Its endpoint error is at most b+sigma_v+e_v, where e_v is the fourth peeled offset on the physical source side providing that next magnitude. This deliberately loose bound is used in A0. For A1, activity bounds the error by M_v-r_v+72-c<=M_v-r_v-1. The profile maxima are r_v+1 at m1 and at most r_v+2 at m2,m3, so the sharper heavy error is at most e_v+sigma_v+b-1. Together with the other two light errors, this is the stated aggregate3b-1 below. The native deficiency of a heavy resource is at least73, and delta>=e+c.

14.2 A0: physical-target capacity replaces the old total-mass transport

Let s be the actual sole pair1 owner, and e=8, epsilon=delta-e. The original unit holes are beta-pair,O_s,O_t. Target t receives exactly its leading and seven retained ordinary resources H,U,V,C5,C6,C7,alpha, with capacity K+539. Their grouping gives

```
|F_t|>=K+539-epsilon,
|F_s|<=K+700-e,
S_s<=K+700-e+w/2.
```

The literal identity is

```
H_s=P+2alpha_s+alpha_t+C5_t+4M+4X+2G.
```

Use endpoint resources P,alpha_s,alpha_t,C5_t,H_s with clipping/error weights1,2,1,1,1. Their nominal lower/upper constants give

```
S_s>=233+4M+4X+2G-E_clip-B_err.
```

On each target v the retained tree is leading_v--V_v--U_v, with the extra edge leading_v--C7_v. Its three gaps are X_v,g_v,X_o. For the positive edge restrict the full leading to lambda_v+M_v+Q_v; for the negative edge use lambda_v-M_o+Q_v, where Q_s=Q61 and Q_t=-Q61. Ordinary vertices are restricted to their matching Q_v translates. Each edge contributes at least f minus its gap. Both restricted copies remain subsets of one full leading capacity K. Across the two targets these six edges have total gap2X+G, so their grouped deficiency satisfies D_F>=6f-(2X+G). They are disjoint from the five endpoint resources, and E_clip+2D_F<=2epsilon. Combining the last three displays with K<=M+R+21 yields the master inequality

```
3M+12f+3e<=R+488+w/2+2delta+B_err.
```

All target holes are already in these native/grouped deficiencies.

Let l,D,Mcap,chi0,b be the following constants:

m	l	D=3(4f constant)-488	Mcap	chi0	b
1	11	10	148	I	1
2	44	-5	115	chi	2
3	75	-20	84	I	2

The master uses M>=R+L, L>=l-w+chi0+sigma+delta, and R>=70,71,72 respectively.

If every endpoint is light, B_err<=6b+2epsilon. Direct substitution forces

```
5R+4chi0+sigma+5e <= Mcap+6b-3l-D+(9/2)w.
```

The maximum right sides are138,27,-82 respectively; the left side is at least350+40,355+40, 360+40. Thus no all-light case survives.

If there is one heavy endpoint of weight q in{1,2}, then

```
B_err<=6b+2epsilon-qc+q sigma+q e_v.
```

Cancel q e_v against qL and use the mass cap. The exact terminal row is

```
5R+(3+q)chi0+(q-1)delta+5e+qc+(3-q)l+D-6b
    <=2q Mcap+(7/2+q)w.
```

For m1 a heavy forces w<=4. For q1, left>=489 and right<=314. For q2, delta>=e+c>=81 gives left>=632 and right<=614. For m2 the q1/q2 left floors are539,649 and right ceilings257,493. For m3 they are591,670 and ceilings195,369. All are strict contradictions. Hence all A0 first gaps are excluded.

14.3 A1: the same ten-edge forest applies to all three profiles

Let s be physical beta owner, t actual pair1 owner, r=r_s, e=11 and epsilon=delta-e. Reserve b_*=Gamma_beta_s+Gamma_P_t and h_*=Gamma_H_t.

The ten-edge forest consists of these retained arms:

```
on s: leading_s--V_s--U_s;
      leading_s--C7_s--C6_s--C5_s;
on t: leading_t--V_t--U_t--H_t;
      leading_t--C7_t--C6_t.
```

Its literal edge inventory is:

target	edge	gap
s	leading_s,V_s	X_s
s	V_s,U_s	g_s
s	leading_s,C7_s	X_t
s	C7_s,C6_s	g_t
s	C6_s,C5_s	h_t
t	leading_t,V_t	X_t
t	V_t,U_t	g_t
t	U_t,H_t	h_t
t	leading_t,C7_t	X_s
t	C7_t,C6_t	g_s

Their gaps total2X_s+2X_t+2g_s+2g_t+2h_t. Restricted positive and negative hub copies are lambda_v+M_v+Q_v and lambda_v-M_o+Q_v; each stays inside the same full leading vertex. Ordinary vertices use Q_v translates. Omitting both paired resources leaves deficit at most epsilon-b_*, and hence

```
X_s+X_t+g_s+g_t+h_t>=5f-(epsilon-b_*)/2.
```

The four-edge subforest uses leading_s--V_s--U_s and leading_t--C7_t--C6_t. It omits both pairs and H_t, so

```
X_s+g_s>=2f-(epsilon-b_*-h_*)/2.
```

The two stars leading_s--V_s and leading_t--C7_t use the full core E69, whose span is R, and give

```
X_s>=137-R-(epsilon-b_*-h_*)/2.
```

Add the ten-edge row, twice the four-edge row and three times the two-star row. The resulting positive scalar combination gives

```
W=6X_s+X_t+3g_s+g_t+h_t
  >=9f+411-3R-3epsilon+3b_*+(5/2)h_*.
```

The center identity H_t=2d_s+beta_s+P_t+2s4+t4+g_s+g_t+h_t and its three endpoint rows, followed by the leading-s bound twice, give

```
S_s+3g_s+4X_s+2s4+t4+g_t+h_t
  <=1089+w+r+delta-2e+E_err.
```

All-light E_err<=3b+b_*+h_*. One-heavy E_err<=e_u+sigma_u+(3b-1)+G_light, and

```
3b_*+(5/2)h_*-G_light>=365/2,
L-e_u>=4, sigma_s-sigma_u>=-sigma.
```

The physical source lower rows are exactly Tr+6r+19 at m1, Tr+6r+18 for m2,k=r, Tr+7r+24 for m2,k=68-r, and Tr+7r+22 at m3, plus sigma_s+R+L+6X_s+X_t. Hence the resulting scalar right constants are571/2,1191/4,1167/4,305 respectively, in the shared form

```
Tr+c_r r+sigma_s-9sigma/4+R/4+L+5e+3b_*+5h_*/2
  <=constant+w+4delta+E_err,
```

with c_r=5,5,6,6 respectively.

14.3.1 m1

All-light substitution gives

```
Tr+5r+4I+7sigma/4+sigma_s+R/4+5e<=1443/2+5w.
```

For integer r, Tr+5r+4(r-34)^2>=679. R>=70 and5e55 give a left floor1503/2=751.5. For w<=4 the right is<=741.5. If w6, three low holes include1 and force tau>=176, so sigma>=108 and the left is strictly larger than751.5, whereas the right is at most751.5.

One-heavy gives

```
Tr+5r+4I+3sigma/4+R/4+5e<=693+5w<=723,
```

contradicting the same left floor751.5.

14.3.2 m2: both physical unique-1 owners

If k=r, all-light gives

```
Tr+4r+I+3(r-34)(r-35)+7sigma/4+sigma_s+R/4+5e
  <=2283/4+5w<=2403/4.
```

Its quadratic part is at least661. If k=68-r the corresponding row is

```
Tr+7r+I+3(r-34)(r-33)+7sigma/4+sigma_s+R/4+5e
  <=2531/4+5w<=2651/4,
```

whose quadratic part is at least723. Both contradict even without the other positive terms.

One-heavy gives the two rows

```
Tr+5r+4(r-34)(r-35)+3sigma/4+R/4+5e<=2305/4+5w<=2425/4,
Tr+6r+4(r-34)(r-33)+3sigma/4+R/4+5e<=2281/4+5w<=2401/4.
```

Their quadratic parts have minima695 and689 respectively.

14.3.3 m3

All-light gives

```
Tr+6r+4I+7sigma/4+sigma_s+R/4+5e<=488+5w<=518.
```

One-heavy gives

```
Tr+6r+4I+3sigma/4+R/4+5e<=919/2+5w<=979/2.
```

Here Tr+6r+4(r-34)^2>=709. Thus neither alternative survives. Together with the normal-endpoint m4 exclusions, A1 is impossible.

Every use of repeated scalar inequalities is a positive linear combination of separately valid budget consequences. Each forest vertex keeps only its original full resource capacity.

15 Fresh P4 proof and cards for n85

These calculations use the fresh structure, the proved sharp raw caps, and the universal source theorem T=m0+A<=165. With j4,w2 one has m0=26, A=I+tau+delta, I=(r_s-34)^2, and

```
A<=139, S=2K+1164-delta,
2K=26+A+4R+U+2C+J,
L=M-R>=18+A,
K<=M+R+21.
```

This note includes the source-seed argument, all six resulting core-hole profiles, the exhaustive classification, and the terminal cards using their actual fresh constants. The shared positive-case prefix notation is in `fresh_positive_notes.md`.

15.1 Source localization and profile reduction

Originally z<=14,w<=4z. If the minimum side size is<=20 then I>=196,rho>=49 and the structural T_floor48 imply delta<=244-I-rho<0. At sizes21..23, rho>=58,I>=121 and the structural floor144-2z>=116 give delta<=118-I<0. Thus both sizes>=24.

The structural H bound is H<=90+4z-I-delta<=146<T_24. The fresh inner/outer layer rows give central-strip join margins>=154 and456. Therefore w<=z, both even. In particular the loose old source bound A<=144+w<=158 gives core mass<=1348 and core maxima<=201; these intentionally conservative bounds suffice for the seven-seed argument in the notes. That argument gives w<=z<=4 and the exact six profiles.

Here is that seed argument explicitly. The greedy first-gap rule leaves total m-1 below m, with all later entries>m and no entry equal to m. First gaps>=4 have minimal I+tau>=199 (m4),228 (m5), or tau>=248 (m>=6), contradicting A<=158. At m1, two values u,v from{2, ...,6} absent on both signs force tau>=214-2u-2v>=192, so at most one is absent. At m2, an absent q in{3,4,5} forces tau>=208-r-2q, where r owns the unique1; the quadratic I+tau has minimum174-2q>=164. At m3, an absent q in{4,5,6} forces tau>=206-2q>=194. Thus the following actual labelled seeds are available:

For the generic m>=6 bound, let k_v count the low occurrences on sign v. The core maximum<=201 gives m<=201 and T_(k_v)<=m-1, hence k_v<=19. At m6 the total low count is<=3, giving tau>=4(68-3)=260. At m>=7, k_v<=3 gives the side floor4(r_v-3). Otherwise m-k_v>=k_v+3; as r_v>=24 and k_v<=19, the difference (k_v+3)(r_v-k_v)-4(r_v-3) is at least k_v+7>0. Summing again gives tau>=248. The m4/m5 exact profile summations are displayed in the shared prefix lemmas.

condition	seed	seed mass	lower holes	minimum mass of any9 core occurrences
m1,2 absent	3,4,5,6	18	1,2	43
m1,3 absent	2,4,5,6	17	1,3	41
m1,4 absent	2,3,5,6	16	1,4	39
m1,2,3,4,5 present	2,3,4,5	14	1	34
m1,2,3,4,6 present	2,3,4,6	15	1	34
m2	1,3,4,5	13	2	37
m3	1,1,4,5,6	17	3	39

Direct bitset addition, or a short enumeration of each seed's sums, gives the full support except the stated lower holes and their reflections. The last column follows by taking at most two occurrences of each smallest permitted magnitude. Adjoin the remaining labelled core entries in increasing order. A failed central-interval join at next entry x means x>=T-2h, where h is the largest current lower seed-hole position. Any N remaining magnitudes at least x, multiplicity<=2, have total>=Nx+floor((N-1)^2/4). For n<=8 selected entries,N>=60; the seven resulting core-mass lower bounds are1728,1547,1366,1604,1665,1423,1547. For9<=n<=32,N>=36 and the last column applies; the bounds are1753,1607,1461,1492,1492,1531,1533. All exceed1348. For n>=33, T>=289 and x<=201<T-2h-1. Thus all joins succeed and the absolute core has at most four holes.

In the first three rows the profile baselines tau are136,134,132 respectively. For the missing4 row, missing2 on one sign would add r+1 above the baseline and force I+tau>=167>A<= 148. Both opposite2 labels are therefore present, filling4; only1 remains as a lower hole. In the other rows the sole displayed first gap persists. Every peeled magnitude exceeds4, so all lower core holes persist to the source. The exact (tau,z=w) pairs are therefore(68+sigma,2) for ordinarym1, (135-r+sigma,2) for m2, (132+sigma,2) for m3, (136+sigma,4) for missing1, 2, (134+sigma,4) for missing1,3, and (132+sigma,2) for missing1,4.

Every target-hole center is active: its hole offset is at most4 and subtracting any of the at least25 opposite core magnitudes produces distinct positive hull images well below the top two holes. Nonpositive centers in[1-M_v,0] use source M_v. Hence every integral center in[1-M_v,S_v] is active before any endpoint license.

Both leading resources are active by K-delta>=166-w+I+tau>0. If both O were active, their sharp caps give d_s+d_t<=2M+44+w. Set Y=2M-2K+44+w+delta. The leading and high-tail counts imply 2Y>488, while K-M>=R-132-w+A and R>=70,tau>=68,w<=4 give Y<=44, contradiction.

If one O_v were active, its sharp cap and leading row imply

```
2X_v+g_v<=154+3w/2-R-I-tau.
```

An active lower-coordinate witness makes that coordinate integral. The two full-core overlaps with the two leading resources (allowing an empty mate with zero capacity) then give q>=137-R-delta/2. Since q<=2X_v+g_v, this would force I+tau<=17+3w/2+delta/2<=63, using the conservative delta<=80-I. This contradicts tau>=68. Thus all four lower-coordinate witnesses would be empty, forcing both O active, already excluded. Consequently both O are empty.

Their two known holes leave only two others. A nonintegral lower coordinate forces two witness holes and additionally H_v or alpha_v nonintegral, impossible. After lower integrality, a nonintegral v2 forces H_v,alpha_v and also C5_o in A or beta_v in B missing. Thus all outer coordinates are integral. O_v>-M_v by positive leading activity, so each empty O is HIGH. Its four fixed labels C11,C12,C13,C14 are HIGH and outside every baseline group. These eight labels license credit8: delta>=8 and epsilon=delta-8.

The universal source cap A<=137+w now eliminates all extra high-mass profiles immediately. First-gap3 and missing1,4 have tau>=132,w2, hence delta<=7. Missing1,3 has tau>=134, w4, delta<=7. Missing1,2 has tau>=136,w4, delta<=5. Each contradicts credit8.

Only the ordinary first-gap1 profile and first-gap2 profile remain, with exact target support F_v=[1,S_v] minus{S_v-m}. At first gap1, tau=68+sigma and I+sigma+delta<=71, so both physical sizes lie in[26,42]. At first gap2, unique1 owner h satisfies h in[29,40] and true profile excess sigma obeys I+sigma+delta<=h+4. In both profiles delta<=71 and epsilon<=63.

15.2 Empty-center rule, endpoints and CORE8 classification

Source U obeys U<=350-2T, so U<=162. The first opposite peeled magnitude is<=R_o+ U<=304. On both profiles S_v>=469 and R<=142. The points M_v,0,-R_o give activity throughout[1-M_v,S_v+R_o], except possibly at S_v+R_o-m. There the first opposite peel produces a positive hit below S_v-m (at least163). Thus an empty integral center is LOW<= -M_v or HIGH>=S_v+R_o+1.

An empty own center cannot be LOW by leading activity. A retained singleton above any HIGH center loses at least73 source points: all positive points, zero, and all opposite core points. Such a loss exceeds epsilon<=63. An empty cross center cannot be HIGH, since it would put all77 points of lambda_v+D_v above the target. Cross activity is upward closed C4=>C5= >C6=>C7.

Before CORE8 retention, the target-independent separated-arm forests give

```
2X_v+X_o+g_v>=252-tau-delta,
2X_v+X_o+G>=305-tau-delta.
```

The arm choices involve at most one empty vertex at each point, and positive/negative restricted supports are disjoint because M>R. If beta_v,H_v are active, the sharp beta/H raw cap is d_v-M<=21+w/2<=22. Activity of C7_v or C6_v would instead give d_v-M>=253-tau-delta>=114 or306-tau-delta>=167. Thus neither triple beta_v,H_v,C7_v nor beta_v,H_v,C6_v

can be wholly active.

These rules reproduce the exhaustive whole-unit classification with the fresh native-loss threshold73:

- A1 retained beta: two contributing betas would require one further hole in each{H_v,C6_v}; then both C5 singletons are active and the four-center identity beta_s+beta_t+C5_s+C5_t=-2W contradicts their activity lower bounds. Thus beta is sole on s. Active H_s would force both C6_s,C5_s empty, leaving C7_s active and contradicting the forbidden triple. Hence the known holes are O_s,H_s,O_t, with one further Z. CORE8 is retained by the cross chain and the higher-retained73-loss rule. Z lies in{H_t,alpha_s,alpha_t,C5_s,C5_t,pair1}. Pair1 is sole if retained because its two centers sum<-3M. If it is empty, use T5 below. If retained with opposite owner, use the opposite-owner card. If same owner s, the triple C4_s,beta_s,C5_t is impossible, so Z=C5_t and alpha_s is retained; use same-owner card.
- B: U,V must be retained by the higher-retained rule, and the forbidden triples plus cross chain force C6,C7 retained. The two further holes are one from each{H_s,beta_s},{H_t,beta_t}. Both alphas and both pairs remain. Empty beta is LOW. They cannot both be empty because beta_s+beta_t>0. Pair1 and C5 pair have sole matching owners by the literal center identities. Any active beta on the opposite owner is excluded by the opposite-owner card, and the remaining active beta on the common owner with retained alpha is excluded by the same-owner card.
- A0 beta-pair empty: CORE8 is retained. Its one further hole Z belongs to{H_s,H_t,alpha_s,alpha_t,C5_s,C5_t,pair1}. If pair1 is retained it is sole; normalize owner s. Then C5_s is retained and Z lies in{H_s,H_t,alpha_s,alpha_t,C5_t}. At least one of H_v,alpha_v is retained above each empty beta, so both beta components are LOW. V6_s,V6_t are LOW and meet one baseline group. With the eight HIGH O labels the fixed credit is9.

First-gap1 charged endpoints have uniform allowance1: I+sigma+epsilon<=63<73 makes the next core rank exact. First-gap2 has delta<=38 and epsilon<=30, while an outside-hull center clips at least33 source points, so charged endpoints are normal. With a=1 for m1,a=0 for m2, both are

```
Z>=73-r_v-l_Z-a,
Z<=S_v+r_o-72+h_Z+a.
```

15.3 Common source/rank bounds for the terminal cards

Here sigma=tau-68 is the generic excess; at m1 each sigma_v>=0, at m2 sigma_v>=-1 and sigma>=27. Put eta=0 at m1 and eta=1 at m2. The penultimate core estimate gives

```
K<=C+312+sigma+eta,
2R+I+delta<=224+eta<=225.
```

Q61 overlap floor is f_eta=(166-eta-sigma+R)/4. Full-core69 overlap floor is137-R. Ordinary gaps and stars can use any actual common trimmed core, with no change to capacities. At m1, writing xi=R-70, three available overlap floors are

```
F0=67-xi, F1=65-sigma+xi, F2=63-sigma/2+xi/2.
```

At m2 using the true profile excess sigma' and xi'=R-71 they are

```
F0=66-xi', F1=64-sigma'+xi', F2=62-sigma'/2+xi'/2.
```

The same full resources certify all these floors; each scalar inequality may use the maximum F.

15.4 Opposite-owner card

Suppose beta_s and P_t=C4_t are licensed capacity77 native resources and CORE8 is retained. The identity V_t=2d_s+beta_s+P_t+2s4+t4+g_s and three endpoint rows followed by the leading-s inequality twice give

```
S_s+3g_s+4X_s+2s4+t4<=1014+3a+r_s+delta-2e.
```

Two ordinary gap-g_s pairs give3g_s>=3f_eta-3epsilon/2. The stars give6X_s+X_t>=959-7R-3epsilon. Combining these with the source mass, L>=86+I+sigma+delta and the radial bound yields

```
T_r+5r+29I/8+sigma/4+sigma_s+13e/2
  <=827/2+27eta/8+15delta/8+3a.
```

Use delta<=71-I, eta<=1,a<=1 and sigma_s>=-1. Then

```
T_r+5r+11I/2+sigma/4+sigma_s+13e/2<=553.
```

The global integral quadratic minimum T_r+5r+11(r-34)^2/2 is1401/2. Credit e>=8 and sigma_s>=-1 give left>=1503/2>553. Thus opposite owners are impossible.

15.5 Same-owner card with active alpha

Suppose alpha_s,P_s,beta_s are licensed distinct native resources. The literal identity and outer order give

```
alpha_s-P_s-beta_s> d_t+2s4+t4+G.
```

Three endpoints and the distinct leading-t loss give

```
S_t+g_t+2X_t+2s4+t4+G<=K+1082+3a-r_t-e.
```

Three disjoint ordinary pairs give g_s+2g_t>=3f_eta-epsilon, while stars give2X_s+3X_t>=685-5R-3epsilon/2. Source and radial substitution gives

```
T_r+5r+25I/8+sigma/4+sigma_t+5delta/8+7e/2
  <=1349/2+23eta/8+3a.
```

At m1 the right is677.5, sigma_t>=0; at m2 it is677.375, sigma_t>=-1. The quadratic part has exact integer minimum5265/8=658.125. Credit8 makes the respective left floors5489/8=686.125 and5481/8=685.125. Both exceed the right side.

15.6 T5: sole retained beta, empty pair1

The holes are O_s,H_s,O_t,pair1, beta sole owner s. Credit13 follows by an exact label ledger. Leading K-delta>=164+I+tau>R gives d_v>g_v, so F3,F4,F5 are LOW. Empty beta_t is LOW by retained alpha_t. Both empty C4 are LOW, giving C55,C56,C57,C58 LOW on each sign. Genuine beta_s,C5_t imply d_t>d_s; C5_s gives t2<d_s, so V3_t,V4_t LOW. There are20 LOW labels meeting18 baseline groups, and11 HIGH labels from O_s,H_s,O_t outside those groups. Thus e=11+20-18=13, with no extra hole allowance.

Upper endpoints alpha_s,H_t, lower endpoints alpha_t and twice beta_s, plus C5_t, together with the leading-s inequality twice give

```
2M+6X_s+2X_t+3g_s+g_t<=932+6a+delta-2e.
```

The weighted native losses are<=2epsilon and the six endpoint allowances are exactly6a. Stars and pairs using overlap F give6X_s+2X_t>=8F-3epsilon and3g_s+g_t>=4F-3epsilon/2. Since M>=R+86+I+sigma+delta and I+sigma+delta<=71,

```
2R+11I/2+11sigma/2+12F<=924+6a.
```

At m1, R=70+xi and13F0/2+11F1/2 implies12F+2xi+11sigma/2>=793+xi, so the left>=933>930. At m2, use generic sigma,xi=R-70, F0=67-xi,F1=64-sigma+xi; the same combination gives a left floor927.5+xi>=928.5>924. This excludes T5.

15.7 A0 case1: pair1 empty

All H,alpha,C5 and CORE8 are retained, with credit9. Pair-sum joint-loss counting and the two H identities, using uniformly weakened endpoints72-r_v and S_v+r_o-71, give

```
S>=4M+4X+2G+302+2F-3epsilon.
```

The source upper bound S<=2M+2R+1206-delta and the stars/pairs with overlap F then give

```
14F+2I+2tau<=814+3delta.
```

At m1, mass I+sigma+delta<=71 forces14F+5I+5sigma<=891. But5F0+F1+8F2 gives14F+5sigma>=904, contradiction.

At m2 with true excess sigma' and unique1 owner h, tau=135-h+sigma' and I+sigma'+delta<=h+4. Thus14F+5sigma'+5I<=556+5h. Yet5F0+F1+8F2 gives14F+5sigma'>=890, forcing I-h<=-334/5, while min[(h-34)^2-h]=-34. Contradiction.

15.8 A0 case2: retained pair1, opposite alpha retained

The conservative transport being used is

```
T_k+4k+128-tau_s<=7R+3delta,
```

where k=r_t is the side opposite the pair1 owner. Its ordinary first-gap1 proof uses the sharper actual fringe bound B_s+B_t<=min(20,183-T); the root's common-source certificate gives positive margins3,2,1 for T<=163,T164,T165. This is a separate licensed source lemma, not an assumed129 transport.

At m2, use R<=71+sigma' and the unique1-owner mass cap. If h=k, tau_s=136-2k+sigma_s and I+sigma'+delta<=k+4. The conservative row forces

```
T_k-2k+8I+5delta<=537.
```

Its quadratic part is at least496, and delta>=9 makes the left>=541. If h=68-k, tau_s=67-k+sigma_s and the cap is I+sigma'+delta<=72-k. The resulting row is

```
T_k+13k+8I+5delta<=1012.
```

Its quadratic minimum is971, and delta>=9 gives left>=1016. Thus both physical unique1 owners are excluded with margin4.

15.9 A0 case3: retained pair1, opposite alpha empty

Every other H,alpha_s,C5 and CORE8 resource is retained. The same seven weighted endpoint uses as the old identity, now with the fresh weakened constants, give

```
S>=5s4+4t4+2g_s+3g_t+2h_t+298-r_s-E,
E<=2epsilon.
```

Hence

```
2M+M_s+5X_s+4X_t+2g_s+3g_t+2h_t
  <=2R+908+r_s-delta+2epsilon.
```

Use f=(233+R-tau)/4, which is valid for both profiles. The disjoint ordinary groups give2g_s+3g_t+2h_t>=5f+122-2epsilon. Stars give5X_s+4X_t>=1233-9R-5epsilon/2. Also2M+M_s>=2R+41+r_s+2I+2tau+2delta. Thus

```
488+2I+2tau+5f+3delta<=9R+13epsilon/2,
3351/4+2I+3tau/4<=31R/4+7delta/2.
```

Use2R+I+delta<=225. This forces

```
-273+47I+6tau+3delta<=0.
```

Since tau>=68, the left is at least135. Contradiction.

All original P4 configurations are exhausted by these owner/empty-pair cases and the A0 three-case split.

15.10 An analytic exclusion of the first-gap-one A0 cross case

Consider the normalized cardinality-85 source with $d = 77$ and $E = P_s \cup \{0\} \cup (-P_t)$, where $r = |P_s|$, $k = |P_t|$ and $r + k = 68$. Assume that the first missing positive subset sum of the absolute core is 1, that $z = w = 2$, and that the A0 classification gives the retained centers $P = C_{4s}$, α_t, C_{6t} and U_s and all eight CORE8 resources. The selected P is an actual contributor of its sole-owned paired unit. The fixed-label credit is 9; write $\varepsilon = \delta - 9 \geq 0$.

The inputs used here are the proved source bound

$$T = 26 + I + \tau + \delta \leq 165, \qquad I = (k - 34)^2,$$

the nominal first-gap-one endpoint bounds

$$Z \geq r_o + 4 - \ell_Z, \qquad Z \leq S_v - r_v - 3 + h_Z,$$

and the actual native/grouped loss budgets. All subset copies used below belong to the specified retained resources; no capacity is assigned to an empty center.

Write the four increasing source-tail offsets as $a_v < b_v < c_v < e_v$. Let $U = a_s + b_s + a_t + b_t$, $J = e_s - c_s + e_t - c_t$, and let B_v count the distinct sums of two offsets on sign v that exceed e_v. Set $B = B_s + B_t$ and let p be the number of odd integers $a_v + b_v$. The two source envelopes and the quartet fringe bound give

$$J + U \leq 332 - T, \quad J \geq T - 18 + p, \quad 2B \leq U + 16 - p.$$

Thus $2T + U + p \leq 350$ and

$$B \leq \min(20, 183 - T). \tag{5}$$

The full-source central interval gives $K \leq M + R + 1 + B$, where $M = M_s + M_t$ and $R = R_s + R_t$. Consequently the exact original mass identity yields

$$S_s + S_t = 2K + 1164 - \delta \leq 2M + 2R + 1166 + 2B - \delta. \tag{6}$$

The endpoint transport, including every constant. Put $a = s_4 = M_s + X_s$, $b = t_4 = M_t + X_t$ and $G = g_s + g_t$. The literal center identity is

$$U_s = P + \alpha_t + C_{6t} + 2a + 4b + 2G.$$

Three lower endpoints and the upper endpoint of U_s imply

$$S_s \geq 219 - 2k + 2a + 4b + 2G - E_{\text{end}}, \qquad E_{\text{end}} = \ell_P + \ell_{\alpha_t} + \ell_{C_{6t}} + h_{U_s} \leq \varepsilon.$$

Indeed their constant is $(k + 4) + 2(r + 4) + (r + 3) = 219 - 2k$. The opposite source satisfies $S_t \geq T_k + \tau_t + 4R_t + 10$, while $R_t \geq k + 1$ and $M_t \geq k + 5$. Combining with (6) therefore gives

$$T_k + 4k + \tau_t + 243 + 2X_s + 4X_t + 2G \leq 2R + 1166 + 2B - \delta + E_{\text{end}}. \tag{7}$$

Here $243 = 219 + 10 + 14$, and 14 comes from $2M_t + 4R_t - 2k \geq 4k + 14$.

Admissible asymmetric core copies. Put

$$E_v = \tau_v - R_v + 1 \geq 0, \quad \zeta = \tau_t - k \geq 0, \quad D = E_s + E_t = \tau - R + 2.$$

Every core magnitude is at least its rank plus 1. After deleting the maximum, its remaining rank deviations sum to E_v. Deleting the j_v largest core entries therefore leaves maximum at most $r_v - j_v + 1 + \lfloor E_v / j_v \rfloor$. For the signed trimmed core Q, this proves

$$|Q \cap (Q + q)| \geq 67 - j_s - j_t - \lfloor E_s / j_s \rfloor - \lfloor E_t / j_t \rfloor - q.$$

Choose $j_s = 3$ and choose $j_t \in \{1, 2, 3\}$ minimizing $j_t + \lfloor E_t / j_t \rfloor$. Define

$$\phi_s = 3 + \lfloor E_s / 3 \rfloor, \quad \phi_t = \min_{j=1,2,3} \big(j + \lfloor E_t / j \rfloor \big), \quad f = 67 - \phi_s - \phi_t, \quad F = \max(137 - R, f).$$

The full core gives the bound $137 - R - q$; the trimmed core gives $f - q$.

For a leading-to-own3 edge, the leading resource contains λ_v+M_v+Q and the ordinary resource contains λ_v+v_4+Q, with gap $X_v=v_4-M_v$. The reflected leading-to-C_7 edge uses the reflected copies at the same gap. Thus these are actual subsets of the retained resources. The rooted forest count, separately for an individual two-edge star and for the four-edge star, gives

$$2X_t \ge 2F-\varepsilon, \qquad 2X_s+2X_t \ge 4F-\varepsilon,$$

and hence $2X_s+4X_t \ge 6F-2\varepsilon$. The four disjoint CORE8 ordinary pairs similarly give $2G \ge 4f-\varepsilon$. Substituting into (7) and using $E_{\text{end}} \le \varepsilon$ and $\varepsilon=\delta-9$ yields

$$T_k+4k+\tau_t+6F+4f \le 2R+887+2B+3\delta. \tag{8}$$

In particular the conservative old-$B=20$ residual is

$$L=T_k+4k+\tau_t-927-2R-3\delta+6F+4f.$$

The actual residual in (8) is $L+2(20-B)$; this is exactly the fringe correction, not an additional credit against δ.

A short integer certificate. Put $q=T_k+5k$. Since $T=26+I+\tau+\delta$,

$$L+3T=q+3I-855+\zeta+3(E_s+E_t)+R+6F+4f.$$

The two inequalities $F \ge 137-R$ and $F \ge f$ give $6F \ge 137-R+5f$, so

$$L+3T \ge q+3I-115+(3E_s-9\phi_s)+(\zeta+3E_t-9\phi_t).$$

First, $3E_s-9\phi_s \ge -27$. If $E_t=0$, the second parenthesis is at least -9. If $E_t>0$, then $R_t-k-1 \ge 1$: equality would force the entire k-element core to be $\{2,\dots,k+1\}$, for which $E_t=0$. Hence $\zeta=E_t+(R_t-k-1) \ge E_t+1$. For $E_t \ge 6$, the choice $j_t=3$ gives

$$\zeta+3E_t-9\phi_t \ge E_t-26 \ge -20.$$

For $E_t=1,2,3,4,5$, the respective bounds are $-13,-18,-14,-19,-15$, by using $j_t=1,2,2,2,3$. Thus the second parenthesis is always at least -20. Finally, for every integer k, with $x=k-28$,

$$q+3I=T_k+5k+3(k-34)^2=654+\frac{7x^2-5x}{2} \ge 654.$$

We have proved $L \ge 492-3T$. By (5),

$$L+2(20-B) \ge 492-3T+2\max(0,T-163)>0 \qquad (T \le 165).$$

For $T \le 163$ the margin is at least 3; at $T=164,165$ it is at least $2,1$, respectively. This contradicts (8) and excludes the conditional A0 cross case with first gap 1.

16 Complete Z5 A0 alpha-hole finish at size 85

This is the A0 family with both paired units empty and exact whole-unit holes {pair1,pair2,O_s,O_t,alpha_s}. There is no assigned paired owner. All thirteen remaining ordinary singleton units are retained. The fresh structural and retention arguments have already given integral outer coordinates, leading activity, w=0, and the source notation

$$r_s + r_t = 68, \quad R_v = r_v + \xi_v, \quad 0 \le \xi_v \le \tau_v, \quad \tau_s + \tau_t = \tau, \quad A = I + \tau + \delta \le 60.$$

Write a=s_4=M_s+X_s, b=t_4=M_t+X_t, G=g_s+g_t, B=alpha_t, C=C5_s, D=C5_t. The alpha hole is LOW, so h_s<=X_s. The generic retained-core-hit argument and complete-target rank bounds apply before normality; they do not import the old Z5 normality bound. The refined source lemma gives e_v>=A+16 on both signs.

16.1 Fixed credit and the order in which normality is established

The following stages are the fresh tail/order deductions proved in the zero-case companion. They are listed here with their actual labels and incidence so the final loss budget can be checked.

Initially all retained singleton deficiencies are at most60, giving at least17 target hits, more than the eight peeled source entries; each has a core hit. All selected native clips are below73, so the generic rank errors are bounded by the appropriate xi_v. Both paired components and the alpha hole are LOW.

The initial ledger gives e=14. Its source support obtained by trimming four core ranks on both signs has q=61 and overlap at least f=61-(tau-xi_s-xi_t)/4. The stars, g pairs, leading, and the retained {H_s,U_s} pair then give

$$X_v, g_v \ge 38, \quad h_s \ge 15, \quad d_v \ge 107 + g_v + 2X_v.$$

These inequalities and the exact32-type upper-tail inventory give S_s<=H_s+M_s+2 and S_t<=H_t+M_t+16, yielding the additional13 HIGH labels below. The two places requiring cross-centre lower endpoints use the proved core-hit lower bound C5_v>=1-R_v; they have strict margins larger than26 and16. The consecutive-tail argument needs h_s>2, supplied by15. Thus this step gives e=27 without assuming normality.

Now a BAD ordinary centre has deficiency at least 73-max(r_s,r_t). Put k=|r_s-34|, so I=k2. Its excess over the available budget epsilon=delta-27 is at least

$$73 - (34 + k) - (60 - k^2 - \tau - 27) = k^2 - k + 6 + \tau > 0.$$

Hence every retained centre is normal. The native upper endpoints and unused-loss rebates prove B,D<a; the fixed-only prefix count proves C<b. All three retained centres B,C,D are now positive.

The complete HIGH set is

$$\mathcal{H} = \{C_{11}, C_{12}, C_{13}, C_{14}\}^{s,t} \cup \{C_{22}, C_{23}, C_{24}, C_{16}, C_{17}, C_{18}\}^{s,t} \cup \{C^s_{33}\}, \qquad |\mathcal{H}| = 21.$$

It meets no baseline group. The LOW stages and exact incidence are:

Stage	Added LOW labels	H	L	Groups touched	Credit
Initial	C55,C56,C57, C58,V5,V6, V7,V8 on both signs; U5,U6, U7,U8,V2,V3, V4 on s; F3, F4,F5 on both signs	8	29	23	14

Stage	Added LOW labels	H	L	Groups touched	Credit
Tail	No new LOW labels;13 HIGH additions	21	29	23	27
B,D<a	U5,U6,U7,U8, V3,V4,C66, C67,C68 on t	21	38	25	34
C<b	C66,C67,C68, C35 on s	21	42	25	38
Two cross labels	C25 on both signs	21	44	26	39
Remaining three	C35_t,C45_s, C45_t	21	47	26	42

For the last two stages the literal identities are

$$C25_s = C - D - 2b - g_t, \quad C25_t = D - C - 2a - g_s,$$
$$C35_t = C25_t - h_t, \quad C45_s = C35_s - g_s, \quad C45_t = C35_t - g_t.$$

All are negative because 1<=C<b, 1<=D<a, and the outer gaps are positive. The C25_s,C25_t group is the sole previously untouched unstarred baseline row1, so two new labels cost one new touched group. C35_t is in the reflected row9 already touched by V3_t; the two C45 labels are in the two row12 groups already touched by V8_s,V8_t. Thus the final compatible fixed credit is **42**, and all native losses below share epsilon=delta-42>=0.

16.2 One actual adaptive core support

The final credit and A<=60 imply I+tau<=18. Let E_v=tau_v-xi_v. For j_v in 0,1,2,3,4, remove that many largest positive core coefficients on sign v. For zero removals use the excess xi_v. For positive removals, monotonicity of the core deviations gives the retained maximum at most

$$r_v - j_v + \lfloor E_v/j_v \rfloor.$$

Choose on each sign a minimizing removal count for

$$\phi_v = \min\left(\xi_v,\ \min_{1\le j\le 4}\{j + \lfloor E_v/j \rfloor\}\right).$$

This determines one **actual** subset Q of the core, containing zero. It has q=69-j_s-j_t>=61. Its integer displacement overlap is at least F-u, where

$$F = 69 - \phi_s - \phi_t.$$

Both physical targets use this same subset and its reflection. All group inequalities below use this chosen support; no overlaps of different subsets are added or charged to the same budget. The triple bound keeps q>=61 explicitly, so replacing the old overlap by F does not discard its source cardinality condition.

16.3 Disjoint groups and exact transport

Use the following original retained singleton groups, and no paired capacity:

Group	Resources	Deficiency
Pair	B,D	Gamma
Triple	H_s,U_s,V_s	Psi
Two ordinary pairs	U_t,V_t and C6_s,C7_s	Summed Omega
Singleton	C=C5_s	Gamma_C

Group	Resources	Deficiency
Singleton	H_t	Gamma_Ht

They contain eleven distinct retained singleton units. Their actual union deficiencies sum to at most epsilon. The joint counting lemma gives

$$2B + D \geq 3(73 - r_t) + F - 3\Gamma, \quad g_s + h_s \geq q + F - \Psi, \quad 2g_t \geq 2F - \Omega.$$

The first row combines B+D>=2(73-r_t)+F-2Gamma with the normal native lower bound B>=73-r_t-Gamma; it uses the same Gamma throughout. Also l_C<=Gamma_C, u_Ht<=Gamma_Ht, and u_Hs<=Psi. The literal centre identities and normal upper endpoints are

$$H_s = a + 2b + G + h_s + B + C, \quad H_t = a + 2b + G + h_s + B + D, \quad H_v \leq S_v + r_o - 72 + u_{H_v}.$$

Use the exact source mass S=2K+1085-delta and K<=M+R+21. Adding the upper endpoints and subtracting the packed lower endpoints gives precisely

$$2M_t + 2X_s + 4X_t + 2(G + h_s) + F \leq 2R + 827 + 2r_t - \delta + 3\Gamma + l_C + u_{H_s} + u_{H_t}.$$

Insert the two gap inequalities. The loss terms are bounded by 3Gamma+3Psi+Omega+Gamma_C+Gamma_Ht<=3epsilon, so

$$2M_t + 2X_s + 4X_t + 2q + 5F \leq 2R + 827 + 2r_t - \delta + 3\epsilon.$$

Separately valid weighted stars on the same support give 2X_s+4X_t>=6F-2epsilon. Combining these scalar inequalities does not assert that the star grouping is disjoint from the displayed eleven-unit grouping. Finally use q>=61 to obtain

$$2M_t + 122 + 11F \leq 2R + 827 + 2r_t - \delta + 5\epsilon. \tag{*}$$

16.4 Exact finite scalar contradiction

Since M_t=r_t+xi_t+e_t and e_t>=A+16, (*) would require

$$2I + 2\tau - 2\delta - 2\xi_s + 11F + 5 \cdot 42 - 809 \leq 0.$$

Using delta<=60-I-tau, its left side is at least

$$4I + 4\tau - 2\xi_s + 11F - 719.$$

The nonnegative 4I may be omitted. The complete remaining finite domain is 0<=tau<=18, tau_s+tau_t=tau, 0<=xi_s<=tau_s, 0<=xi_t<=tau_t, all integers. It has7315 tuples. The exact checker finds

$$\min(4\tau - 2\xi_s + 11F - 719) = 6,$$

attained in the relaxed domain at (tau,tau_s,tau_t,xi_s,xi_t,F) equal to (16,11,5,5,3,61). Relaxing unrealizable core parameters only enlarges the domain, so a positive minimum on this domain suffices. This contradicts (*), and excludes the final A0 alpha-hole family.

The companion z5_alpha_final85.py reconstructs every baseline group, checks all staged incidence counts and literal new-label identities, selects one actual trimming prescription per tuple, and checks this exact finite minimum. It does not substitute arithmetic for the preceding actual-cover, normality, and joint-deficiency arguments.

17 Fresh size-85 Z5: all A1 and B alternatives

This proof uses the fresh structure and the actual-retention classification in `zero85_z4_z5_cards.md`. Its source has 77 points, core 69 points, physical positive side sizes `r_s+r_t=68`, and no target holes. It does not transplant the old Z5 normal-endpoint assumption.

Write

```
I=(r_s-34)^2, A=I+tau+delta<=60,
R_v=r_v+xi_v, xi=xi_s+xi_t<=tau, R=68+xi,
M_v=R_v+e_v, L=e_s+e_t, M=R+L,
a=s4=M_s+X_s, b=t4=M_t+X_t, G=g_s+g_t.
```

The source rows used below are

```
S_s+S_t=2K+1085-delta, K<=M+R+21,
L>=99+A, e_v>=A+16.
```

Several deliberately weaker computations use only `e_v>=A+8`. On each physical side,

```
S_v>=T_(r_v)+tau_v+4R_v+6+e_v.
```

All retained ordinary resources and licensed sole pair components have a core hit, since their deficiencies are at most 60, less than 69. With lower and upper clipping counts l_Z,u_Z and native deficiency Gamma_Z,

```
Z>=73-r_v-l_Z-xi_v,
Z<=S_v+r_o-72+u_Z+xi_o,
Gamma_Z>=l_Z+u_Z.
```

These rows include centers outside the target hull. Monotonicity of `x_i-i` bounds every core rank displacement by xi_v. No assertion that the centers themselves are inside the target is made.

The eight O HIGH labels give fixed credit e>=8. Put epsilon=delta-e. All compatible group deficiencies and unused native losses share this one budget. The four extreme leaves are retained. Each of the ordinary gap pairs contains at most one empty translate; their intersections with an empty translate are off target and are charged only to its retained partner's actual loss.

Delete the four largest core magnitudes on each side. The Q61 support gives

```
f=61-(tau-xi)/4,
c_s X_s+c_t X_t >= (c_s+c_t)f-max(c_s,c_t)*epsilon/2,
c_s g_s+c_t g_t >= (c_s+c_t)f-max(c_s,c_t)*epsilon/2.
```

The second row uses the two disjoint ordinary pairs for each physical gap. These star and ordinary-group inequalities are separately valid scalar budget consequences. When endpoint resources are omitted from a grouping, their native deficiencies can be rebated from epsilon. The full-core star also gives `2X_v>=274-2R-epsilon`.

17.1 Four reusable transport exclusions

17.1.1 HIGH alpha_s

Suppose alpha_s is HIGH, while H_t,alpha_t,C5_t are retained. Since alpha_s>=S_s+R_t+1, the identity

```
alpha_s+C5_t=d_t-2(a+b)-G
```

and the endpoints of C5_t and H_t-alpha_t=d_t give

```
2S_s+R_t+2X+G<=2R+976+r_t+xi+xi_t.
```

The three endpoint losses share epsilon. Using `S_s>=T_r+tau_s+4R_s+10`, `2X>=4f-epsilon`, and discarding G>=0 gives

```
2T_r+8r+I+2tau_s+6xi_s-2xi_t+e<=928.
```

Because xi_t<=tau_t and tau+delta<=60-I,

```
2T_r+8r+3I+2delta+e<=1048.
```

For every integer r, `2T_r+8r+3(r-34)2^=1092` (minimum at r=24). Thus this HIGH-alpha alternative is impossible.

17.1.2 Both pairs retained, opposite owners

Let beta_s and P_t be the actual sole owners. Use their lower endpoints, the upper endpoint of V_t, and the leading-s inequality twice. The literal identity is

```
V_t=2d_s+beta_s+P_t+2a+b+g_s.
```

It gives

```
S_s+3g_s+4X_s+2a+b
   <=935+r_s+delta-2e+xi+xi_s.
```

The source lower row and the separately valid bounds `6X_s+X_t>=7f-3epsilon` and `3g_s>=3f-3epsilon/2` yield, using the weaker e_v>=A+8,

```
T_r+4r+3I+tau_s+tau/2+4xi_s+5xi/2
   <=136+5delta/2-13e/2.
```

The source mass cap therefore implies

```
T_r+4r+11I/2+tau_s+3tau+4xi_s+5xi/2
   <=286-13e/2.
```

Its left side is at least `T_r+4r+3I>=626`, whereas its right side is at most 234. Opposite owners are impossible.

17.1.3 Both pairs retained, same owner, alpha_s retained

Let both sole owners be s and put r=r_t. Since alpha_t=h_t-b>-b, the identity

```
alpha_s-P_s=beta_s+beta_t+2d_t+2(a+b)+G
```

gives the strict lower relation `alpha_s-P_s>beta_s+d_t+2a+b+G`. Use the endpoints of the three retained s-resources and one leading-t inequality. The transport is

```
S_t+M_s+2X_s+3X_t+g_s+2g_t
   <=R+1023-r-e+2xi_s+xi_t.
```

Here the source row is `S_t+M_s>=T_r+3r+tau_t+4xi_t+xi_s+173+A`. Together with `2X_s+3X_t>=5f-3epsilon/2`, `g_s+2g_t>=3f-epsilon`, and e_v>=A+8 imply

```
T_r+4r+5I/2+tau_t+tau/2+4xi_t<=520-7e/2.
```

The left side is at least `T_r+4r+5(r-34)2/2>=608`, while the right side is at most 492. This alternative is impossible.

17.1.4 The branch-independent four-endpoint exclusion

Suppose pair1 has sole owner s, alpha_t is retained, and all eight lower own/cross singletons are retained. The literal identity

```
U_s=P_s+alpha_t+C6_t+2a+4b+2G
```

uses four distinct endpoints. Their losses share epsilon, giving

```
S_t+2M_t+2X_s+4X_t+2G
   <=904+2r_t+2R-e+xi_s+3xi_t.
```

Use `S_t+2M_t>=T_r+tau_t+6R_t+18`, `2X_s+4X_t>=6f-2epsilon`, and `2G>=4f-epsilon`. Then

```
T_r+4r+3I+tau_t+tau/2-xi_s/2+7xi_t/2<=592-4e.
```

Since xi_s<=tau_s, its remaining excess terms are nonnegative. Again the left side is at least 626 and the right side at most 560. This exclusion uses no beta capacity, endpoint, or ownership.

17.2 A1 with pair1 empty

The sole beta owner is s. The base holes are O_s,H_s,O_t,pair1 and the remaining hole eta is exactly one of

```
alpha_s, H_t, alpha_t, C5_s, C5_t, U_s.
```

For eta=alpha_s, emptiness above an active beta makes alpha_s HIGH; the HIGH-alpha exclusion applies. Explicitly beta_s has a core hit, so beta_s>=1-R_s. The leading inequality and A<=60 imply `d_s-R_s>=93-r_s+I>0`, hence alpha_s=d_s+beta_s>0. Since `S_s>R_t`, an empty alpha_s cannot have `S_s<alpha_s<=S_s+R_t`: its image at source point -R_t would be a target hit. Thus alpha_s>=S_s+R_t+1, as the exclusion requires.

For eta=C5_s or U_s, the retained endpoints are alpha_s,beta_s,H_t,alpha_t,C5_t. The identities alpha_s-beta_s=d_s, H_t-alpha_t=d_t, and beta_s+C5_t=d_t-d_s-2(a+b)-G, followed by two leading-s bounds, give

```
2M+6X_s+2X_t+3g_s+g_t
   <=853+delta-2e+3xi.
```

All endpoint and leading loss coefficients are at most two on distinct resources. The weighted star and gap bounds give `12f-9epsilon/2` for the gap part, so the source cap implies

```
3+11I/2+5tau/2+13e/2<=0.
```

This is impossible. The ordinary gap pairs may contain the empty U_s; its intersection is paid solely by the retained V_s loss.

For eta=H_t, use lower alpha_t,C5_s,beta_s,C5_t and upper alpha_s,V_t. Their identities give

```
4M_s+3M_t+6X_s+5X_t+3g_s+2g_t
   <=2R+744+r_s+delta-2e+4xi+xi_s.
```

The source lower bound is `4M_s+3M_t>=3R+305+r_s+xi_s+4A`, using e_v>=A+8. The gap part is at least `16f-9epsilon/2`. Consequently

```
537+R+4I-3delta/2+13e/2<=0,
```

contradicting R>=68 and delta<=60.

For eta=alpha_t, use upper alpha_s,H_t and lower beta_s,C5_t,C5_s. The latter beta and C5_s clipping coefficients are two. The exact transport is

```
4M_s+M_t+6X_s+3X_t+3G+3h_t
   <=2R+754+3r_s+delta-2e+5xi_s+2xi_t.
```

Group the two g_s pairs with total deficiency Omega and the triple {H_t,U_t,V_t} with deficiency Psi. The triple has `g_t+h_t>=61+f-Psi`; hence `3G+3h_t>=6f+183-3epsilon`. Separately the stars give `6X_s+3X_t>=9f-3epsilon`. The physical source lower row `4M_s+M_t>=R+123+3r_s+3xi_s+4A` then implies

```
219+7I+13tau/4+3xi/4+8e<=0,
```

again impossible. No LOW-alpha endpoint assumption is needed.

17.3 The final A1 hole eta=C5_t

The exact holes are O_s,H_s,O_t,pair1,C5_t. All eight lower own/cross singletons, alpha_s, alpha_t,H_t,C5_s and sole beta_s are retained.

Initially the eight O labels and C22_s,C23_s,C24_s are HIGH. The LOW labels are C55,C56, C57,C58 on both signs and C66_t,C67_t,C68_t. There are 11 HIGH, 11 LOW, and 9 baseline groups touched by LOW, giving compatible fixed credit e=13.

17.3.1 The beta cutoff, with actual physical-target capacity

We first prove `beta_s+s3<d_s-X_s`. Suppose the reverse. Since `d_s=2(a+b)+G+C5_s+alpha_t`, it forces

```
alpha_s>=3M_s+4M_t+2X_s+4X_t+g_s+2g_t
             +2C5_s+2alpha_t.
```

Target t has its leading and exactly six retained ordinary resources H_t,alpha_t,U_t,V_t,C6_t, C7_t. Their total capacity is K+462. Their grouped deficiency is at most epsilon, so the exact two-target mass gives `S_s<=K+623-e`. The endpoint inequality above therefore gives

```
2M_s+3M_t+2X_s+4X_t+g_s+2g_t
   <=R+416+r_t-e+E+2xi_s+3xi_t,
```

where `E=2l_(C5_s)+2l_(alpha_t)+u_(alpha_s)`. The source gives `2M_s+3M_t>=2R+R_t+214+3A`. The weighted stars give `2X_s+4X_t>=6f-2epsilon`. If Omega is the weighted ordinary-gap deficiency, then `g_s+2g_t>=3f-Omega` and `E+Omega<=2epsilon`, because those ordinary groups avoid the three endpoints. Thus

```
3A+9f<=134-e+4epsilon+xi.
```

The source cap turns this into

```
355+4I+7tau/4+5xi/4+5e<=0,
```

an immediate contradiction. This proves the beta cutoff.

17.3.2 The complete actual upper-tail inventory

Put t0=d_s-X_s>0 and add the four s-outer magnitudes to a target value u>t0. A four-term representation with at most one top-two occurrence is at most s1+3s3. With exactly two and not both s1, it is at most s1+s2+2s3. Both lie below the required total, since `d_s-g_s-X_s>=K-native_leading_loss+X_s>0`. If both are s1, both remaining entries must be s3 or s4; otherwise the total is at most the threshold. This gives three types C33,C34,C44. Exactly three top-two occurrences give the four source families O_s+D_s,H_s+D_s,alpha_s+D_s,beta_s+D_s, plus 24 free-outer types. Exactly four give five further fixed types.

The complete 32 non-source types, in the original fixed inventory, are

```
C13,C14,C15,C16,C17,C18,
C23,C24,C25,C26,C27,C28,
U3,U4,U5,U6,U7,U8,
V3,V4,V5,V6,V7,V8,
V2,U2,C22,C12,C11,C44,C34,C33,
```

all on target s. O_s and H_s source families are empty. The cutoff removes beta_s+D_s and the six V3..V8 fixed types. Thus only the retained alpha_s source family and at most 26 fixed types cover the tail. Writing Gamma_alpha for its native deficiency,

```
S_s<=d_s-X_s+103-Gamma_alpha.
```

The inequality is automatic when S_s<=t0.

17.3.3 Four extra HIGH types and the sharper tail

The core hit of beta_s gives beta_s>=1-R_s. Since `h_s=d_s+a+beta_s`, the preceding tail bound and the full-core star give

```
h_s-S_s>=e_s+1+2X_s-103+Gamma_alpha
        >=52+I+tau-2xi+e
        >=-8+2I+delta+e>0.
```

Thus U4_s=h_s and U3_s=h_s+g_s are HIGH. Also `h_s-d_s=a+beta_s>=e_s+X_s+1>0`, so V2_s=2h_s+g_s-d_s and U2_s=2h_s+g_s are HIGH. These four distinct labels are disjoint from the previous HIGH ledger, which now has 15 HIGH, 11 LOW, touched 9: the credit is e=17.

All four labels belong to the 26 remaining types in the displayed tail inventory. Removing them leaves at most 22 fixed types, giving

```
S_s<=d_s-X_s+99-Gamma_alpha.
```

The generic endpoint rows of beta_s and alpha_s now imply

```
X_s<=22+l_beta-l_alpha+xi.
```

17.3.4 The final trim-one stars, with native rebates

Remove the largest core magnitude on each sign. The actual Q67 support has span at most 66+tau-xi, so

|Q67 intersect (Q67+q)|>=f1-q, f1=67-tau+xi.

The two physical-s stars {L_s,V_s} and {L_t,C7_t} are retained and avoid beta_s,alpha_s. Their original-budget inequality is

2X_s>=2f1-epsilon+Gamma_beta+Gamma_alpha.

Combine this with the upper bound on X_s and use Gamma_beta>=l_beta, Gamma_alpha>=l_alpha, and Gamma_beta+Gamma_alpha<=epsilon. It yields

2f1<=44+2xi+2epsilon,
tau+delta>=45+e=62.

This contradicts tau+delta<=60-I. All ranks, outside-hull endpoints, and possible numerical coincidences are included; no normality assumption is needed.

17.4 Routing the retained-pair1 A1 alternatives

First exclude the initially split beta case using only its whole paired capacity. Both beta components active force the holes O_s,O_t,H_s,H_t,C5_t, with pair1's actual sole owner s and every lower own/cross resource and both alphas retained. The whole-pair total hit count N satisfies

77-delta<=N<=77, beta_s+beta_t>=N-M.

Use the endpoints of P_s,alpha_s and the identity alpha_s-P_s=beta_s+beta_t+2d_t+2(a+b)+G. Two leading-t inequalities then give

S_t+M+2X_s+6X_t+g_s+3g_t<=931+2delta+xi.

The native endpoint and leading losses have coefficient at most two; no selected beta component receives its own capacity or deficiency. The weaker physical-source row S_t+M>=T_r+4r+R+109+A+tau_t+4xi_t, together with 2X_s+6X_t>=8f-3epsilon and g_s+3g_t>=4f-3epsilon/2, implies

T_r+4r+13I/2+7tau/2+tau_t+3xi+4xi_t<=316.

But T_r+4r+13(r-34)2ℓ2>=1357/2 for all integers r. This excludes split beta before any sole-beta native loss is licensed.

If pair1 is retained, opposite owners are excluded above. Same owners with alpha_s retained are excluded above. If alpha_s is empty, the owner identity forces C5_t empty; the exact remaining holes are O_s,H_s,O_t,alpha_s,C5_t. Pair1,alpha_t and all eight lower resources are retained, so the branch-independent four-endpoint exclusion applies. Together with the six pair1-empty placements this exhausts A1.

17.5 Branch B

In B, beta_s,beta_t are singleton units, and C5_s,C5_t form a retained paired unit with actual sole owner s. Pair1, when retained, has the same owner. The classification proves integrality and extreme-leaf retention before any native endpoint is used.

If alpha_s is retained, branch B's strict sign gives

alpha_s-C5_s> d_t+2(a+b)+G.

Use its two endpoints and one leading-t inequality. The transport is

S_t+M+2X_s+4X_t+g_s+2g_t
   <R+1029-e+xi.

The physical source lower row and the star/gap bounds, with the weaker e_v>=A+8, imply

T_r+4r+3I+tau_t+3tau/4+4xi_t+5xi/4<427-4e.

The left side is at least 626 and the right side at most 395.

If alpha_s is empty, it is the only extra hole outside the mandatory beta/H choices. Consequently pair1,alpha_t and all eight lower resources are retained, and pair1 has sole owner s. The branch-independent four-endpoint exclusion applies. It borrows no beta-pair capacity from A. These alternatives exhaust B.

The exact arithmetic checker is `check_zero85_z5_a1_b.py`. It checks the displayed integral minima and terminal margins; the physical-target, inventory, endpoint, and budget deductions are the written proof above. The independent final-family audit is `../scalar/z5_c5t_cutoff_audit.md`, with standalone checker `../scalar/check_z5_c5t_cutoff_audit.py`. It independently confirms the physical-target cutoff, all four additional HIGH types, the 26-to-22 tail pruning and the generic-endpoint trim-one contradiction.

18 Independent Z5 C5_t-final cutoff and tail audit

This is a conditional local argument using the cardinality-85 structural/activity conclusions. Exact holes are O_s,H_s,O_t,pair1,C5_t. The retained beta pair has sole actual contributor beta_s. Every resource below is its original full native resource; neither empty pair nor an inactive component is assigned capacity.

Notation: |D|=77, r_s+r_t=68, R=68+xi, A=I+tau+delta<=60, epsilon=delta-e. Source bounds used are K<=M+R+21, M>=R+99+A, and M_t>=R_t+16+A. The fixed ledger initially gives e=13. Generic endpoints have lower73-r_v-l-xi_v and upperS_v+r_o-72+u+xi_o. All selected clips are light because delta<=60<73.

18.1 A direct physical-union proof of the beta cutoff

The exact physical t side has six retained native singletons H_t,U_t,V_t,C6_t,C7_t,alpha_t and its full leading. Grouping them gives actual deficiency at most epsilon, so

`S_t >= K+6*77-epsilon = K+462-epsilon`.

From the unchanged mass identity S_s+S_t=2K+1085-delta,

`S_s <= K+623-e`.

Write z=C5_s,q=alpha_t and beta=beta_s. Suppose beta+s3>=d_s-X_s. The literal identity d_s=2(a+b)+G+z+q then gives

`beta>=M_s+2M_t+2X_t+g_t+z+q`,

`alpha_s=d_s+beta>=3M_s+4M_t+2X_s+4X_t+g_s+2g_t+2z+2q`.

The two lower endpoints of z,q have summed nominal constant

`(73-r_s)+(73-r_t)=78`.

Using their weight2, the upper alpha_s endpoint, the physical cap, and K<=M+R+21 yields

`2M_s+3M_t+2X_s+4X_t+g_s+2g_t <= R+416+r_t-e+E+2xi_s+3xi_t`,

where **E=2l_z+2l_q+u_alpha_s**. The constant is exactly623+21-72-2*78=416. The endpoints z,q,alpha_s are distinct and avoid all CORE8 ordinary pairs.

Let Omega be the combined deficiencies of the four CORE8 pairs. Since the endpoint weights are at most2 and those endpoint resources are disjoint from the ordinary pairs,

`E+Omega<=2epsilon`.

Select the actual four-trim core Q61, with overlap f=61-(tau-xi)/4. Independently, the retained extreme-star forests give

`2X_s+4X_t>=6f-2epsilon`.

The four ordinary pairs give 2g_s>=2f-Omega_s and 2g_t>=2f-Omega_t; hence

`g_s+2g_t>=3f-Omega`.

The source lower bound is

`2M_s+3M_t=2M+M_t>=2R+R_t+214+3A`.

Combine these rows and R_t=r_t+xi_t to obtain

`3A+9f <=134-e+4epsilon+xi`.

This deliberately permits the ordinary gap groups and stars separate copies of the same deficit. It makes no claim that they are disjoint. Substitution gives

`415+3I+3tau/4+5xi/4-delta+5e<=0`.

But delta<=60-I-tau, so

`355+4I+7tau/4+5xi/4+5e<=0`.

Its left side is at least420 at e=13. Thus beta+s3<d_s-X_s, strictly, for every position of all selected endpoints. No normality assumption is used.

18.2 Independent audit of the four-label tail pruning

The actual 32-type upper-tail inventory in the original report explicitly contains U2_s,U3_s, U4_s,V2_s. The six removed beta fixed types are V3_s,...,V8_s, so these four labels remain distinct inventory types after the beta cutoff.

The cutoff and the initial 32-type classification leave only the alpha_s source family and at most26 fixed types in (d_s-X_s,S_s]. Its source capacity is77, so

`S_s<=d_s-X_s+103-Gamma_alpha`.

Since beta_s has a core hit, beta_s>=1-R_s. Thus h_s=beta_s+d_s+M_s+X_s satisfies

`h_s-S_s>=e_s+1+2X_s-103+Gamma_alpha`.

The full-core two-edge physical-s stars give2X_s>=274-2R-epsilon. With e_s>=A+16 and xi<=tau,

`h_s-S_s>=52+I+tau-2xi+e>=-8+2I+delta+e>0`.

Therefore U4_s=h_s and U3_s=h_s+g_s are HIGH. Also

`V2_s=beta_s+s2>=h_s+e_s+X_s+g_s+1>h_s`,

so V2_s is HIGH. Actual beta activity implies U2_s>H_s, so U2_s is HIGH independently. These four additional HIGH labels are distinct from the eleven selected LOW labels. The LOW set still touches nine baseline groups, so the ledger becomes HIGH15, LOW11, touched9 and raises e13 to e17.

Pruning these four HIGH types leaves at most22 fixed types in the same tail. Hence

`S_s<=d_s-X_s+99-Gamma_alpha`.

Generic complete-target endpoints for alpha_s=d_s+beta_s, with l_alpha+u_alpha<=Gamma_alpha, now give

`X_s<=22+xi+l_beta-l_alpha`.

Choose the actual one-trim core Q67, with overlap f1=67-tau+xi. The two physical-s stars are disjoint from the native beta and alpha resources, so

`2X_s>=2f1+Gamma_beta+Gamma_alpha-epsilon`.

Combining, then using l_beta<=Gamma_beta and Gamma_beta<=epsilon, gives

`2f1<=44+2xi+2epsilon`.

Therefore tau+delta>=45+e=62, contradicting A<=60. This closes the conditional final family without a normality bootstrap.

Scope: this independently checks the local transport, resource-disjoint loss use, constants, and listed tail types. It relies on the structural/source/activity conclusions and the exhaustive 32-type representation classification.

18.3 Final independent skim of the other A1/B transports

The newly written `../zero/zero85_z5_a1_b.md` was independently reconstructed against its literal center identities and generic source endpoints. Opposite owners, the pair1-empty C5_s/U_s placements, H_t placement, alpha_t placement, and branch B with retained alpha_s have correct nominal constants, source decompositions, and separately valid star/gap budgets. HIGH-alpha's displayed scalar bounds are conservative; retaining its displayed +10 source-mass term actually strengthens928 to908, so its contradiction remains valid.

Two nonblocking transcription errors were found and corrected by the source author before packaging:

* Same-owner alpha_s-retained transport leaves **M_s**, not M, after K<=M+R+21. Its corrected scalar row is `T_r+4r+(5/2)I+tau_t+tau/2+4xi_t<=520-7e/2`. The integer quadratic minimum608 at r28 exceeds492 at e8, margin116. * The branch-independent four-endpoint row has radial terms **-xi_s/2+7xi_t/2**, replacing the originally stronger positive terms. Its correction still closes: `tau/2-xi_s/2>=tau_t/2>=0`, so the left side remains at least Q3>=626 versus upper560.

The updated proof/checker therefore receives an independent **PASS**, conditional on the actual retention grammar and structural/source prerequisites. Its routed A1 and B alternatives use only retained native resources and already licensed sole pair components; the branch-

independent card assigns no beta capacity.

19 Independent final audit of the Z4 retention cards

Verdict: **PASS after the Card S light-row sign correction already applied by the zero-hole author.** This is an analytical audit of actual native resources, endpoint constants, and loss budgets, in addition to the scalar replay. It is conditional on the proved cardinality-85 structural/activity classification and source lemmas.

Common fresh parameters are |D|=77, r_s+r_t=68, S=2K+1162-delta, R=68+xi, A=I+tau+delta, epsilon=delta-e, K<=M+R+21, L>=A+20 and A<=137. The card calculations deliberately retain the weaker valid cap143. Generic endpoints have lower73-r_v and upperS_v+r_o-72, with their actual clipping/rank errors. Choose the actual four-trim Q61 with f=61-(tau-xi)/4.

19.1 Physical target capacities

Every physical target bound uses a group of its original retained full leading and ordinary resources. It is a union-deficiency bound and requires no disjointness of their target images.

Card	Opposite physical t ordinary native resources	Capacity	Consequent cap on S_s
E-H	H_t,U_t,V_t,C5_t,C6_t,C7_t,alpha_t	K+539	K+623-e
Owner alpha_s missing	Same seven	K+539	K+623-e
Opposite alpha_t missing	H_t,U_t,V_t,C5_t,C6_t,C7_t	K+462	K+700-e
E-C5	H_t,U_t,V_t,C6_t,C7_t, alpha_t	K+462	K+700-e
S	H_t,U_t,V_t,C6_t,C7_t, alpha_t	K+462	K+700-e

In Card S, C5_t is inactive by the same-owner identity. In branch A beta has actual sole owner s; in branch B beta_t is the fourth whole hole and the second paired capacity belongs to its actual s owner. Thus no unused paired component is included in the physical t count. All fixed credit is already in epsilon.

19.2 Card F

The endpoint identity in the first two applications is

`H_s=P_s+2alpha_s+alpha_t+C5_t+4(M+X)+2G`.

Its exact fresh nominal constant is

`3(73-r_s)+2(73-r_t)+(72-r_t)=233`.

The third application has the reflected identity with sole native beta_s and no use of the empty pair1. The endpoint constant is again233. The six-edge forest has W=2X+G, D_F>=6f-W and N+D_F<=epsilon. Its weighted clipping E<=2N, so W*2-E>=12f-2epsilon. Endpoints and retained forest vertices are distinct.

If the one U/V vertex is empty, the actual separated positive/negative restricted zones (M>R) license the same charge to retained full resources, including an empty middle V, with zero capacity assigned to that vertex. This is the literal one-empty-vertex argument, not restoration of its capacity.

Adding the opposite source mass T_r+tau_res+4R_r+10 to the endpoint side and using the global source bound gives

`T_r+4r+tau_res+2L-3tau+2e<=229+delta+B`,

after dropping nonnegative radial terms. All-light B<=2epsilon yields the stated Q3 row332. A unique heavy endpoint of weight q=1,2 gives B<=2epsilon-qc+q*tau+q*e_v and e_v<=L; the stated upper332+306q follows exactly. The minima626/649/665 and native c>=73,delta>=8+c give all claimed strict contradictions.

19.3 Empty-beta E cards

The following constants were reconstructed directly from the literal endpoint identities, the physical capacities above, and K<=M+R+21:

Card	Fresh nominal endpoint constant	Body after source magnitudes	Final master constant
E-H	228-r_t	3a+4b+g_s+2g_t	416
Owner alpha_s missing	225-2r_t	2a+4b+2G, using raw alpha_s>=1-a	419
Opposite alpha_t missing	161+r_t	4a+3b+2G, using raw alpha_t>=1-b	492
E-C5	228-r_t	3a+4b+2G	493

For example E-H gives S_s>=228-r_t+3M_s+4M_t+W-E-B. Physical cap<=M+R+644-e removes M and leaves2M_s+3M_t. Using2M_s+3M_t=2R+2L+r_t+xi_t+e_t then produces constant416, without any residual source-mass estimate.

The forests and native endpoint losses have precisely these budgets:

Card	Actual forest edges	Endpoint clipping/forest budget
E-H	Six edges plus four stars; stars contain V_s only	E+D6+Dstar<=2epsilon-2N+h
Owner alpha missing	Eight double-arm edges plus two opposite stars; all four endpoints avoided	E+D8+D2<=2epsilon-N
Opposite alpha missing	Six edges plus five edges; all weighted endpoints avoided	E+D6+D5<=2epsilon
E-C5	Seven edges plus four edges; all five endpoints avoided	E+D7+D4<=2epsilon-N

These are separate valid inequalities with the same original deficit. Overlapping forest systems are not declared one disjoint group. The E-H upper endpoint's overlap with the star system is exactly the +h correction in Jstar=E+B-2N+h. Its light lower terms are nonpositive because b<=clip; the upper term is at most its actual rank error. A heavy lower term contributes at most xi_v+e_v-4-c.

All published E scalar finishes are valid. Two clarifications are useful:

* For owner-alpha heavy on t, the exact cancellation gives RHS-9/2; the published RHS-1/2 is merely weaker and remains contradictory. * In opposite-alpha, the only weight2 heavy endpoint is alpha_s, whose error source is s. Thus2L+e_s-2e_s=L+e_t>=A+24. For weight1, either physical source leaves at least A+24. The bound would not hold for an invented weight2/source-t endpoint, but that endpoint is absent from the literal list.

19.4 Cards O and S

The external/source reviewer independently rederived Card O's full endpoint master1006, its native rebates and all four endpoint-heavy flags. The nominal constant in the V_t identity is150; the weighted leading rebate retains E_clip+2ell_s<=2epsilon-(B+H). The actual CORE8 forest vertices and endpoints are retained and disjoint exactly where rebates are used. The scalar minima Q4>=649 and4tau+10f>=1990/3 give the displayed strict margins. No empty component is used.

For Card S I independently reconstructed nominal constant155 and the physical-union master `2M+9f+4e+2N<=545+3delta+B`.

The leading t and the six/three-edge forests avoid all four endpoints. Hence E+ell_t+D6+D3<=3epsilon-2N, exactly as needed. The heavy scalar row is correct.

The all-light row originally printed a wrong sign/coefficient on delta. Its correct form is `(9/4)I-(3/4)delta+4e+2N<=-577/4`.

It still strictly contradicts e11,delta<=143: its left side is at least-253/4, a margin81. The source author has applied this correction in both the proof notes and aggregate checker.

All exact A-E/A-H/B retention rows dispatched to these cards are therefore excluded, and the separately audited ownerless certificate completes Z4. This receipt checks local analytical applicability; it does not treat scalar replay as a configuration classifier.

The added two-beta-hole B pre-dispatch row was also independently checked. Insert the fresh source masses, K-M>=R-132+A, M_v>=r_v+4, and the retained three-arm row2X_v+g_v>=189-3tau/4-delta into its displayed master1161. With R>=68,A<=143 the exact final row is `2T_r+9r+4I+7tau/4+2tau_v<=1063`. Its global integer minimum1192 at r26 gives margin129. The owner-side HIGH beta alternative contradicts2d_v<=M_o-1 versus actual pair1 activity d_v>M. Thus this extra B form does not leave an undispatched case.